

Task – Students will perform security posture assessment

Lab requirement

- 1 linux server (192.168.88.155/24)
 - Internet active: yes
- 1 windows server (192.168.88.158/24)
 - Internet active: yes
- Using NMAP as CLI for tracing details on linux.
- Installing & Using Nessus web-based tool for gathering information about target.

On linux server:

```
[root@server ~]# nmap -sn 192.168.88.0/24
Starting Nmap 7.92 ( https://nmap.org ) at 2026-01-07 22:49 IST
Nmap scan report for 192.168.88.1
Host is up (0.00036s latency).
MAC Address: 00:50:56:C0:00:08 (VMware)
Nmap scan report for 192.168.88.2
Host is up (0.00014s latency).
MAC Address: 00:50:56:F4:DC:E9 (VMware)
Nmap scan report for 192.168.88.155
Host is up (0.00018s latency).
MAC Address: 00:0C:29:49:A3:C0 (VMware)
Nmap scan report for 192.168.88.254
Host is up (0.00014s latency).
MAC Address: 00:50:56:FE:45:49 (VMware)
Nmap scan report for 192.168.88.151
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 2.03 seconds
[root@server ~]#
```

Command

- “-sn” – it checks only hosts are up or not. It sends ICMP echo ping request to port 443.

Scanning port and service using “nmap -sS -sV -O 192.168.88.155”

```
[root@server ~]# nmap -sS -sV -O 192.168.88.155
Starting Nmap 7.92 ( https://nmap.org ) at 2026-01-08 11:49 IST
Nmap scan report for 192.168.88.155
Host is up (0.00025s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
80/tcp    open  http           Microsoft IIS httpd 10.0
135/tcp   open  msrpc          Microsoft Windows RPC
139/tcp   open  netbios-ssn    Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds   Microsoft Windows Server 2008 R2 - 2012 microsoft-ds
MAC Address: 00:0C:29:49:A3:C0 (VMware)
Device type: general purpose
Running: Microsoft Windows 2016
OS CPE: cpe:/o:microsoft:windows_server_2016
OS details: Microsoft Windows Server 2016 build 10586 - 14393
Network Distance: 1 hop
Service Info: OSs: Windows, Windows Server 2008 R2 - 2012; CPE: cpe:/o:microsoft:windows

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 18.47 seconds
```

Command:

- “-sS” = tcp syn scan
- “-sV” = version detection
- “-O” = detect OS version

After installing DNS role on Windows server, run same command "nmap -sS -sV -O 192.168.88.156"

```
[root@server ~]# nmap -sS -sV -O 192.168.88.155
Starting Nmap 7.92 ( https://nmap.org ) at 2026-01-08 12:13 IST
Nmap scan report for 192.168.88.155
Host is up (0.00034s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
53/tcp    open  domain       Simple DNS Plus
80/tcp    open  http         Microsoft IIS httpd 10.0
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds Microsoft Windows Server 2008 R2 - 2012 microsoft-ds
MAC Address: 00:0C:29:49:A3:C0 (VMware)
Device type: general purpose
Running: Microsoft Windows 2016
OS CPE: cpe:/o:microsoft:windows_server_2016
OS details: Microsoft Windows Server 2016 build 10586 - 14393
Network Distance: 1 hop
Service Info: OSs: Windows, Windows Server 2008 R2 - 2012; CPE: cpe:/o:microsoft:windows
```

Task – Students will use OSINT to gather information

Tools

- Using WHOIS database on wipro.com domain.

```
(kali@kali)-[~]  
$ whois wipro.com  
Domain Name: WIPRO.COM  
Registry Domain ID: 2559888_DOMAIN_COM-VRSN  
Registrar WHOIS Server: whois.registrar.amazon  
Registrar URL: http://registrar.amazon.com  
Updated Date: 2025-10-28T11:30:36Z  
Creation Date: 1992-12-08T05:00:00Z  
Registry Expiry Date: 2030-12-07T05:00:00Z  
Registrar: Amazon Registrar, Inc.  
Registrar IANA ID: 468  
Registrar Abuse Contact Email: trustandsafety@support.aws.com  
Registrar Abuse Contact Phone: +1.2024422253  
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited  
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited  
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited  
Name Server: NS-1145.AWSDNS-15.ORG  
Name Server: NS-1575.AWSDNS-04.CO.UK  
Name Server: NS-326.AWSDNS-40.COM  
Name Server: NS-659.AWSDNS-18.NET  
DNSSEC: signedDelegation  
DNSSEC DS Data: 31918 13 2 0074B6D159450F1E12E2243EFEA93235A69CBDB77A1CD6A795EEA4F87E7A5C9E  
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/  
>>> Last update of whois database: 2026-01-08T14:28:14Z <<<
```

Other useful information:

```
Registrant City: Hayes  
Registrant State/Province: Middlesex  
Registrant Postal Code: UB3 9TR  
Registrant Country: GB  
Registrant Phone: +44.1483307527  
Registrant Phone Ext:  
Registrant Fax: +44.1483304031  
Registrant Fax Ext:  
Registrant Email: 7649f398-8b17-4de3-9427-9d9020cca728@identity-protect.org  
Registry Tech ID: Not Available From Registry  
Tech Name: On behalf of wipro.com owner  
Tech Organization: Identity Protection Service  
Tech Street: PO Box 786  
Tech City: Hayes  
Tech State/Province: Middlesex  
Tech Postal Code: UB3 9TR  
Tech Country: GB  
Tech Phone: +44.1483307527  
Tech Phone Ext:  
Tech Fax: +44.1483304031  
Tech Fax Ext:  
Tech Email: 7649f398-8b17-4de3-9427-9d9020cca728@identity-protect.org  
Name Server: NS-659.AWSDNS-18.NET  
Name Server: NS-326.AWSDNS-40.COM  
Name Server: NS-1575.AWSDNS-04.CO.UK  
Name Server: NS-1145.AWSDNS-15.ORG
```

Using dig command for domain Wipro.com for fetching Public IP address.

```
(kali㉿kali)-[~]  
$ dig wipro.com  
  
; <<>> DiG 9.20.9-1-Debian <<>> wipro.com  
;; global options: +cmd  
;; Got answer:  
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 63955  
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1  
  
;; OPT PSEUDOSECTION:  
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 4096  
;; QUESTION SECTION:  
;wipro.com.                IN      A  
  
;; ANSWER SECTION:  
wipro.com.                 5       IN      A      18.205.78.10  
  
;; Query time: 44 msec  
;; SERVER: 192.168.88.2#53(192.168.88.2) (UDP)  
;; WHEN: Thu Jan 08 09:30:46 EST 2026  
;; MSG SIZE rcvd: 54
```

Verifying it using PING:

```
C:\Users\Jeetu>ping wipro.com  
  
Pinging wipro.com [18.205.78.10] with 32 bytes of data:  
Request timed out.  
Request timed out.  
Request timed out.  
Request timed out.  
  
Ping statistics for 18.205.78.10:  
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Using DNSDUMPSTER to gather global information:

kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB

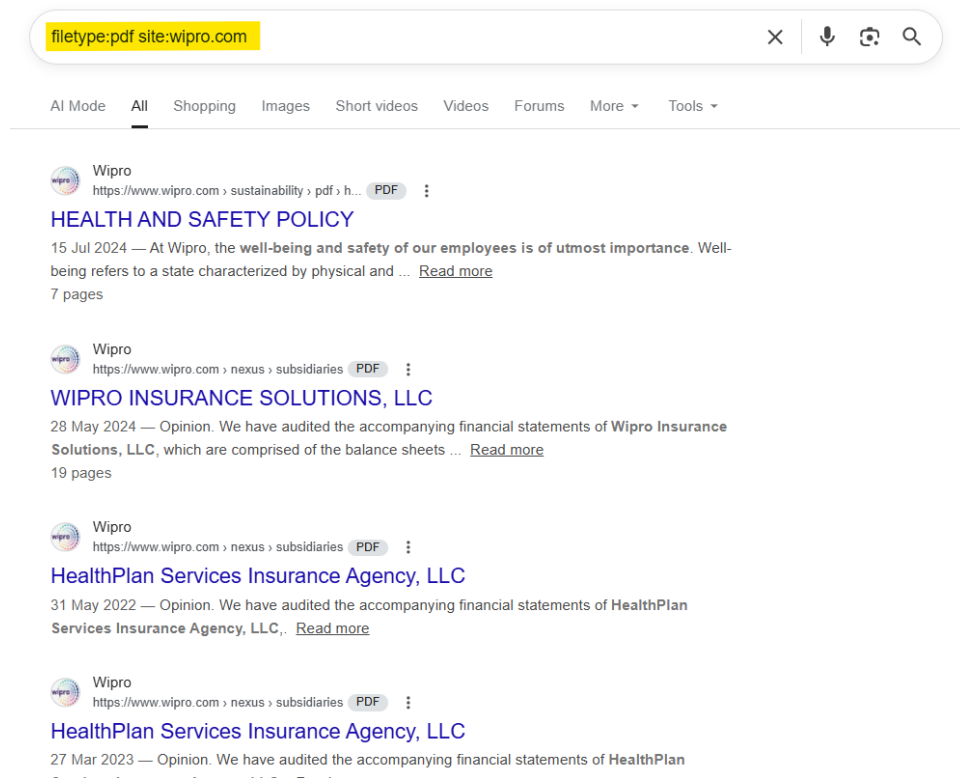
https://dnsdumpster.com 130% ☆

A Records (subdomains from dataset)

Host	IP	ASN	ASN Name	Open Services (from DB)	RevIP
apiaccountservices.wipro.com	203.91.197.71	ASN: 23664 203.91.197.0/24	WIPRO-TECH-AS-AP Wipro Technologies, IN India		1 ⋮
autodiscover.wipro.com	203.91.193.61	ASN: 23664 203.91.193.0/24	WIPRO-TECH-AS-AP Wipro Technologies, IN India		1 ⋮
azeu-p-mcafgnt.wipro.com	52.224.135.151	ASN: 8075 52.224.0.0/11	MICROSOFT-CORP-MSN-AS-BLOCK United States		1 ⋮
base.wipro.com	203.91.193.54	ASN: 23664 203.91.193.0/24	WIPRO-TECH-AS-AP Wipro Technologies, IN India		1 ⋮
blrweb-ext.wipro.com	203.91.193.123	ASN: 23664 203.91.193.0/24	WIPRO-TECH-AS-AP Wipro Technologies, IN India		1 ⋮
cellentvpn.wipro.com	62.154.234.100	ASN: 3320 62.154.0.0/15	DTAG Internet service provider operations, DE Germany		1 ⋮
cloudcar.wipro.com	3.136.185.207 ec2-3-136-185-207.us-east-2.compute.amazonaws.	ASN: 16509 3.136.0.0/13	AMAZON-02 United States		2 ⋮

Task – Students will use Google dorks for performing various operations

Google Dork: **filetype:pdf site:wipro.com**



The screenshot shows the Google search interface with the query "filetype:pdf site:wipro.com" entered in the search bar. The search results are filtered to show PDF files from the wipro.com domain. The first result is titled "HEALTH AND SAFETY POLICY" and is dated 15 Jul 2024. The second result is titled "WIPRO INSURANCE SOLUTIONS, LLC" and is dated 28 May 2024. The third result is titled "HealthPlan Services Insurance Agency, LLC" and is dated 31 May 2022. The fourth result is also titled "HealthPlan Services Insurance Agency, LLC" and is dated 27 Mar 2023.

filetype:pdf site:wipro.com

AI Mode All Shopping Images Short videos Videos Forums More Tools

Wipro
https://www.wipro.com › sustainability › pdf › h... PDF
HEALTH AND SAFETY POLICY
15 Jul 2024 — At Wipro, the well-being and safety of our employees is of utmost importance. Well-being refers to a state characterized by physical and ... [Read more](#)
7 pages

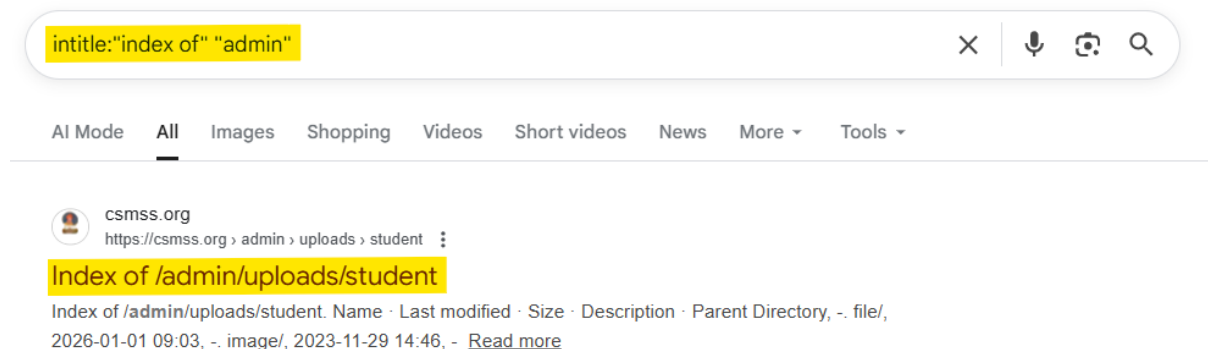
Wipro
https://www.wipro.com › nexus › subsidiaries PDF
WIPRO INSURANCE SOLUTIONS, LLC
28 May 2024 — Opinion. We have audited the accompanying financial statements of Wipro Insurance Solutions, LLC, which are comprised of the balance sheets ... [Read more](#)
19 pages

Wipro
https://www.wipro.com › nexus › subsidiaries PDF
HealthPlan Services Insurance Agency, LLC
31 May 2022 — Opinion. We have audited the accompanying financial statements of HealthPlan Services Insurance Agency, LLC,. [Read more](#)

Wipro
https://www.wipro.com › nexus › subsidiaries PDF
HealthPlan Services Insurance Agency, LLC
27 Mar 2023 — Opinion. We have audited the accompanying financial statements of HealthPlan

Another example of Google Dork

intitle:"index of" "admin"



The screenshot shows the Google search interface with the query "intitle:\"index of\" \"admin\"" entered in the search bar. The search results are filtered to show pages with the title "index of" and the word "admin" in the title. The first result is titled "Index of /admin/uploads/student" and is from the domain csmss.org. The result shows a list of files and directories, including "2026-01-01 09:03, -, image/, 2023-11-29 14:46, -".

intitle:"index of" "admin"

AI Mode All Images Shopping Videos Short videos News More Tools

csmss.org
https://csmss.org › admin › uploads › student
Index of /admin/uploads/student
Index of /admin/uploads/student. Name · Last modified · Size · Description · Parent Directory, -, file/,
2026-01-01 09:03, -, image/, 2023-11-29 14:46, - [Read more](#)

Index of /admin/uploads/student/file

Name	Last modified	Size	Description
Parent Directory		-	
2-student.pdf	2023-11-25 18:15	26K	
3-student.pdf	2023-11-25 18:15	26K	
4-student.pdf	2023-11-25 18:14	756K	
10-student.jpg	2023-11-25 18:17	289K	
10-student.pdf	2023-11-25 18:17	397K	
11-student.pdf	2023-11-25 18:17	207K	

Using Exiftool to get the information of an image, video or PDFs.

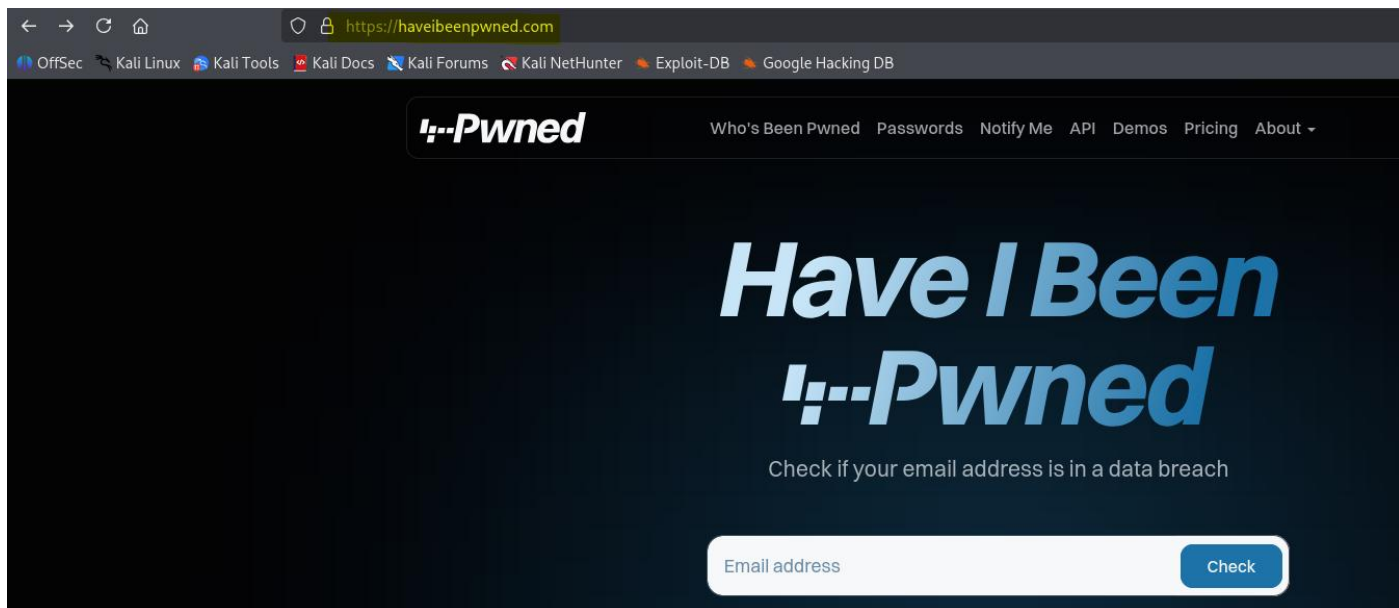
- First you need to download any image from internet. (ex - <https://melangeoftales.com/wp-content/uploads/2024/11/AdiyogiAarthy2-2.jpeg>)
- Then go to the downloaded location and use "exiftool <filename>.extension"

```
(kali㉿kali)-[~/Downloads]
$ exiftool AdiyogiAarthy2-2.jpeg
ExifTool Version Number      : 13.25
File Name                    : AdiyogiAarthy2-2.jpeg
Directory                    : .
File Size                    : 144 kB
File Modification Date/Time   : 2026:01:08 10:02:36-05:00
File Access Date/Time        : 2026:01:08 10:02:36-05:00
File Inode Change Date/Time   : 2026:01:08 10:02:36-05:00
File Permissions              : -rw-rw-r--
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
JFIF Version                 : 1.01
Resolution Unit              : None
X Resolution                  : 72
Y Resolution                  : 72
Image Width                   : 1024
Image Height                  : 768
Encoding Process              : Progressive DCT, Huffman coding
Bits Per Sample               : 8
Color Components              : 3
Y Cb Cr Sub Sampling         : YCbCr4:2:0 (2 2)
Image Size                    : 1024x768
Megapixels                    : 0.786
```

I have tried it on an image that I created using Adobe Photoshop CS6

```
(kali㉿kali)-[~/Desktop]
$ exiftool jeetu-creative3.jpg
ExifTool Version Number      : 13.25
File Name                    : jeetu-creative3.jpg
Directory                    : .
File Size                    : 1275 kB
File Modification Date/Time   : 2022:08:20 01:25:14-04:00
File Access Date/Time        : 2026:01:08 10:07:26-05:00
File Inode Change Date/Time   : 2026:01:08 10:07:25-05:00
File Permissions              : -rw-----
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
Exif Byte Order               : Big-endian (Motorola, MM)
Photometric Interpretation    : RGB
Orientation                   : Horizontal (normal)
```

You can check if an email ID has been ever breached or not using “<https://haveibeenpwned.com/>”



The screenshot shows the homepage of the 'Have I Been Pwned' website. The browser's address bar displays 'https://haveibeenpwned.com'. The website has a dark blue background. At the top, there is a navigation bar with the 'Have I Been Pwned' logo on the left and a list of links on the right: 'Who's Been Pwned', 'Passwords', 'Notify Me', 'API', 'Demos', 'Pricing', and 'About'. The main content area features the large text 'Have I Been Pwned' in a light blue font. Below this, a subtitle reads 'Check if your email address is in a data breach'. At the bottom, there is a search form with a text input field labeled 'Email address' and a blue 'Check' button.

Task – Students will use shodan.io for analysis

Set up your environment

- Create a **Shodan account** (free tier is enough for basic queries).
- Use a **test domain or IP range you own** (or a lab VM with internet exposure).

Query: 'port:80 country:IN org:"amazon"'

The screenshot shows the Shodan search interface. The search bar contains the query 'port:80 country:IN org:"amazon"'. The results show 320,542 total results. The top cities are listed as Mumbai (309,014), Hyderabad (8,383), Kolkata (3,140), and Delhi (4). The top result is for 'ec2-3-7-99-76.ap-south-1.compute.amazonaws.com', which is an Amazon Data Services India instance in Mumbai. The result details include the IP address, the organization name, the location, and the server information (HTTP/1.1 200 OK, Date: Thu, 08 Jan 2026 14:39:03 GMT, Content-Type: application/json, Content-Length: 21, Connection: keep-alive, server: uvicorn).

TOTAL RESULTS
320,542

TOP CITIES

Mumbai	309,014
Hyderābād	8,383
Kolkata	3,140
Delhi	4

View Report View on Map Advanced Search

Product Spotlight: We've Launched a new API for Fast Vulnerability Lookups. Check out [CVEDB](#)

3.7.99.76

ec2-3-7-99-76.ap-south-1.compute.amazonaws.com
Amazon Data Services India
India, Mumbai
cloud

HTTP/1.1 200 OK
Date: Thu, 08 Jan 2026 14:39:03 GMT
Content-Type: application/json
Content-Length: 21
Connection: keep-alive
server: uvicorn

Note – You need license for in-depth scanning

Another query "apache 2.4.49"

The screenshot shows the Shodan search interface. The search bar contains the query 'apache 2.4.49'. The results show 1,394 total results. The top countries are listed as United States (Cedar Knolls). The top result is for '172.104.208.100', which is a Linode server in the United States. The result details include the IP address, the organization name, the location, and the server information (HTTP/1.1 301 Moved Permanently, Date: Thu, 08 Jan 2026 14:16:01 GMT, Server: Apache/2.4.49 (Debian), Location: https://mailadmin.nerdyit.com/, Content-Length: 319, Content-Type: text/html; charset=iso-8859-1).

TOTAL RESULTS
1,394

TOP COUNTRIES

United States, Cedar Knolls

View Report View on Map Advanced Search

Product Spotlight: Free, Fast IP Lookups for Open Ports and Vulnerabilities using [InternetDB](#)

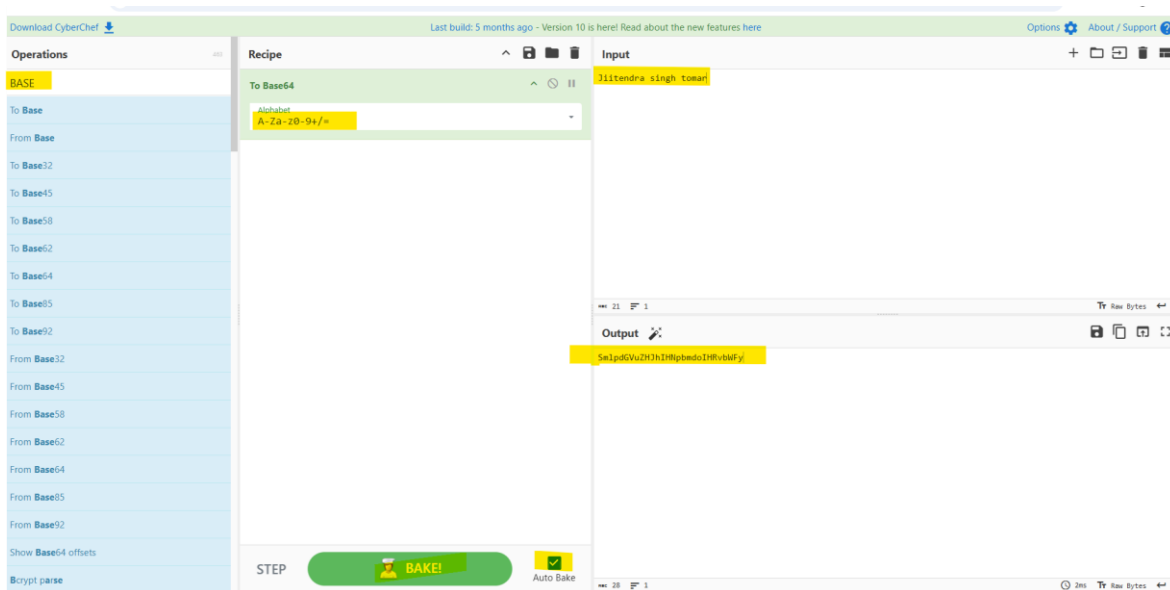
301 Moved Permanently

172.104.208.100
apollo.nerdyit.com
Linode
United States, Cedar Knolls
cloud

HTTP/1.1 301 Moved Permanently
Date: Thu, 08 Jan 2026 14:16:01 GMT
Server: Apache/2.4.49 (Debian)
Location: https://mailadmin.nerdyit.com/
Content-Length: 319
Content-Type: text/html; charset=iso-8859-1

1. Go to web browser and type “CyberChef” and select the 1st link.
2. Once you reach <https://gchq.github.io/CyberChef/>
3. You can see 4 sections
 - a. **Operations** – what to do?
 - b. **Recipe** – users arrange and configure the sequence of operations called recipes.
 - c. **Input** – data that needs to be processed.
 - d. **Output** – displays the results of the operation applied to the input.

- Go to operations, and search “Base64”
- Once you see “To Base64”, drag and drop it to the recipe section.
- In the input section, give the string you want to convert
- In the output section, you can see the converted data.



The screenshot shows the CyberChef web application. On the left is a sidebar with a list of recipes: 'Analyse hash', 'Generate all checksums', 'Generate all hashes', 'MD2', 'MD4', 'MD5' (highlighted in yellow), 'MD6', 'SHA0', 'SHA1', 'SHA2', 'SHA3', 'SM3', 'Keccak', and 'Chacha'. The main area is divided into three sections. The top section is the 'Recipe' editor, which has a header bar with icons for undo, redo, and delete, and a dropdown menu. Below this is a list of recipes, with 'MD5' selected and highlighted in yellow. The middle section is the 'Input' field, which contains the text 'pass@word1' and is highlighted in yellow. The bottom section is the 'Output' field, which displays the result '29821b0e714352428d6bc19a9cb595ad' and is also highlighted in yellow. At the top of the interface, there is a green banner with the text 'Download CyberChef' and a download icon, and a status bar indicating 'Last build: 5 months ago - Version 10 is here! Read about the new features here'.

Download CyberChef

PGP Verify

PGP Encrypt and Sign

PGP Decrypt and Verify

Generate RSA Key Pair

RSA Sign

RSA Verify

RSA Encrypt

RSA Decrypt

Generate ECDSA Key Pair

ECDSA Signature Conversion

ECDSA Sign

ECDSA Verify

Parse SSH Host Key

Parse CSR

Public Key from Certificate

Public Key from Private Key

SM2 Encrypt

SM2 Decrypt

Last build: 5 months ago - Version 10 is here! Read about the new features here

Recipe

Generate RSA Key Pair

RSA Key Length

2048

Output Format

PEM

5

1

Output

```

-----BEGIN PUBLIC KEY-----
MIIBIjANBgkqhkiG9w0BAQFAAQCAQ8AMIIBCgKCAQEAE4fJ3Tuj4GZHyXo9GInQV
ZLB1okBjVmo5Tzb/XutgpFCyAKI7JTRcFchNP6pGTN9v26A4Gcy3DB4czkncmJW1
2L4n4tArexciv4htuu6yqfJEkr2AKS5vrydum3bUoM4TF/Mo1WsGh9Mxqb+ag23
utkgbdp6rRkpng06GFfXbkvJUux6yZAA1cqEakgnVW+w8LFU8WmhBKgvGfM9uJot
XHRpzRbT5aVuDY2s1BtwMw+aog5I2VfDv5HhgSy45z84tj/uCA4Bk6eTatyTmGid
4evX44QK+E/AGfGcIIoJCBoo/BmPPTY457mVYEElnUEDP+w1bDUB4a3TNaqD8kQD
BQIDAQAB
-----END PUBLIC KEY-----
-----BEGIN RSA PRIVATE KEY-----
MIIEowIBAAKCAQEAE4fJ3Tuj4GZHyXo9GInQVZLB1okBjVmo5Tzb/XutgpFCyAKI7
JTRcFchNP6pGTN9v26A4Gcy3DB4czkncmJW12L4n4tArexciv4htuu6yqfJEkr2
AKS5vrydum3bUoM4TF/Mo1WsGh9Mxqb+ag23utkgbdp6rRkpng06GFfXbkvJUux6
yZAA1cqEakgnVW+w8LFU8WmhBKgvGfM9uJotXHRpzRbT5aVuDY2s1BtwMw+aog5I
2VfDv5HhgSy45z84tj/uCA4Bk6eTatyTmGid4evX44QK+E/AGfGcIIoJCBoo/BmP
PTY457mVYEElnUEDP+w1bDUB4a3TNaqD8kQDQIDAQABaoIBAG5LdWHRJ+cXgugc
6WAVKYJkUvINWu0cQeQbs9vd/AN9bASDWmmaJMoYdGgtaV95rUfu3L9es6b/nTG
wnvrjvo+QGs5MiV2kdxOBzCDVt3mNR3vYlHEpGueWCJy0P6w1nEVzKcrygTAZBK
EtuDUB5Qkk99uKe54kJoc4UBk8HeTcXg3I14VFTdY/pjUUUwCcjK6/LL7Mq6ZTeu
CZzraEYm50s4K5uP28EpKb1xaBWzd4UjPRRA+4L2aBgDVJGTxaHPa/0pW3kVdN
lNmWdhsGIZ659f0E12PbCzwnCzF28an2khIFcETnf3GrUr7cXkMowf1oAaCpFY
Qb6qKUsCgYEY/HnrrCLtaZgZvJuoI0DKEEv0tA5U8aSpqbqabR1rYB1u3JPNYKXz
uIM/HUL444T1lodmFL2Ya02uVBtU10cD4jYReddudxHlUGbTscvKFE1ZIR75gZ
nPLMKFXGfAZe1gtPu1e11c7o/P4b50ciurD2W78agLr7WbtTnh9cGcYEASRnC
T06f4z78z23uY6z0P8z6CvK4a4p456z0DEK4z0B5T7M34w00P8z7z3z07z

```

Task 4 – Creating encoded URL (by hackers)

Download CyberChef [↓](#) Last build: 5 months ago - Version 10 is here! Read about the new features [here](#)

Operations	Recipe	Input
URL	To Base64 Alphabet A-Za-z0-9+/=	https://google.com
Fang URL	URL Encode ☐ Encode all special chars	
Defang URL		
URL Decode		
URL Encode		
Extract URLs		
Split Colour Channels		
Randomize Colour Palette		
Image Hue/Saturation/Lightness		
To Quoted Printable		
From Quoted Printable		
Extract domains		
Fernet Decrypt		
Fernet Encrypt		
Parse URL		

Output

aHR0cHM6Ly9nb29nbGUuY29t

Task 5 – decoding the encoded URL (using CyberChef, used by SOCs)

Input - aHR0cHM6Ly9nb29nbGUuY29t

Operations	Recipe	Input
URL	From Base64 Alphabet A-Za-z0-9+/= ☒ Remove non-alphabet chars	aHR0cHM6Ly9nb29nbGUuY29t
Fang URL	☐ Strict mode	
Defang URL	URL Encode ☐ Encode all special chars	
URL Decode		
URL Encode		
Extract URLs		
Split Colour Channels		
Randomize Colour Palette		
Image Hue/Saturation/Lightness		
To Quoted Printable		
From Quoted Printable		
Extract domains		
Fernet Decrypt		

Output

https://google.com

Task – Scan a website with the help of a website scanner from Qualys and submit your analysis report on the same

Checking the SSL using URL: <https://www.ssllabs.com/ssltest/>

1st website

ssllabs.com/ssltest/analyze.html?d=www.techademy.com

**Qualys. SSL Labs**[Home](#)[Projects](#)[Qualys Free Trial](#)[Contact](#)

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SSL Report: **www.techademy.com**

Assessed on: Thu, 08 Jan 2026 17:43:05 UTC | [Hide](#) | [Clear cache](#)

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	Server	Test time	Grade
1	66.33.60.193 Ready	Thu, 08 Jan 2026 17:41:32 UTC Duration: 45.706 sec	A+
2	76.76.21.241 Ready	Thu, 08 Jan 2026 17:42:18 UTC Duration: 46.973 sec	A+

SSL Report v2.4.1

2nd website

You are here: [Home](#) > [Projects](#) > [SSL Server Test](#) > www.rpsconsulting.in

SSL Report: **www.rpsconsulting.in** (172.184.162.67)

Assessed on: Thu, 08 Jan 2026 17:44:26 UTC | [Hide](#) | [Clear cache](#)

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Summary

Overall Rating

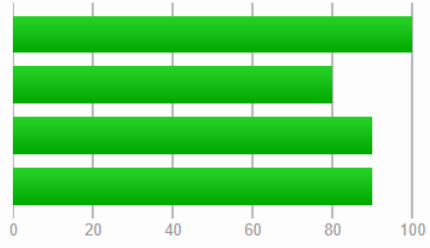


Certificate

Protocol Support

Key Exchange

Cipher Strength



Visit our [documentation page](#) for more information, configuration guides, and books. Known issues are documented [here](#).

This server does not support TLS 1.3. [MORE INFO »](#)

HTTP Strict Transport Security (HSTS) with long duration deployed on this server. [MORE INFO »](#)

Certificate #1: EC 256 bits (SHA384withECDSA)



Server Key and Certificate #1



Subject	rpsconsulting.in Fingerprint SHA256: 01efb8c5dd4c0f291e1efd1996f2ab27712920bece77bae1cbb2d0a1d6b306be Pin SHA256: H+DR1fBBHjLZnyYn6suwd1xqt/v22llUhEPgSkw8B1s=
Common names	rpsconsulting.in
Alternative names	rpsconsulting.in www.rpsconsulting.in
Serial Number	057f30e287a211a4d0fec47c88bf5d7e55c8
Valid from	Tue, 18 Nov 2025 08:09:26 UTC
Valid until	Mon, 16 Feb 2026 08:09:25 UTC (expires in 1 month and 7 days)
Key	EC 256 bits
Weak key (Debian)	No
Issuer	E7 AIA: http://e7.i.lencr.org/
Signature algorithm	SHA384withECDSA
Extended Validation	No
Certificate Transparency	Yes (certificate)
OCSP Must Staple	No
Revocation information	CRL CRL: http://e7.c.lencr.org/73.crl
Revocation status	Validation error CRL ERROR: IOException occurred
DNS CAA	No (more info)
Trusted	Yes Mozilla Apple Android Java Windows

Task – Students will practice different networking commands

S.No	Usage	Windows Cmd	Kali Cmd
1	Listing IP Address	ipconfig ipconfig /all	ip addr ifconfig
2	Connectivity Testing (ICMP)	ping google.com ping -t google.com	ping google.com
3	Route Verification	route print	ip route route -n
4	DNS Resolution Testing	nslookup example.com	dig example.com nslookup example.com
5	ARP Cache	arp -a	arp -a ip neigh
6	Port & Service Testing	netstat -ano	ss -tulnp netstat -tulnp
7	Tracing Network Path	tracert google.com	traceroute google.com
8	Remote Login & Access	mstsc	ssh user@192.168.1.20
9	Packet Capture	netsh trace start capture=yes netsh trace stop	tcpdump -i eth0
10	Bandwidth & Performance Testing	iperf3 -c 192.168.1.10	iperf3 -s

Map commands to OSI layers

Layer No.	Layer Name	Windows	Kali	Focus on
1	Physical Layer	ipconfig	ethtool eth0 mii-tool	Physical connectivity (cables, signals, NIC status)
2	Data Link Layer	arp -a nbtstat -n	arp -a ip neigh nmblookup	MAC addressing, ARP, switching
3	Network Layer	ipconfig route print tracert google.com	ip addr ip route traceroute google.com	IP addressing and routing
4	Transport Layer	netstat -ano	ss -tulnp netstat -tulnp	TCP/UDP ports and connections
5	Session Layer	mstsc net use	ssh user@host smbclient	Session establishment and management
6	Presentation Layer	certutil -hashfile file.txt SHA256	openssl s_client -connect example.com:443	Encryption, encoding, compression
7	Application Layer	nslookup example.com curl http://example.com ftp	dig example.com curl http://example.com wget http://example.com ftp	User-facing network services

1. Install docker commands: (run as root user) → whoami

```
# yum install -y yum-utils
```

```
# yum-config-manager --add-repo https://download.docker.com/linux/centos/docker-ce.repo
```

- docker-ce
- docker-ce-cli
- containerd.io
- docker-buildx-plugin
- docker-compose-plugin

```
# yum repolist
```

```
# yum install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
```

```
# which docker
```

2. verify if docker is running, else start it.

```
# systemctl status docker
```

```
# systemctl start docker
```

```
# systemctl enable docker
```

```
# systemctl status docker
```

3. verify docker image & container status

```
# docker info
```

4. checking images

```
listing images
```

```
[root@docmaster ~]# docker images
```

```
REPOSITORY TAG IMAGE ID CREATED SIZE
```

```
[root@docmaster ~]# docker image ls
```

```
REPOSITORY TAG IMAGE ID CREATED SIZE
```

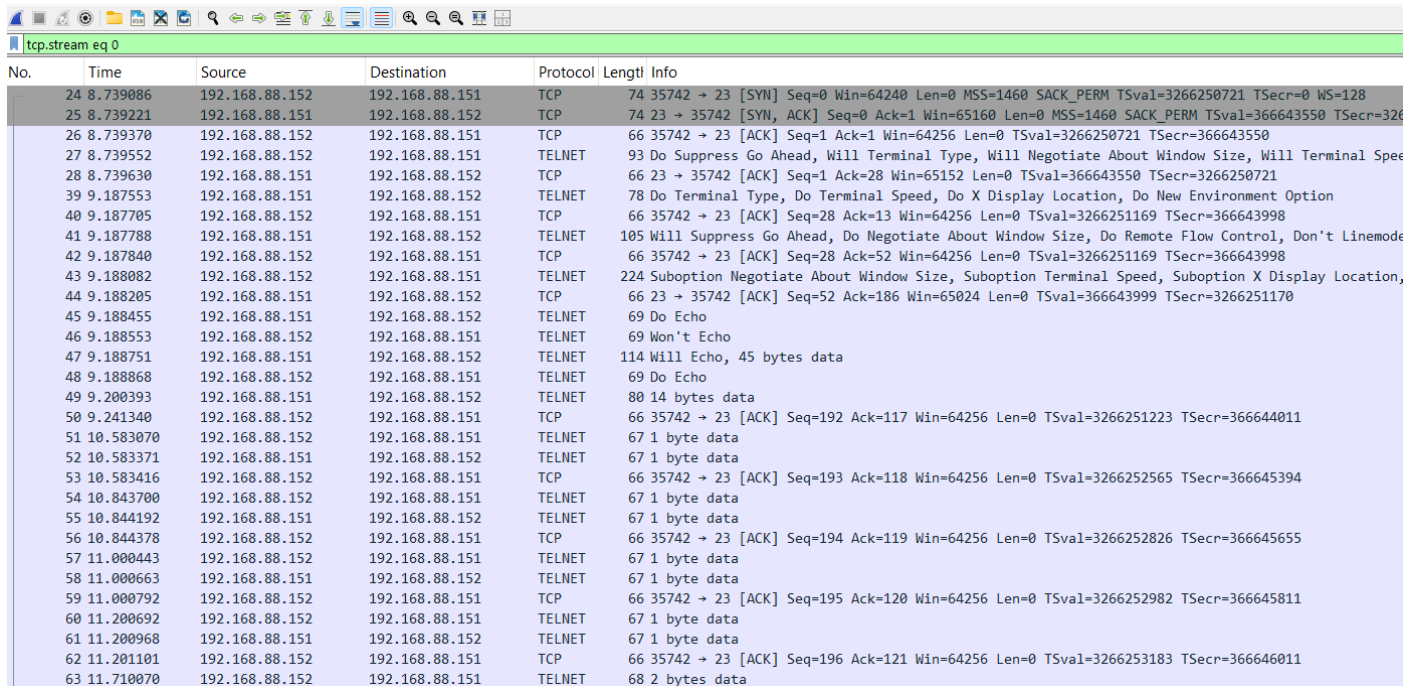
5. downloading the image(s)

```
# docker pull ubuntu //this downloads latest ubuntu version
```

```
# docker pull ubuntu:18.04 //this downloads specific ubuntu version
```

Task – Capture packets with Wireshark and ensure that you get the password

Start the packet analysis using Wireshark:



No.	Time	Source	Destination	Protocol	Length	Info
24	8.739086	192.168.88.152	192.168.88.151	TCP	74	35742 → 23 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3266250721 TSecr=0 WS=128
25	8.739221	192.168.88.151	192.168.88.152	TCP	74	23 → 35742 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=366643550 TSecr=3266250721
26	8.739370	192.168.88.152	192.168.88.151	TCP	66	35742 → 23 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=3266250721 TSecr=366643550
27	8.739552	192.168.88.152	192.168.88.151	TELNET	93	Do Suppress Go Ahead, Will Terminal Type, Will Negotiate About Window Size, Will Terminal Speed
28	8.739630	192.168.88.151	192.168.88.152	TCP	66	23 → 35742 [ACK] Seq=1 Ack=28 Win=65152 Len=0 TSval=366643550 TSecr=3266250721
39	9.187553	192.168.88.151	192.168.88.152	TELNET	78	Do Terminal Type, Do Terminal Speed, Do X Display Location, Do New Environment Option
40	9.187705	192.168.88.152	192.168.88.151	TCP	66	35742 → 23 [ACK] Seq=28 Ack=13 Win=64256 Len=0 TSval=3266251169 TSecr=366643998
41	9.187788	192.168.88.151	192.168.88.152	TELNET	105	Will Suppress Go Ahead, Will Negotiate About Window Size, Do Remote Flow Control, Don't Linemode
42	9.187840	192.168.88.152	192.168.88.151	TCP	66	35742 → 23 [ACK] Seq=28 Ack=52 Win=64256 Len=0 TSval=3266251169 TSecr=366643998
43	9.188082	192.168.88.152	192.168.88.151	TELNET	224	Suboption Negotiate About Window Size, Suboption Terminal Speed, Suboption X Display Location,
44	9.188205	192.168.88.151	192.168.88.152	TCP	66	23 → 35742 [ACK] Seq=52 Ack=186 Win=65024 Len=0 TSval=366643999 TSecr=3266251170
45	9.188455	192.168.88.151	192.168.88.152	TELNET	69	Do Echo
46	9.188553	192.168.88.152	192.168.88.151	TELNET	69	Won't Echo
47	9.188751	192.168.88.151	192.168.88.152	TELNET	114	Will Echo, 45 bytes data
48	9.188868	192.168.88.152	192.168.88.151	TELNET	69	Do Echo
49	9.200393	192.168.88.151	192.168.88.152	TELNET	80	14 bytes data
50	9.241340	192.168.88.152	192.168.88.151	TCP	66	35742 → 23 [ACK] Seq=192 Ack=117 Win=64256 Len=0 TSval=3266251223 TSecr=366644011
51	10.583070	192.168.88.152	192.168.88.151	TELNET	67	1 byte data
52	10.583371	192.168.88.151	192.168.88.152	TELNET	67	1 byte data
53	10.583416	192.168.88.152	192.168.88.151	TCP	66	35742 → 23 [ACK] Seq=193 Ack=118 Win=64256 Len=0 TSval=3266252565 TSecr=366645394
54	10.843700	192.168.88.152	192.168.88.151	TELNET	67	1 byte data
55	10.844192	192.168.88.151	192.168.88.152	TELNET	67	1 byte data
56	10.844378	192.168.88.152	192.168.88.151	TCP	66	35742 → 23 [ACK] Seq=194 Ack=119 Win=64256 Len=0 TSval=3266252826 TSecr=366645655
57	11.000443	192.168.88.152	192.168.88.151	TELNET	67	1 byte data
58	11.000663	192.168.88.151	192.168.88.152	TELNET	67	1 byte data
59	11.000792	192.168.88.152	192.168.88.151	TCP	66	35742 → 23 [ACK] Seq=195 Ack=120 Win=64256 Len=0 TSval=3266252982 TSecr=366645811
60	11.200692	192.168.88.152	192.168.88.151	TELNET	67	1 byte data
61	11.200968	192.168.88.151	192.168.88.152	TELNET	67	1 byte data
62	11.201101	192.168.88.152	192.168.88.151	TCP	66	35742 → 23 [ACK] Seq=196 Ack=121 Win=64256 Len=0 TSval=3266253183 TSecr=366646011
63	11.710070	192.168.88.152	192.168.88.151	TELNET	68	2 bytes data

Connect through Telnet client to server:

Install telnet on server and client → **yum install -y telnet**, start, enable and check the status of telnet.socket.

```
[root@client ~]# telnet 192.168.88.151
```

```
Trying 192.168.88.151...
```

```
Connected to 192.168.88.151.
```

```
Escape character is '^['.
```

```
Kernel 5.14.0-582.el9.x86_64 on an x86_64
```

```
server login: root
```

```
Password:
```

```
Last login: Wed Dec 10 11:15:05 from ::ffff:192.168.88.152
```

```
[root@server ~]#
```

Analyze the captured packets

The screenshot shows the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The toolbar contains icons for various functions. The filter bar at the top left shows the filter 'tcp.stream eq 0'. The packet list pane on the left shows a list of captured packets with columns for No., Time, and Size. The packet details pane on the right shows the selected packet's structure, including Protocol, Length, and Info. The packet bytes pane at the bottom shows the raw data of the selected packet.

No.	Time	Size	Protocol	Length	Info
24	8.739086	19	TCP	74	35742 → 23 [SYN] S
25	8.739221	19	TCP	74	23 → 35742 [SYN, A
26	8.739370	19	TCP	66	35742 → 23 [ACK] S
27	8.739552	19	TELNET	93	Do Suppress Go Ahe
28	8.739630	19	TCP	66	23 → 35742 [ACK] S
39	9.187553	19	TELNET	78	Do Terminal Type,
40	9.187705	19	TCP	66	35742 → 23 [ACK] S
41	9.187788	19	TELNET	105	Will Suppress Go A
42	9.187840	19	TCP	66	35742 → 23 [ACK] S
43	9.188082	19	TELNET	224	Suboption Negotiat
44	9.188205	19	TCP	66	35742 → 23 [ACK] S
45	9.188455	19	TCP	66	35742 → 23 [ACK] S
46	9.188553	19	TCP	66	35742 → 23 [ACK] S
47	9.188751	19	TCP	66	35742 → 23 [ACK] S
48	9.188868	192	TCP	66	35742 → 23 [ACK] S
49	9.200393	192	TCP	66	35742 → 23 [ACK] S
50	9.241340	192	TCP	66	35742 → 23 [ACK] S
51	10.583070	192	TCP	66	35742 → 23 [ACK] S
52	10.583371	192	TCP	66	35742 → 23 [ACK] S
53	10.583416	192	TCP	66	35742 → 23 [ACK] S
54	10.843700	192	TCP	66	35742 → 23 [ACK] S
55	10.844192	192	TCP	66	35742 → 23 [ACK] S
56	10.844378	192	TCP	66	35742 → 23 [ACK] S
57	11.000443	192	TCP	66	35742 → 23 [ACK] S
58	11.000663	192	TCP	66	35742 → 23 [ACK] S
59	11.000792	192	TCP	66	35742 → 23 [ACK] S

And you can see the server's Root password

```
.....!.."'.#
...#..'.....#.....'.....
...o......38400,38400....#.client.localdomain:0....'..XAUTHORITY./run/user/0/.mutter-Xwaylandauth.9078G3.DISPLAY.client.localdomain:0.....XTERM-256COLOR
...
...
...
Kernel 5.14.0-582.el9.x86_64 on an x86_64
...
server login:
r
r
o
o
o
o
t
t
.
.
Password:
centos
.
Last login: Wed Dec 10 11:15:05 from ::ffff:192.168.88.152
.]0;root@server:~.[?2004h[root@server ~]#
```

16 client pkts, 16 server pkts, 19 turns.

Entire conversation (442 bytes) Show as ASCII No delta times Stream 0

Find: Case sensitive Find Next

Filter Out This Stream Print Save as... Back Close Help

Red & Blue

- Blue color-coding showing output displayed by terminal.

Wireshark · Follow TCP Stream (tcp.stream eq 0) · VMware Network Adapter VMnet8

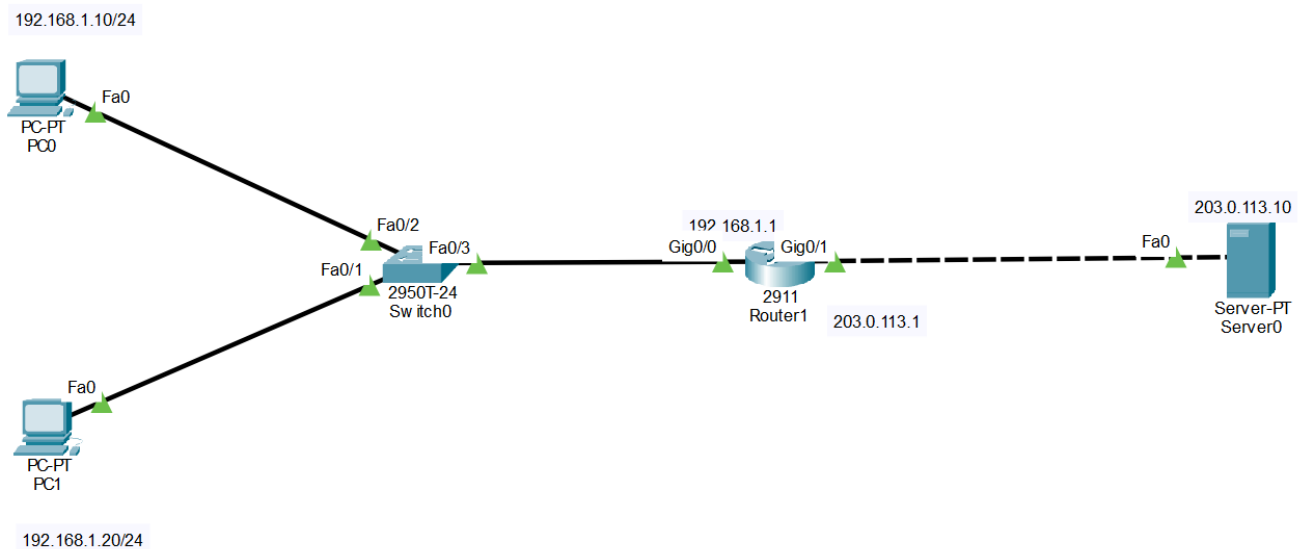
```
.....!.."'.#.....#.....'.....
Kernel 5.14.0-582.el9.x86_64 on an x86_64
server login: root
Password:
Last login: Wed Dec 10 11:15:05 from ::ffff:192.168.88.152
.]0;root@server:~.[?2004h[root@server ~]#
```

- Red color-coding shows the input given to the terminal.

Wireshark · Follow TCP Stream (tcp.stream eq 0) · VMware Network Adapter VMnet8

```
.....!.."'.#.....o......38400,38400....#.client.localdomain:0....'..XAU
THORITY./run/user/0/.mutter-Xwaylandauth.9078G3.DISPLAY.client.localdomain:0.....XTERM
-256COLOR.....root
.centos
```


Task – Configure NAT Policy (Source NAT) – using Cisco Packet Tracer



Setup:

- ✓ 1st PC – IP = 192.168.1.10/24
- ✓ 2nd PC – IP = 192.168.1.20/24
- ✓ Router interface G0/0 = 192.168.1.1/24
- ✓ Router interface G0/1 = 203.0.113.1/24
- ✓ Server – IP = 203.0.113.10/24

Now configuring inside and outside on the router on router's interface g0/0 & g0/1

```
Router#int
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int
Router(config)#interface g0/0
Router(config-if)#ip nat inside
Router(config-if)#no shut
Router(config-if)#exit
Router(config)#int
Router(config)#interface g0/1
Router(config-if)#ip nat outside
Router(config-if)#no shut
Router(config-if)#exit
Router(config)#access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)#ip nat inside source list 1 interface g0/1 overload
Router(config)#exn
```

% Invalid input detected at '^' marker.

```
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Router#wr
Building configuration...
[OK]
```

Now ping from PC1 to Server0:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 203.0.113.10

Pinging 203.0.113.10 with 32 bytes of data:

Reply from 203.0.113.10: bytes=32 time<1ms TTL=127
Reply from 203.0.113.10: bytes=32 time<1ms TTL=127
Reply from 203.0.113.10: bytes=32 time<1ms TTL=127
Reply from 203.0.113.10: bytes=32 time<1ms TTL=127
```

Now validating if NAT worked:

```
Router#sho
Router#show ip nat st
Router#show ip nat statistics
Total translations: 0 (0 static, 0 dynamic, 0 extended)
Outside Interfaces: GigabitEthernet0/1
Inside Interfaces: GigabitEthernet0/0
Hits: 0 Misses: 0
Expired translations: 0
Dynamic mappings:
Router#sh
Router#show ip nat tra
Router#show ip nat translations
Pro  Inside global      Inside local      Outside local      Outside global
icmp 203.0.113.1:4        192.168.1.10:4    203.0.113.10:4    203.0.113.10:4
icmp 203.0.113.1:5        192.168.1.10:5    203.0.113.10:5    203.0.113.10:5
icmp 203.0.113.1:6        192.168.1.10:6    203.0.113.10:6    203.0.113.10:6
icmp 203.0.113.1:7        192.168.1.10:7    203.0.113.10:7    203.0.113.10:7

Router#show
Router#show runn
Router#show running-config | sec
Router#show running-config | section nat
ip nat inside
ip nat outside
ip nat inside source list 1 interface GigabitEthernet0/1 overload
```

Validating:

```
Router#sh
Router#show acce
Router#show access-lists
Standard IP access list 1
10 permit 192.168.1.0 0.0.0.255 (8 match(es))

Router#sh
Router#show ip inter
Router#show ip interface bre
Router#show ip interface br
Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0 192.168.1.1    YES manual up          up
GigabitEthernet0/1 203.0.113.1    YES manual up          up
GigabitEthernet0/2 unassigned      YES unset  administratively down down
Vlan1           unassigned      YES unset  administratively down down

Router#deb
Router#debug ip nat
IP NAT debugging is on
Router#wr
Building configuration...
[OK]
```

Task – Create a firewall policy – on windows server/client

Allowing PING (ICMP), while firewall is on Windows Server

- ✓ Number of VMs: 2
- ✓ Windows Server (192.168.88.155) → firewall policy to be applied.
- ✓ Centos 9 (192.168.88.151) → client

Go to client machine and test ping to server (before) → its pinging while firewall is OFF.

```
[root@server ~]# ping 192.168.88.155
PING 192.168.88.155 (192.168.88.155) 56(84) bytes of data.
64 bytes from 192.168.88.155: icmp_seq=1 ttl=128 time=2.44 ms
64 bytes from 192.168.88.155: icmp_seq=2 ttl=128 time=0.288 ms
64 bytes from 192.168.88.155: icmp_seq=3 ttl=128 time=0.198 ms
64 bytes from 192.168.88.155: icmp_seq=4 ttl=128 time=0.208 ms
^C
--- 192.168.88.155 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3077ms
rtt min/avg/max/mdev = 0.198/0.782/2.437/0.955 ms
```

Now go to server machine and turning ON firewall, the ping fails.

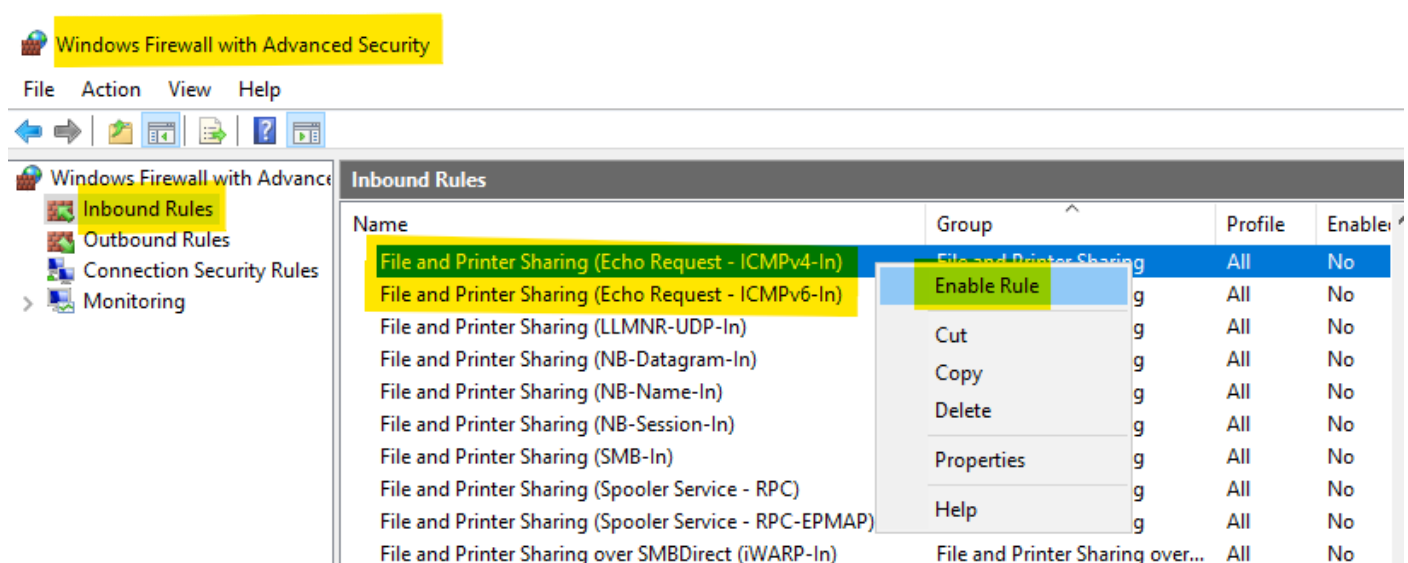
```
[root@server ~]# ping 192.168.88.155
PING 192.168.88.155 (192.168.88.155) 56(84) bytes of data.

^C
--- 192.168.88.155 ping statistics ---
176 packets transmitted, 0 received, 100% packet loss, time 179207ms

[root@server ~]#
```

Now to windows server and open “wf.msc” for configuring the firewall rule.

- Now look for the rule
 - File and Printer Sharing (Echo Request – ICMPv4-In)
 - File and Printer Sharing (Echo Request – ICMPv6-In)
- And right-click on both (one by one) and select “Enable Rule”



Now go to client and ping again to test if it's working.

```
[root@server ~]# ping 192.168.88.155
PING 192.168.88.155 (192.168.88.155) 56(84) bytes of data.
64 bytes from 192.168.88.155: icmp_seq=6 ttl=128 time=0.389 ms
64 bytes from 192.168.88.155: icmp_seq=7 ttl=128 time=0.186 ms
64 bytes from 192.168.88.155: icmp_seq=8 ttl=128 time=0.303 ms
64 bytes from 192.168.88.155: icmp_seq=9 ttl=128 time=0.319 ms
64 bytes from 192.168.88.155: icmp_seq=10 ttl=128 time=0.319 ms
64 bytes from 192.168.88.155: icmp_seq=11 ttl=128 time=0.293 ms
64 bytes from 192.168.88.155: icmp_seq=12 ttl=128 time=0.276 ms
64 bytes from 192.168.88.155: icmp_seq=13 ttl=128 time=0.323 ms
^C
--- 192.168.88.155 ping statistics ---
13 packets transmitted, 8 received, 38.4615% packet loss, time 12319ms
rtt min/avg/max/mdev = 0.186/0.301/0.389/0.053 ms
[root@server ~]# █
```

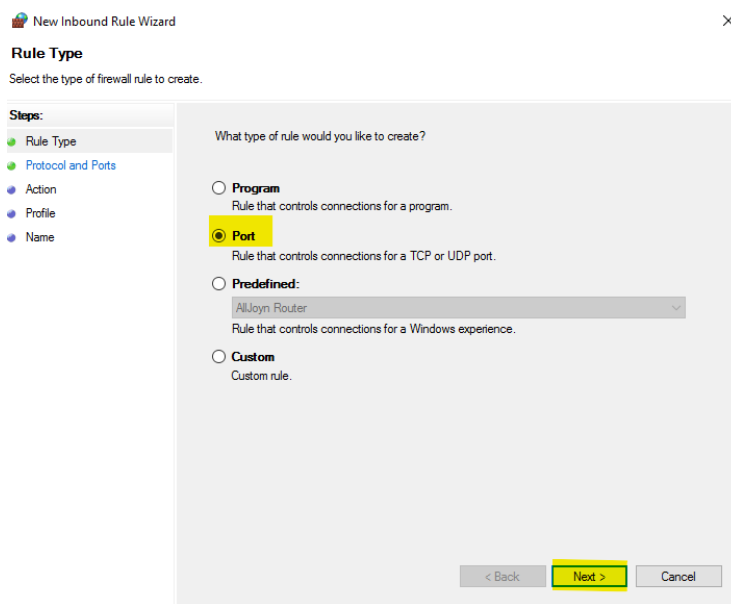
Task – Allow Port 80 (HTTP) on Windows Server Firewall

Step 1: Open Firewall Console

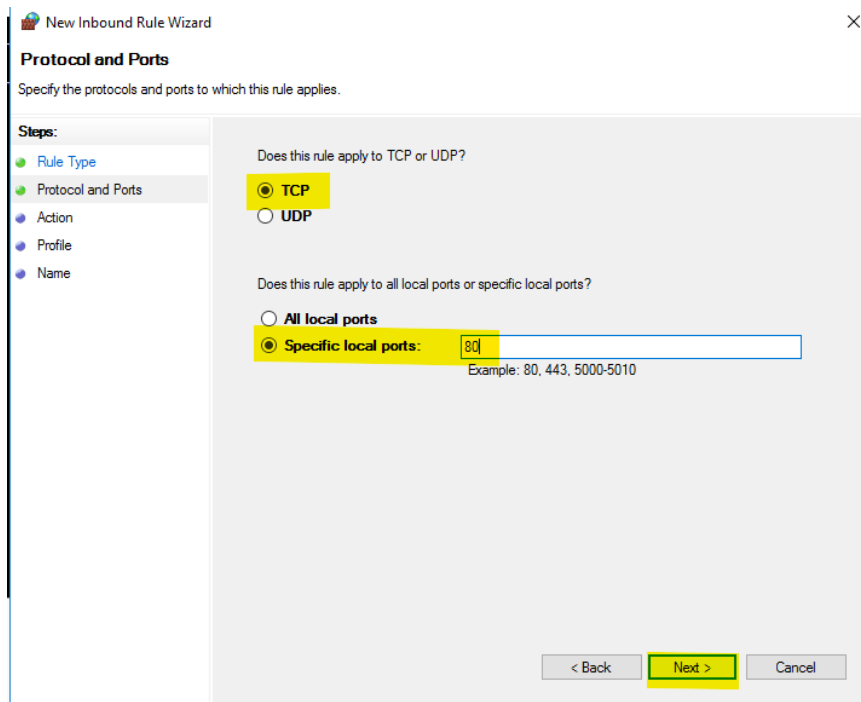
- Press **Win + R**
- Type wf.msc
- Press **Enter**

Step 2: Create Inbound Rule for Port 80

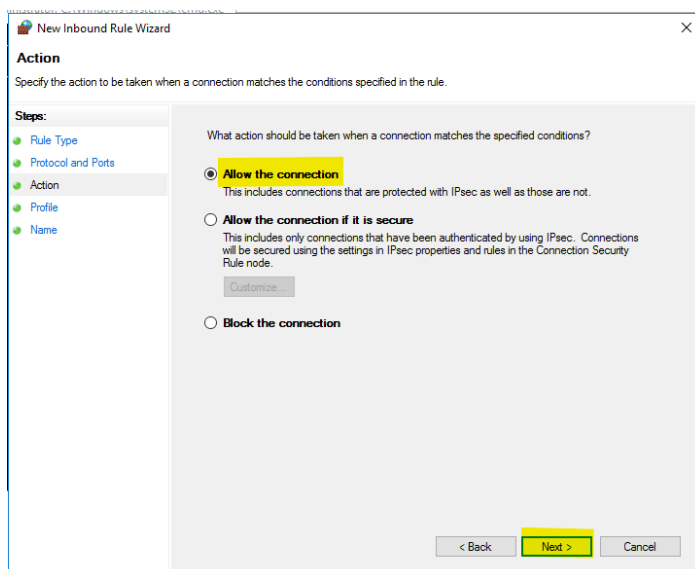
1. Click **Inbound Rules**
2. Click **New Rule...**
3. Select **Port** → Click **Next**



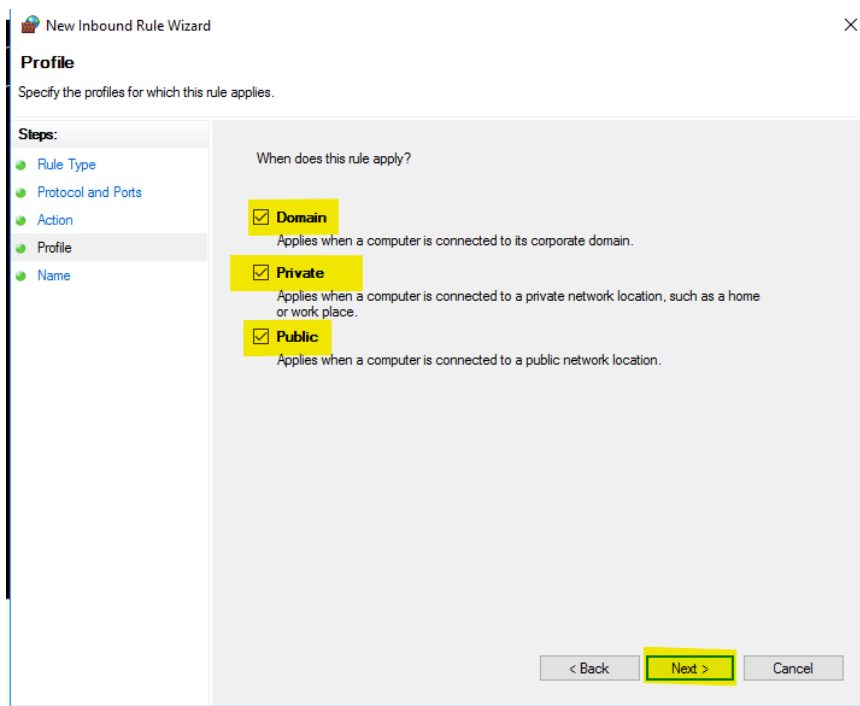
4. Select **TCP**
5. Specify **Specific local ports** → 80



6. Click **Next**
7. Select **Allow the connection**
8. Click **Next**



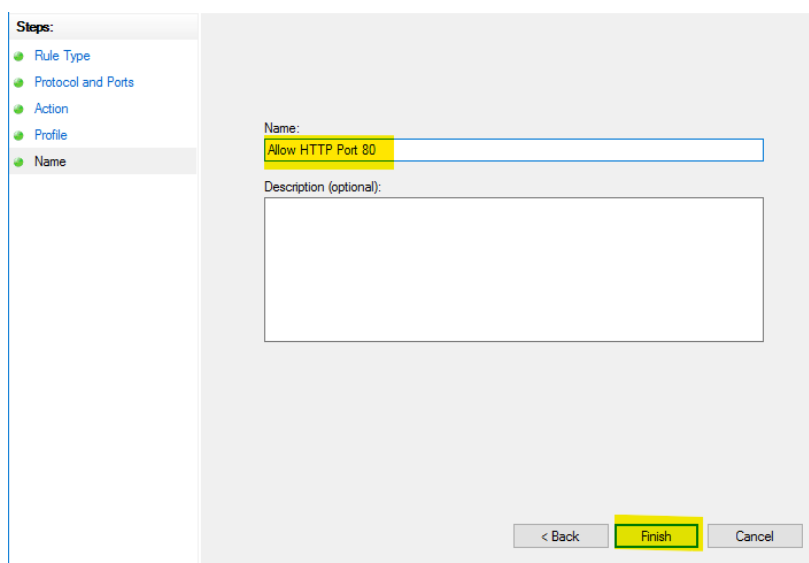
9. Choose Profiles:
 - Domain (recommended for servers joined to AD)
 - Private (optional)
 - Public (only if required)



10. Click **Next**

11. Name the rule:

- **Allow HTTP Port 80**



12. Click **Finish**

Result: HTTP traffic is now allowed on port 80.

Windows Firewall with Advanced Security

File Action View Help

Windows Firewall with Advanced Security

- Inbound Rules
- Outbound Rules
- Connection Security Rules
- Monitoring

Name	Group	Profile	Enabled
Allow HTTP Port 80		All	Yes
Alloyn Router (TCP-In)	Alloyn Router	Domain...	Yes
Alloyn Router (UDP-In)	Alloyn Router	Domain...	Yes
BranchCache Content Retrieval (HTTP-In)	BranchCache - Content Retr...	All	No
BranchCache Hosted Cache Server (HTTP-In)	BranchCache - Hosted Cach...	All	No
BranchCache Peer Discovery (WSD-In)	BranchCache - Peer Discove...	All	No
Cast to Device functionality (qWave-TCP-In)	Cast to Device functionality	Private...	Yes
Cast to Device functionality (qWave-UDP-In)	Cast to Device functionality	Private...	Yes
Cast to Device SSDP Discovery (UDP-In)	Cast to Device functionality	Public	Yes
Cast to Device streaming server (HTTP-Streaming-In)	Cast to Device functionality	Private	Yes
Cast to Device streaming server (HTTP-Streaming-In)	Cast to Device functionality	Public	Yes
Cast to Device streaming server (HTTP-Streaming-In)	Cast to Device functionality	Domain	Yes
Cast to Device streaming server (RTCP-Streaming-In)	Cast to Device functionality	Private	Yes
Cast to Device streaming server (RTCP-Streaming-In)	Cast to Device functionality	Domain	Yes
Cast to Device streaming server (RTCP-Streaming-In)	Cast to Device functionality	Public	Yes
Cast to Device streaming server (RTSP-Streaming-In)	Cast to Device functionality	Public	Yes
Cast to Device streaming server (RTSP-Streaming-In)	Cast to Device functionality	Domain	Yes
Cast to Device streaming server (RTSP-Streaming-In)	Cast to Device functionality	Private	Yes
Cast to Device UPnP Events (TCP-In)	Cast to Device functionality	Public	Yes
COM+ Network Access (DCOM-In)	COM+ Network Access	All	No

Actions

Inbound Rules

- New Rule...
- Filter by Profile
- Filter by State
- Filter by Group
- View
- Refresh
- Export List...
- Help

Allow HTTP Port 80

- Disable Rule
- Cut
- Copy
- Delete
- Properties
- Help

Task – Allowing PING (ICMP), while firewall is on Centos 9

1^{stly} go to centos and ensure firewall service is active and running.

```
[root@server ~]# systemctl enable firewalld
Created symlink /etc/systemd/system/dbus-org.fedoraproject.FirewallD1.service → /usr/lib/systemd/system/firewalld.service.
Created symlink /etc/systemd/system/multi-user.target.wants/firewalld.service → /usr/lib/systemd/system/firewalld.service.
[root@server ~]#
[root@server ~]# systemctl start firewalld
[root@server ~]#
[root@server ~]# systemctl status firewalld
● firewalld.service - firewalld - dynamic firewall daemon
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-01-09 23:09:43 IST; 3s ago
     Docs: man:firewalld(1)
    Main PID: 113439 (firewalld)
      Tasks: 2 (limit: 22763)
     Memory: 30.5M
        CPU: 348ms
    CGroup: /system.slice/firewalld.service
            └─113439 /usr/bin/python3 -s /usr/sbin/firewalld --nofork --nopid

Jan 09 23:09:43 server systemd[1]: Starting firewalld - dynamic firewall daemon...
Jan 09 23:09:43 server systemd[1]: Started firewalld - dynamic firewall daemon.
[root@server ~]# █
```

List current allowed port using command

```
[root@server ~]# firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: ens160
  sources:
  services: cockpit dhcpv6-client ssh
  ports: 443/tcp 22/tcp 7878/tcp 162/udp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[root@server ~]# █
```

Now allowing port 80

```
[root@server ~]# firewall-cmd --add-port=80/tcp --permanent
success
[root@server ~]# firewall-cmd --reload
success
[root@server ~]# firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: ens160
  sources:
  services: cockpit dhcpv6-client ssh
  ports: 443/tcp 22/tcp 7878/tcp 162/udp 80/tcp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[root@server ~]# █
```

Disable Ping (Block ICMP Echo Requests)

```
[root@server ~]# firewall-cmd --get-icmp-types
address-unreachable bad-header beyond-scope communication-prohibited destination-unreachable echo-reply echo-request failed-policy fragmentation-needed host-precedence-violation host-prohibited host-redirect host-unknown host-unreachable ip-header-bad mld-listener-done mld-listener-query mld-listener-report mld2-listener-report neighbour-advertisement neighbour-solicitation network-prohibited network-redirect network-unknown network-unreachable no-route packet-too-big parameter-problem port-unreachable precedence-cutoff protocol-unreachable redirect reject-route required-option-missing router-advertisement router-solicitation source-quench source-route-failed time-exceeded timestamp-reply timestamp-request tos-host-redirect tos-host-unreachable tos-network-redirect tos-network-unreachable ttl-zero-during-reassembly ttl-zero-during-transit unknown-header-type unknown-option
[root@server ~]#
[root@server ~]# firewall-cmd --permanent --add-icmp-block=echo-request
success
[root@server ~]# firewall-cmd --reload
success
[root@server ~]# █
```

Now go to windows server and ping linux server again.

```
C:\Windows\system32\cmd.exe

C:\Users\Administrator>ping 192.168.88.151

Pinging 192.168.88.151 with 32 bytes of data:
Reply from 192.168.88.151: Destination net unreachable.
Reply from 192.168.88.151: Destination net unreachable.
Reply from 192.168.88.151: Destination net unreachable.
Reply from 192.168.88.151: Destination net unreachable.

Ping statistics for 192.168.88.151:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

C:\Users\Administrator>_
```

Task – Block an application based on AppID

- ✓ Go to Palo-Alto firewall
- ✓ Go to Policies → Security → Add Rule

PA-VM DASHBOARD ACC MONITOR **POLICIES** OBJECTS NETWORK DEVICE

Security

	NAME	TAGS	TYPE	ZONE	ADDRESS	USER	DEVICE
1	intrazone-default	none	intrazone	any	any	any	any
2	interzone-default	none	interzone	any	any	any	any

Policy Optimizer

- New App Viewer 0
- Rules Without App Controls 0
- Unused Apps 0
- Rule Usage
 - Unused in 30 days 2
 - Unused in 90 days 2
 - Unused 2

Object: Addresses + Add Delete Clone Override Revert Enable Disable Move PDF/CSV Highlight Unused

Under General tab, fill name of the rule:

Security Policy Rule

General Source Destination Application Service/URL Category Actions

Name: BlockingApplication

Rule Type: universal (default)

Description:

Tags:

Group Rules By Tag: None

Audit Comment:

[Audit Comment Archive](#)

OK Cancel

Then go to Source tab → select “any”:

Security Policy Rule

General | **Source** | Destination | Application | Service/URL Category | Actions

☒ Any	☒ Any	any	any
☐ SOURCE ZONE ^	☐ SOURCE ADDRESS ^	☐ SOURCE USER ^	☐ SOURCE DEVICE ^
+ Add - Delete	+ Add - Delete	+ Add - Delete	+ Add - Delete

☐ Negate

OK Cancel

Then to ‘destination’

Security Policy Rule

General | Source | **Destination** | Application | Service/URL Category | Actions

any	☒ Any	any
☐ DESTINATION ZONE ^	☐ DESTINATION ADDRESS ^	☐ DESTINATION DEVICE ^
+ Add - Delete	+ Add - Delete	+ Add - Delete

☐ Negate

OK Cancel

Now go to Application → Add → add required application like Facebook, YouTube and so on.

Security Policy Rule

General | Source | Destination | **Application** | Service/URL Category | Actions

☐ Any	7 items → ×
☐ APPLICATIONS ^	DEPENDS ON ^
☐ facebook	☐ google-base
☒ youtube	☐ mqtt-base
	☐ rtcp
	☐ rtp-base
	☐ ssl
	☐ stun
	☐ web-browsing
+ Add - Delete	Add To Current Rule Add To Existing Rule

OK Cancel

Now go to Service/URL Category tab:

The screenshot shows the 'Security Policy Rule' configuration window with the 'Service/URL Category' tab selected. The 'application-default' dropdown is highlighted. The 'SERVICE' and 'URL CATEGORY' sections are empty, each with an 'Add' and 'Delete' button. The 'Any' checkbox is checked under the URL Category section. 'OK' and 'Cancel' buttons are at the bottom right.

Then go to Actions tab:

The screenshot shows the 'Security Policy Rule' configuration window with the 'Actions' tab selected. The 'Action Setting' section has 'Deny' selected. The 'Log Setting' section has 'Log at Session Start' and 'Log at Session End' checked. The 'Profile Setting' section has 'None' selected. The 'Other Settings' section has 'Schedule' and 'QoS Marking' set to 'None'. 'OK' and 'Cancel' buttons are at the bottom right.

The screenshot shows the PA-VM dashboard with the 'Security' section expanded. The 'Security Policy Rules' table is displayed with the following data:

	NAME	TAGS	TYPE	ZONE	ADDRESS	USER	DEVICE	ZONE	ADDRESS	DEVICE	APPLICATION	SERVICE	ACTION	PROFILE	OPTIONS
1	BlockingApplication	none	universal	any	any	any	any	any	any	any	facebook	application...	Deny	none	
2	Intrazone-default	none	intrazone	any	any	any	any	(intrazone)	any	any	any	any	Allow	none	none
3	Interzone-default	none	interzone	any	any	any	any	any	any	any	any	any	Deny	none	none

Then commit the configuration.

The screenshot shows the 'Commit Status' dialog box. The 'Operation' is 'Commit', 'Status' is 'Active', 'Result' is 'Pending', and 'Progress' is 0%. The 'Commit' tab is selected. 'Cancel' and 'Close' buttons are at the bottom.

The screenshot shows the 'Commit Status' dialog box after a successful commit. The 'Operation' is 'Commit', 'Status' is 'Completed', 'Result' is 'Successful', and 'Details' are 'Configuration committed successfully'. The 'Commit' tab is selected. 'Close' button is at the bottom.

To validate the configuration → go to client VM and access sites like Facebook & YouTube.

Task – URL Filtering advanced

Prerequisites Before Configuration

Ensure the following are in place:

1. **Valid URL Filtering License**
 - Device → Licenses → URL Filtering
2. **Security Policy allowing web traffic**
3. **DNS resolution working**
4. (Optional but recommended) **SSL Decryption enabled**
5. User-ID configured if user-based control is required

Step-by-Step: Configure URL Filtering in Palo Alto Firewall

Step 1: Create a URL Filtering Profile

1. Go to
Objects → Security Profiles → URL Filtering
2. Click **Add**
3. Name the profile
Example: URLF-Internet-Policy

Step 2: Configure Category Actions

Category	Recommended Action
malware	block
phishing	block
command-and-control	block
adult	block
hacking	block
gambling	block
social-networking	alert / allow
streaming-media	alert / block
unknown	block (strict environments)
low-risk	allow

Step 3: Configure Advanced Settings (Recommended)

Under the **Settings** tab:

- Enable **Safe Search Enforcement**
- Enable **Log Container Page Only**
- Enable **User Credential Detection**
- Enable **Block Credential Phishing**

Step 4: Attach URL Filtering Profile to a Security Policy

1. Navigate to
Policies → Security
2. Edit the **Internet Access** policy rule
3. Go to the **Actions** tab
4. Under **Profile Setting**:
 - Select **Profiles**
 - Set **URL Filtering** = URLF-Internet-Policy
5. Ensure:
 - Action = **Allow**
 - Application = web-browsing, ssl

URL Filtering does **not work on deny rules**.

Step 5: Commit the Configuration

- Click **Commit**
- Verify commit success

Step 6. Step-by-Step Validation

Test from Client Machine

- Try accessing:
 - <http://www.facebook.com>
 - <http://testphishing.com>
 - <http://exampleadultsite.com>

Check Logs

1. Go to
Monitor → URL Filtering
2. Verify:
 - URL
 - Category
 - Action (block/allow/alert)
 - User (if User-ID enabled)

Task – Students to analyse the given malware analysis link and answer the questions

On Kali:

Create a Test “Malware” File (EICAR - European Institute for Computer Anti-virus Research (EICAR))

- `mkdir ~/malware-demo`
- `cd ~/malware-demo`
- `echo 'X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*' > eicar.txt`
 - This is 68-bytes string of ASCII text.
 - This is a fixed string.
 - EICAR is used for testing anti-virus software functionality without using real Malware.

Verify hash:

- `sha256sum eicar.txt`

Start a Local Web Server (Delivery Simulation)

- `python3 -m http.server 8080`

```
(root@kali)-[~]
# mkdir ~/malware-demo
cd ~/malware-demo

echo 'X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*' > eicar.txt

(root@kali)-[~/malware-demo]
# ls
eicar.txt

(root@kali)-[~/malware-demo]
# sha256sum eicar.txt
131f95c51cc819465fa1797f6ccacf9d494aaaff46fa3eac73ae63ffbfdf8267 eicar.txt

(root@kali)-[~/malware-demo]
# python3 -m http.server 8080

Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
192.168.113.141 - - [31/Jan/2026 13:14:43] "GET /eicar.txt HTTP/1.1" 200 -
192.168.113.141 - - [31/Jan/2026 13:16:51] "GET /eicar.txt HTTP/1.1" 200 -
```

On Windows:

Run below commands on PowerShell as Administrator.

- `Set-MpPreference -DisableRealtimeMonitoring $false`
- `netsh advfirewall set allprofiles state on`

Verify Network Connectivity

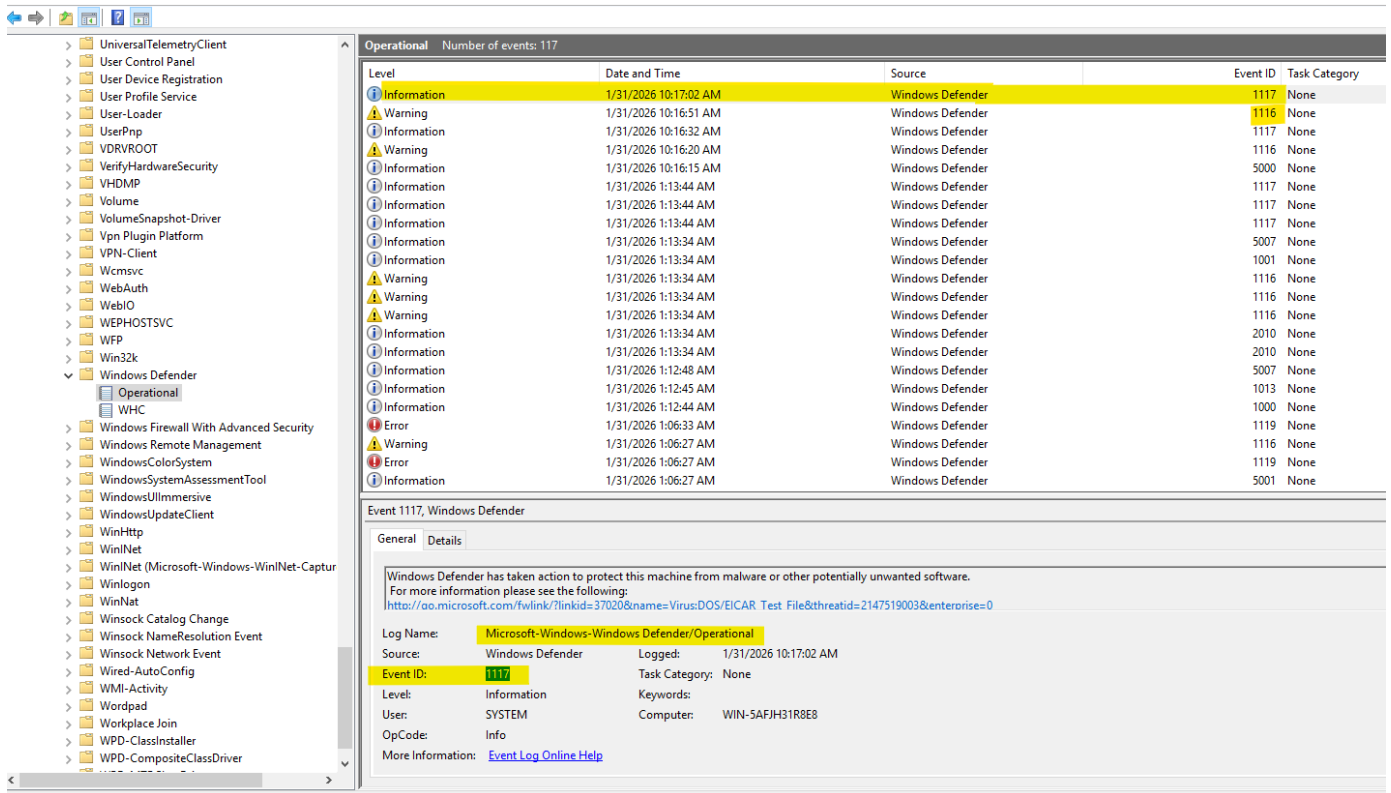
- `ping <KALI_IP>`

Download the File (Simulated Infection Vector)

- `Invoke-WebRequest http://<KALI_IP>:8080/eicar.txt -OutFile C:\Users\Administrator\Desktop\eicar.txt`

Event Logs

Now go to Event Viewer → Application and Service logs → Microsoft → Windows → Windows Defender



Look for:

- Event ID 1116 / 1117

Capture Traffic:

Run as root on kali:

- tcpdump -i eth0 port 8080 -n

Now, go to window server and rerun the same command:

- Invoke-WebRequest http://<KALI_IP>:8080/eicar.txt -OutFile C:\Users\Public\ecar.txt

Go to kali linux again, rerun the tcpdump command:

- tcpdump -i eth0 port 8080 -n

```
(root@kali)~# tcpdump -i eth0 port 8080 -n
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
13:33:35.829366 IP 192.168.113.141.49708 > 192.168.113.136.8080: Flags [SEW], seq 130062625, win 8192, options
0
13:33:35.829409 IP 192.168.113.136.8080 > 192.168.113.141.49708: Flags [S.], seq 706269845, ack 130062626, win
cale 9], length 0
13:33:35.829733 IP 192.168.113.141.49708 > 192.168.113.136.8080: Flags [..], ack 1, win 2053, length 0
13:33:35.829934 IP 192.168.113.141.49708 > 192.168.113.136.8080: Flags [P.], seq 1:174, ack 1, win 2053, lengt
13:33:35.829942 IP 192.168.113.136.8080 > 192.168.113.141.49708: Flags [..], ack 174, win 126, length 0
13:33:35.830781 IP 192.168.113.136.8080 > 192.168.113.141.49708: Flags [P.], seq 1:188, ack 174, win 126, leng
13:33:35.830921 IP 192.168.113.136.8080 > 192.168.113.141.49708: Flags [FP.], seq 188:257, ack 174, win 126, l
13:33:35.831122 IP 192.168.113.141.49708 > 192.168.113.136.8080: Flags [..], ack 258, win 2052, length 0
13:33:35.858011 IP 192.168.113.141.49708 > 192.168.113.136.8080: Flags [F.], seq 174, ack 258, win 2052, lengt
13:33:35.858032 IP 192.168.113.136.8080 > 192.168.113.141.49708: Flags [..], ack 175, win 126, length 0
```

```
(root@kali)-[~/malware-demo]
# ls
eicar.txt

(root@kali)-[~/malware-demo]
# strings eicar.txt
X50!P%AP[4\PZX54(P^7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*

(root@kali)-[~/malware-demo]
# cat eicar.txt
X50!P%AP[4\PZX54(P^7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*

(root@kali)-[~/malware-demo]
# file eicar.txt
eicar.txt: EICAR virus test files

(root@kali)-[~/malware-demo]
#
```

Task – Students to use the MITRE ATT&CK framework and create the defence strategy against a specific attacker group

TEST 1 – T1204: User Execution

Goal

- Confirm **process creation logging** and **command-line capture**.

Simulation

From Windows Server (logged-in test user):

- Check for Event ID: 4688 (new process creation logging)
- Cmdlet:

```
Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4688} -MaxEvents 10 |  
ForEach-Object {  
    [PSCustomObject]@{  
        TimeCreated = $_.TimeCreated  
        User        = $_.Properties[1].Value  
        NewProcess  = $_.Properties[5].Value  
        CommandLine = $_.Properties[8].Value  
    }  
}
```

- Expected output: shows latest 10 new process creation logs.

TEST 2 – T1059.001: PowerShell Execution (Benign)

Goal

Validate **PowerShell Script Block Logging (4104)**

Simulation (Safe)

Run a **non-obfuscated**, harmless PowerShell command:

- No download
- No encoding
- No persistence

Working code – creates log when the suspicious processes are initiated.

```
cls  
# =====  
# CONFIGURATION  
# =====  
$LogName = "Security"  
$EventID = 4688  
  
$SuspiciousProcesses = @(  
    "powershell.exe",  
    "pwsh.exe",  
    "cmd.exe",  
    "wscript.exe",  
    "cscript.exe",  
    "rundll32.exe",  
    "mshta.exe",  
    "certutil.exe",  
    "bitsadmin.exe",  
    "regsvr32.exe"  
)  
  
$AlertLog = "C:\SecurityAlerts\4688-alerts.log"
```

```

# =====
# PREPARE ALERT DIRECTORY
# =====
$AlertDir = Split-Path -Parent $AlertLog
if (-not (Test-Path $AlertDir)) {
    New-Item -ItemType Directory -Path $AlertDir -Force | Out-Null
}

# =====
# INITIALIZE RECORD TRACKING
# =====
$LastRecordId = (Get-WinEvent -LogName $LogName -MaxEvents 1).RecordId

Write-Host "[+] Monitoring Event ID 4688 (Polling Mode)..." -ForegroundColor Cyan

# =====
# LIVE MONITOR LOOP
# =====
while ($true) {

    $Events = Get-WinEvent -FilterHashtable @{
        LogName = $LogName
        Id       = $EventID
    } | Where-Object { $_.RecordId -gt $LastRecordId }

    foreach ($Event in $Events) {

        # Update last processed record
        $LastRecordId = $Event.RecordId

        $Time       = $Event.TimeCreated
        $User        = $Event.Properties[1].Value
        $NewProcess  = $Event.Properties[5].Value
        $Command     = $Event.Properties[8].Value

        foreach ($Suspicious in $SuspiciousProcesses) {

            if ($NewProcess -match [regex]::Escape($Suspicious)) {

                $Alert = @"
[ALERT] Suspicious Process Execution Detected
Time       : $Time
User       : $User
Process    : $NewProcess
CommandLine : $Command
-----
"@

                Write-Host $Alert -ForegroundColor Red
                Add-Content -Path $AlertLog -Value $Alert
                break
            }
        }
    }

    Start-Sleep -Seconds 3
}

```

Task – Students to analyse the given event viewer logs and investigate the login/logout activity

List only enabled logs:

```
Get-WinEvent -ListLog * | Where-Object {$_.IsEnabled -eq $true}
```

List only enabled logs:

```
Get-WinEvent -ListLog * | Where-Object {$_.IsEnabled -eq $true}
```

```
PS C:\Users\Administrator> Get-WinEvent -ListLog * | Where-Object {$_.IsEnabled -eq $true}
```

LogMode	MaximumSizeInBytes	RecordCount	LogName
Circular	1052672	9	Active Directory Web Services
Circular	20971520	355	Application
Circular	15532032	18	DFS Replication
Circular	1052672	38	Directory Service
Circular	104857600	22	DNS Server
Circular	20971520	0	HardwareEvents
Circular	1052672	0	Internet Explorer
Circular	20971520	0	Key Management Service
Circular	134217728	5142	Security
Circular	20971520	1725	System
Circular	15728640	201	Windows PowerShell
Circular	10485760	0	Microsoft-AppV-Client/Admin
Circular	10485760	0	Microsoft-AppV-Client/Operational
Circular	10485760	0	Microsoft-AppV-Client/Virtual Applications
Circular	1052672	36	Microsoft-Client-Licensing-Platform/Admin

List specific log details (example: System):

```
Get-WinEvent -ListLog System | Format-List *
```

List specific log details (example: System):

```
Get-WinEvent -ListLog System | Format-List *
```

```
PS C:\Users\Administrator> # List specific log details (example: System):  
Get-WinEvent -ListLog System | Format-List *
```

```
FileSize           : 1118208  
IsLogFull          : False  
LastAccessTime     : 12/28/2025 9:20:32 PM  
LastWriteTime      : 2/7/2026 9:00:29 AM  
OldestRecordNumber : 1  
RecordCount        : 1725  
LogName            : System  
LogType            : Administrative  
LogIsolation       : System  
IsEnabled          : True  
IsClassicLog       : True  
SecurityDescriptor : 0:BAG:SYD:(A;;0xf0007;;;SY)(A;;0x7;;;BA)(A;;0x3;;;BO)(A;;0x5;;;SO)(A;;0x1;;;IU)(A;;0x3;;;SU)(A;;0x1;;;S-1-5-3)(A;;0x2;;;S-1-5-33)(A;;0x1;;;S-1-5-32-573)  
LogFilePath        : %SystemRoot%\System32\Winevt\Logs\System.evtx  
MaximumSizeInBytes : 20971520  
LogMode            : Circular  
OwningProviderName :  
ProviderNames      : {3ware, ACPI, ADP80XX, AFD...}  
ProviderLevel      :  
ProviderKeywords   :
```

list if DNS is installed

```
Get-WindowsFeature -Name DNS
```

list if DNS is installed

```
Get-WindowsFeature -Name DNS
```

Display Name	Name	Install State
[X] DNS Server	DNS	Installed

listing log names

```
Get-WinEvent -ListLog * | Where-Object {$_.LogName -like "*DNS*"} | Select LogName
```

```
# listing log names
```

```
Get-WinEvent -ListLog * | Where-Object {$_.LogName -like "*DNS*"} | Select LogName
```

```
LogName
-----
DNS Server
Microsoft-Windows-DNS-Client/Operational
Microsoft-Windows-DNSServer/Audit
```

```
# list all DNS server events
```

```
Get-WinEvent -ListLog "Microsoft-Windows-DNSServer*"
```

```
LogMode      MaximumSizeInBytes RecordCount LogName
-----
Circular      104857600      36 Microsoft-Windows-DNSServer/Audit
```

```
# latest 2 successful logins
```

```
Get-WinEvent -FilterHashtable @{
```

```
    LogName='Security'
```

```
    Id=4624
```

```
} -MaxEvents 2 | Format-Table -AutoSize -Wrap
```

```
TimeCreated      Id LevelDisplayName Message
-----
2/7/2026 7:41:26 PM 4624 Information An account was successfully logged on.

Subject:
  Security ID:      S-1-0-0
  Account Name:     -
  Account Domain:   -
  Logon ID:         0x0

Logon Information:
  Logon Type:       3
  Restricted Admin Mode: -
  Virtual Account:  No
  Elevated Token:   Yes

Impersonation Level:      Impersonation

New Logon:
  Security ID:      S-1-5-18
  Account Name:     DC$
  Account Domain:   TRAINING.COM
  Logon ID:         0x147329
```

```
# last 2 Failed login attempts
```

```
Get-WinEvent -FilterHashtable @{
```

```
    LogName='Security'
```

```
    Id=4625
```

```
} -MaxEvents 2 | Format-Table -AutoSize -Wrap
```

TimeCreated	Id	Level	DisplayName	Message
-----	--	-----	-----	-----
2/7/2026 9:08:25 AM	4625	Information		An account failed to log on.
Subject:				
		Security ID:	S-1-0-0	
		Account Name:	-	
		Account Domain:	-	
		Logon ID:	0x0	
		Logon Type:	3	
Account For Which Logon Failed:				
		Security ID:	S-1-0-0	
		Account Name:	WIN-5AFJH31R8E8	
		Account Domain:	TRAINING	
Failure Information:				
		Failure Reason:	Unknown user name or bad password.	
		Status:	0xC000006D	
		Sub Status:	0xC0000064	
Process Information:				
		Caller Process ID:	0x0	
		Caller Process Name:	-	

last 2 logoff events

```
Get-WinEvent -FilterHashtable @{
```

```
LogName='Security'
```

```
Id=4634,4647
```

```
} -MaxEvents 2 | Format-Table -AutoSize -Wrap
```

ProviderName: Microsoft-Windows-Security-Auditing				
TimeCreated	Id	Level	DisplayName	Message
-----	--	-----	-----	-----
2/7/2026 7:53:05 PM	4634	Information		An account was logged off.
Subject:				
		Security ID:	S-1-5-18	
		Account Name:	DC\$	
		Account Domain:	TRAINING	
		Logon ID:	0x1606AD	
		Logon Type:	3	

Show username + time (readable output)

```
Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4624,4634} -MaxEvents 2 |
```

```
Select-Object TimeCreated, Id, LogName, TaskDisplayName, MachineName | Format-Table
```

TimeCreated	Id	LogName	TaskDisplayName	MachineName
-----	--	-----	-----	-----
2/7/2026 7:58:27 PM	4634	Security	Logoff	dc.training.com
2/7/2026 7:58:27 PM	4624	Security	Logon	dc.training.com

Last 24 hours login/logout


```
$start=(Get-Date).AddHours(-24)
Get-WinEvent -FilterHashtable @{
    LogName='Security'
    Id=4624,4634,4647
    StartTime=$start
} | Format-Table -AutoSize -Wrap
#} | Out-GridView
```

```
$start=(Get-Date).AddHours(-24)
Get-WinEvent -FilterHashtable @{
    LogName='Security'
    Id=4624,4634,4647
    StartTime=$start
} -MaxEvents 1 | Format-Table -AutoSize -Wrap
```

ProviderName: Microsoft-Windows-Security-Auditing

TimeCreated	Id	Level	DisplayName	Message
2/7/2026 8:23:27 PM	4634	Information		An account was logged off.

Subject:

Security ID: S-1-5-18

Account Name: DC\$

Account Domain: TRAINING

Logon ID: 0x1A00FC

Logon Type: 3

This event is generated when a logon session is destroyed. It may be used to identify the user who logged off using the Logon ID value. Logon IDs are only unique between

Pull only username + Security ID (SID) for login/logout events in the last 24 hours from the Security log.

```
$start = (Get-Date).AddHours(-24)
Get-WinEvent -FilterHashtable @{
    LogName = 'Security'
    Id = 4624,4634,4647 # login + logoff
    StartTime = $start
} | ForEach-Object {
    [PSCustomObject]@{
        TimeCreated = $_.TimeCreated
        UserName = $_.Properties[5].Value
        SecurityID = $_.Properties[4].Value
        EventID = $_.Id
    }
} | Format-Table -AutoSize
```

TimeCreated	UserName	SecurityID	EventID
2/7/2026 8:24:27 PM		3	4634
2/7/2026 8:24:27 PM	DC\$	S-1-5-18	4624
2/7/2026 8:24:14 PM		3	4634
2/7/2026 8:24:03 PM		3	4634
2/7/2026 8:24:03 PM		3	4634
2/7/2026 8:24:03 PM		3	4634
2/7/2026 8:24:03 PM	DC\$	S-1-5-18	4624
2/7/2026 8:24:03 PM	DC\$	S-1-5-18	4624
2/7/2026 8:24:03 PM	DC\$	S-1-5-18	4624
2/7/2026 8:24:03 PM	DC\$	S-1-5-18	4624
2/7/2026 8:23:39 PM		3	4634
2/7/2026 8:23:39 PM	DC\$	S-1-5-18	4624
2/7/2026 8:23:39 PM		3	4634
2/7/2026 8:23:39 PM	DC\$	S-1-5-18	4624
2/7/2026 8:23:27 PM		3	4634

Exporting to CSV

```
$start = (Get-Date).AddHours(-24)
Get-WinEvent -FilterHashtable @{
    LogName='Security'
    Id=4624,4634,4647
    StartTime=$start
} | ForEach-Object {
    [PSCustomObject]@{
        TimeCreated = $_.TimeCreated
        UserName    = $_.Properties[5].Value
        SecurityID  = $_.Properties[4].Value
        EventID     = $_.Id
    }
} | Export-Csv C:\LoginLogout_Last24hrs.csv -NoTypeInfo
```

```
"TimeCreated","UserName","SecurityID","EventID"
"2/7/2026 8:28:27 PM",,"3","4634"
"2/7/2026 8:28:27 PM","DC$","S-1-5-18","4624"
"2/7/2026 8:27:27 PM",,"3","4634"
"2/7/2026 8:27:27 PM","DC$","S-1-5-18","4624"
"2/7/2026 8:26:27 PM",,"3","4634"
"2/7/2026 8:26:27 PM","DC$","S-1-5-18","4624"
"2/7/2026 8:25:27 PM",,"3","4634"
"2/7/2026 8:25:27 PM","DC$","S-1-5-18","4624"
"2/7/2026 8:24:27 PM",,"3","4634"
"2/7/2026 8:24:27 PM","DC$","S-1-5-18","4624"
"2/7/2026 8:24:14 PM",,"3","4634"
"2/7/2026 8:24:03 PM",,"3","4634"
"2/7/2026 8:24:03 PM",,"3","4634"
"2/7/2026 8:24:03 PM",,"3","4634"
"2/7/2026 8:24:03 PM","DC$","S-1-5-18","4624"
"2/7/2026 8:24:03 PM","DC$","S-1-5-18","4624"
"2/7/2026 8:24:03 PM","DC$","S-1-5-18","4624"
"2/7/2026 8:24:03 PM","DC$","S-1-5-18","4624"
"2/7/2026 8:23:39 PM",,"3","4634"
"2/7/2026 8:23:39 PM","DC$","S-1-5-18","4624"
"2/7/2026 8:23:39 PM",,"3","4634"
"2/7/2026 8:23:39 PM","DC$","S-1-5-18","4624"
"2/7/2026 8:23:27 PM",,"3","4634"
"2/7/2026 8:23:27 PM","DC$","S-1-5-18","4624"
```

AD login logout report:

```
Import-Module ActiveDirectory

# Time window
$start = (Get-Date).AddHours(-24)

# Get all Domain Controllers
$DCs = Get-ADDomainController -Filter * | Select-Object -ExpandProperty HostName

$results = foreach ($dc in $DCs) {

    Get-WinEvent -ComputerName $dc -FilterHashtable @{
        LogName = 'Security'
        Id = 4624,4634,4647 # Logon + Logoff
        StartTime = $start
    } -ErrorAction SilentlyContinue |

    ForEach-Object {

        $eventType = switch ($_.Id) {
            4624 { "Logon" }
            4634 { "Logoff" }
            4647 { "User initiated logoff" }
        }

        [PSCustomObject]@{
            DateTime = $_.TimeCreated
            UserName = $_.Properties[5].Value
            ComputerName = $_.Properties[18].Value
            Event = $eventType
            DomainController = $dc
        }
    }
}
```

```
}
}
```

Display output

\$results | Sort-Object DateTime | Format-Table -AutoSize

2/7/2026	8:29:55	PM	jeetu	192.168.10.11	Logon	dc.training.com
2/7/2026	8:30:27	PM	DC\$	127.0.0.1	Logon	dc.training.com
2/7/2026	8:30:27	PM			Logoff	dc.training.com
2/7/2026	8:30:37	PM			Logoff	dc.training.com
2/7/2026	8:31:27	PM	DC\$	127.0.0.1	Logon	dc.training.com
2/7/2026	8:31:27	PM			Logoff	dc.training.com
2/7/2026	8:31:30	PM			Logoff	dc.training.com
2/7/2026	8:32:27	PM	DC\$	127.0.0.1	Logon	dc.training.com
2/7/2026	8:32:27	PM			Logoff	dc.training.com
2/7/2026	8:32:54	PM	DC\$	::1	Logon	dc.training.com
2/7/2026	8:33:05	PM			Logoff	dc.training.com
2/7/2026	8:33:27	PM	DC\$	127.0.0.1	Logon	dc.training.com
2/7/2026	8:33:27	PM			Logoff	dc.training.com
2/7/2026	8:33:40	PM	DC\$	192.168.10.10	Logon	dc.training.com
2/7/2026	8:33:40	PM			Logoff	dc.training.com

AD login logout report (excluding loopback)

```
Import-Module ActiveDirectory
# Time window (last 24 hours)
$StartTime = (Get-Date).AddHours(-24)
# Get all Domain Controllers
$DCs = Get-ADDomainController -Filter * | Select-Object -ExpandProperty HostName
$Events = foreach ($DC in $DCs) {
    Get-WinEvent -ComputerName $DC -FilterHashtable @{
        LogName = 'Security'
        Id = 4624,4634,4647 # Logon / Logoff
        StartTime = $StartTime
    } -ErrorAction SilentlyContinue |
    ForEach-Object {
        # Event type mapping
        $EventType = switch ($_.Id) {
            4624 { "Logon" }
            4634 { "Logoff" }
            4647 { "User Logoff" }
        }

        # IP Address (loopback filtering)
        $IPAddress = $_.Properties[18].Value
        if ($IPAddress -and $IPAddress -notin @("127.0.0.1", "::1")) {
            [PSCustomObject]@{
                DateTime = $_.TimeCreated
                UserName = $_.Properties[5].Value
                ComputerName = $_.Properties[11].Value
                Event = $EventType
                IPAddress = $IPAddress
                DomainController = $DC
            }
        }
    }
}
}
```

Display results

\$Events | Sort-Object DateTime | Format-Table -AutoSize

2/7/2026	9:38:13	AM	MEMBER2\$	Logon	192.168.10.11	dc.training.com
2/7/2026	9:38:13	AM	jeetu	Logon	192.168.10.11	dc.training.com
2/7/2026	9:39:14	AM	MEMBER2\$	Logon	192.168.10.11	dc.training.com
2/7/2026	9:39:14	AM	MEMBER2\$	Logon	192.168.10.11	dc.training.com
2/7/2026	9:39:14	AM	MEMBER2\$	Logon	192.168.10.11	dc.training.com
2/7/2026	9:39:50	AM	MEMBER2\$	Logon	192.168.10.11	dc.training.com
2/7/2026	9:40:13	AM	jeetu	Logon	192.168.10.11	dc.training.com
2/7/2026	9:40:13	AM	jeetu	Logon	-	dc.training.com
2/7/2026	9:40:13	AM	jeetu	Logon	192.168.10.11	dc.training.com
2/7/2026	9:40:13	AM	jeetu	Logon	192.168.10.11	dc.training.com
2/7/2026	9:40:13	AM	jeetu	Logon	192.168.10.11	dc.training.com
2/7/2026	9:40:50	AM	DC\$	Logon	192.168.10.10	dc.training.com
2/7/2026	9:40:50	AM	DC\$	Logon	192.168.10.10	dc.training.com
2/7/2026	9:41:11	AM	DC\$	Logon	192.168.10.10	dc.training.com
2/7/2026	9:41:11	AM	DC\$	Logon	192.168.10.10	dc.training.com

Check if SQUID is already installed using “rpm -qa | grep squid”

```
#!/bin/bash
```

```
set -e
```

```
# echo "Updating system packages..."
```

```
# dnf update -y
```

```
echo "Installing Squid proxy..."
```

```
dnf install squid -y
```

```
echo "Backing up original squid.conf..."
```

```
cp /etc/squid/squid.conf /etc/squid/squid.conf.bak
```

```
echo "Creating blocked websites list..."
```

```
cat <<EOF > /etc/squid/blocked_sites.txt
```

```
.facebook.com
```

```
.youtube.com
```

```
.yimg.com
```

```
.googlevideo.com
```

```
EOF
```

```
echo "Configuring squid..."
```

```
cat <<'EOF' > /etc/squid/squid.conf
```

```
# Define local network (change if needed)
```

```
acl localnet src 10.0.0.0/8
```

```
acl localnet src 172.16.0.0/12
```

```
acl localnet src 192.168.0.0/16
```

```
# Safe ports
```

```
acl SSL_ports port 443
```

```
acl Safe_ports port 80
```

```
acl Safe_ports port 443
```

```
acl Safe_ports port 21
```

```
acl Safe_ports port 1025-65535
```

```
acl CONNECT method CONNECT
```

```
# Blocked domains
```

```
acl blocked_sites dstdomain "/etc/squid/blocked_sites.txt"
```

```
# Access rules
```

```
http_access deny blocked_sites
```

```
http_access allow localnet
```

```
http_access allow localhost
```

```
http_access deny all
```

```
# Squid listens on this port
```

```
http_port 3128
```

```
# Logging for monitoring
access_log /var/log/squid/access.log squid
cache_log /var/log/squid/cache.log
# Performance settings
cache_mem 256 MB
maximum_object_size 100 MB
cache_dir ufs /var/spool/squid 1000 16 256

coredump_dir /var/spool/squid
EOF

echo "Setting permissions..."
chown -R squid:squid /var/log/squid
chown -R squid:squid /var/spool/squid

echo "Initializing cache directories..."
squid -z

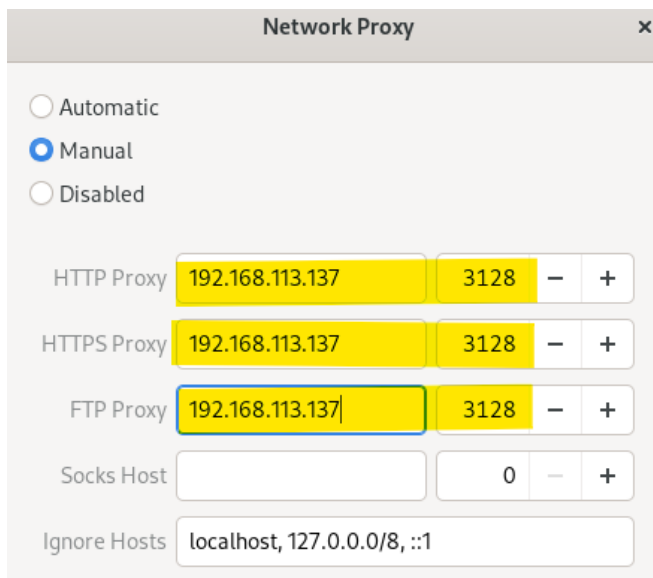
echo "Enabling and starting Squid service..."
systemctl enable squid
systemctl restart squid

echo "Configuring firewall..."
firewall-cmd --permanent --add-port=3128/tcp
firewall-cmd --reload

echo "Verifying service status..."
systemctl status squid --no-pager

echo "=====
echo "SQUID Proxy Installed & Running"
echo "Proxy Port: 3128"
echo "Blocked: Facebook, YouTube"
echo "Logs: /var/log/squid/access.log"
echo "=====
```

Change the proxy IP address in network settings

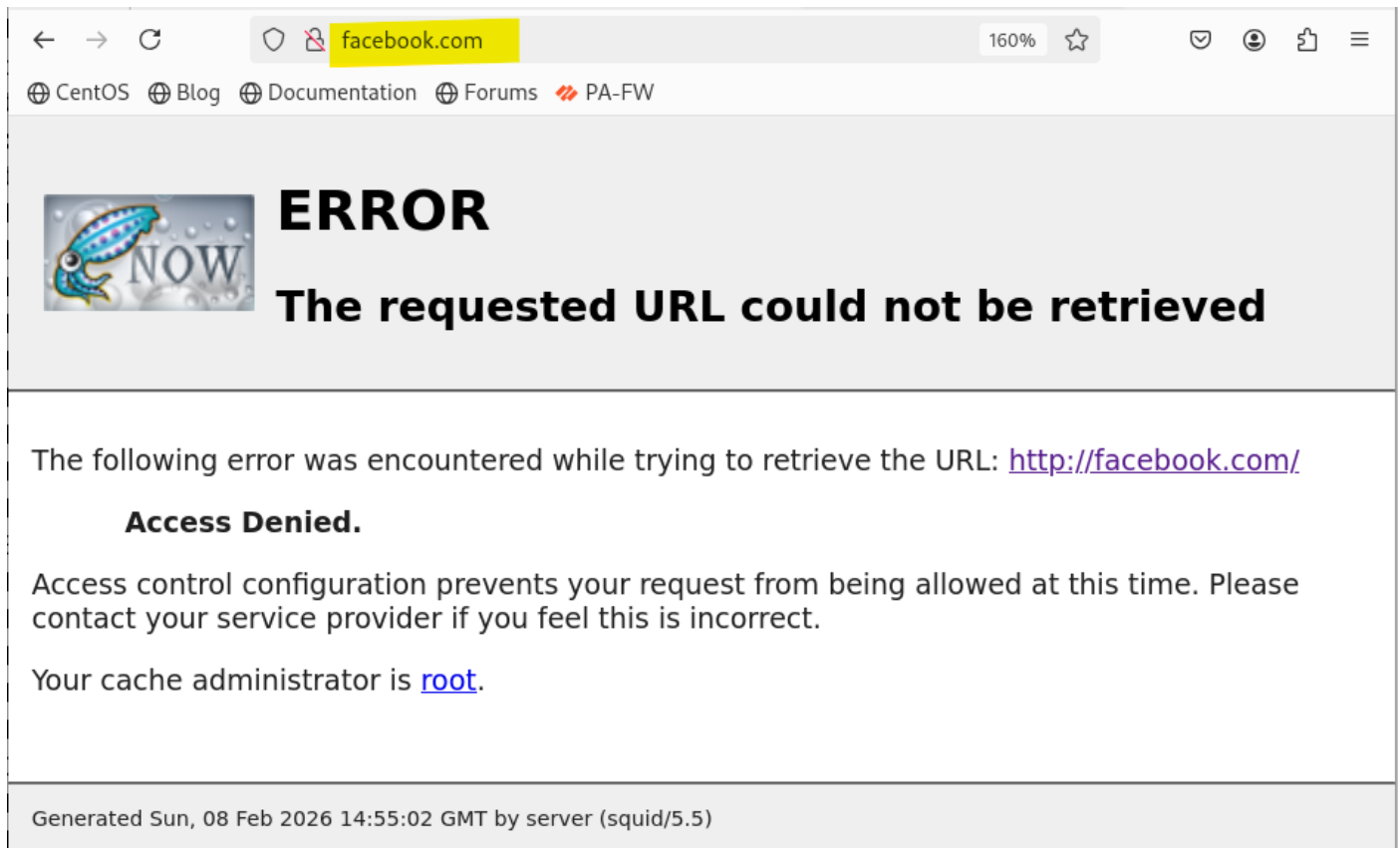


The screenshot shows the 'Network Proxy' settings window. The 'Manual' option is selected. The following table summarizes the proxy configurations shown in the interface:

Proxy Type	IP Address	Port	Controls
HTTP Proxy	192.168.113.137	3128	- +
HTTPS Proxy	192.168.113.137	3128	- +
FTP Proxy	192.168.113.137	3128	- +
Socks Host		0	- +
Ignore Hosts	localhost, 127.0.0.0/8, ::1		

Open Firefox and open following sites and see for which one SQUID block the access.

- Facebook.com
- YouTube.com
- Wikipedia.com



Check the logs:

tail -f /var/log/squid/access.log

```
[root@server ~]# tail -f /var/log/squid/access.log
1770562447.775 0 192.168.113.137 TCP_DENIED/403 3415 CONNECT www.youtube.com:443 - HIER_NONE/- text/html
1770562447.983 0 192.168.113.137 TCP_DENIED/403 3415 CONNECT www.youtube.com:443 - HIER_NONE/- text/html
1770562448.017 0 192.168.113.137 TCP_DENIED/403 3415 CONNECT www.youtube.com:443 - HIER_NONE/- text/html
1770562448.224 286 192.168.113.137 TCP_MISS/200 1222 POST http://o.pki.goog/wr2 - HIER_DIRECT/172.217.26.3 application/ocsp-response
1770562448.504 0 192.168.113.137 TCP_DENIED/403 3415 CONNECT www.youtube.com:443 - HIER_NONE/- text/html
1770562449.005 0 192.168.113.137 NONE_NONE/000 0 - error:transaction-end-before-headers - HIER_NONE/- -
1770562449.589 0 192.168.113.137 TCP_DENIED/403 3415 CONNECT www.youtube.com:443 - HIER_NONE/- text/html
1770562450.865 0 192.168.113.137 TCP_DENIED/403 3415 CONNECT www.youtube.com:443 - HIER_NONE/- text/html
1770562454.355 0 192.168.113.137 TCP_DENIED/403 3415 CONNECT www.youtube.com:443 - HIER_NONE/- text/html
1770562461.291 0 192.168.113.137 TCP_DENIED/403 3415 CONNECT www.youtube.com:443 - HIER_NONE/- text/html
1770562473.045 0 192.168.113.137 TCP_DENIED/403 3415 CONNECT www.youtube.com:443 - HIER_NONE/- text/html
1770562477.103 115 192.168.113.137 TCP_TUNNEL/200 11206 CONNECT img-getpocket.cdn.mozilla.net:443 - HIER_DIRECT/34.120.237.76 -
1770562477.152 163 192.168.113.137 TCP_TUNNEL/200 7462 CONNECT img-getpocket.cdn.mozilla.net:443 - HIER_DIRECT/34.120.237.76 -
1770562477.152 184 192.168.113.137 TCP_TUNNEL/200 10629 CONNECT img-getpocket.cdn.mozilla.net:443 - HIER_DIRECT/34.120.237.76 -
1770562477.152 190 192.168.113.137 TCP_TUNNEL/200 10863 CONNECT img-getpocket.cdn.mozilla.net:443 - HIER_DIRECT/34.120.237.76 -
1770562477.153 169 192.168.113.137 TCP_TUNNEL/200 9774 CONNECT img-getpocket.cdn.mozilla.net:443 - HIER_DIRECT/34.120.237.76 -
1770562477.386 0 192.168.113.137 TCP_DENIED/403 3415 CONNECT www.youtube.com:443 - HIER_NONE/- text/html
1770562488.189 0 192.168.113.137 TCP_DENIED/403 3415 CONNECT www.youtube.com:443 - HIER_NONE/- text/html
1770562488.260 0 192.168.113.137 TCP_DENIED/403 3407 CONNECT i.ytimg.com:443 - HIER_NONE/- text/html
1770562489.201 0 192.168.113.137 TCP_DENIED/403 3415 CONNECT www.youtube.com:443 - HIER_NONE/- text/html
1770562489.662 0 192.168.113.137 TCP_DENIED/403 3415 CONNECT www.youtube.com:443 - HIER_NONE/- text/html
1770562489.714 0 192.168.113.137 TCP_DENIED/403 3415 CONNECT www.youtube.com:443 - HIER_NONE/- text/html
1770562489.719 0 192.168.113.137 TCP_DENIED/403 3415 CONNECT www.youtube.com:443 - HIER_NONE/- text/html
1770562489.720 0 192.168.113.137 TCP_DENIED/403 3415 CONNECT www.youtube.com:443 - HIER_NONE/- text/html
1770562489.721 0 192.168.113.137 TCP_DENIED/403 3415 CONNECT www.youtube.com:443 - HIER_NONE/- text/html
1770562489.921 0 192.168.113.137 TCP_DENIED/403 3415 CONNECT www.youtube.com:443 - HIER_NONE/- text/html
```


Understand access.log format

1770562866.064 0 192.168.113.137 TCP_DENIED/403 3415 CONNECT www.youtube.com:443 - HIER_NONE/- text/html

Field	Meaning
1770562866.064	Unix timestamp
0	Response time (ms)
192.168.113.137	Client IP
TCP_DENIED/403	Result code + HTTP status
3415	Response size
CONNECT/GET	Method
URL	Requested site
HIER_NONE/-	Routing info
text/html	MIME type

Watch only blocked traffic:

```
# tail -f /var/log/squid/access.log | grep TCP_DENIED
```

Monitor YouTube/Facebook attempts:

```
# tail -f /var/log/squid/access.log | egrep -i "facebook|youtube"
```

```
[root@server ~]# tail -f /var/log/squid/access.log | egrep -i "facebook|youtube"
1770562679.478 0 192.168.113.137 TCP_DENIED/403 3415 CONNECT www.youtube.com:443 - HIER_NONE/- text/html
1770562711.797 0 192.168.113.137 TCP_DENIED/403 3415 CONNECT www.youtube.com:443 - HIER_NONE/- text/html
1770562743.906 0 192.168.113.137 TCP_DENIED/403 3415 CONNECT www.youtube.com:443 - HIER_NONE/- text/html
1770562775.046 0 192.168.113.137 TCP_DENIED/403 3415 CONNECT www.youtube.com:443 - HIER_NONE/- text/html
1770562806.168 0 192.168.113.137 TCP_DENIED/403 3415 CONNECT www.youtube.com:443 - HIER_NONE/- text/html
1770562837.251 0 192.168.113.137 TCP_DENIED/403 3415 CONNECT www.youtube.com:443 - HIER_NONE/- text/html
1770562866.064 0 192.168.113.137 TCP_DENIED/403 3415 CONNECT www.youtube.com:443 - HIER_NONE/- text/html
```

Top users consuming internet

```
# awk '{print $3}' /var/log/squid/access.log | sort | uniq -c | sort -nr | head
```

Check blocked websites report

```
# grep TCP_DENIED /var/log/squid/access.log | awk '{print $7}' | sort | uniq -c | sort -nr
```

Bandwidth usage by client

```
# awk '{bytes[$3]+=$5} END {for (ip in bytes) print ip, bytes[ip]/1024/1024 " MB"}' /var/log/squid/access.log | sort -nr -k2
```

Task – Students to launch a brute force attack using Kali linux and observe the events in the event viewer

Kali → Application → 08 Credential Access → Medusa

Use – “medusa -d” to list modules

- medusa -h → for host/IP Address
- -M → list of modules
- -P → password file
- -u → user name

Create a user “demo” and with the password (may be present in a file)

Command

```
# medusa -h <ip-address> -u <username> -P <password-file> -M smbnt
```

```
(root@kali)-[~]
# medusa -h 192.168.113.138 -u demo -P /home/kali/Desktop/1000_pwd_list.txt -M smbnt
Medusa v2.3 [http://www.foofus.net] (C) JoMo-Kun / Foofus Networks <jmk@foofus.net>

2026-02-09 12:43:42 ACCOUNT CHECK: [smbnt] Host: 192.168.113.138 (1 of 1, 0 complete) User: de
```

Wait for some time:

```
2026-02-09 12:43:43 ACCOUNT CHECK: [smbnt] Host: 192.168.113.138 (1 of 1, 0 complete) User: demo (1 of 1, 0 complete) Password: Admin137 (940 of 1000 complete)
2026-02-09 12:43:43 ACCOUNT CHECK: [smbnt] Host: 192.168.113.138 (1 of 1, 0 complete) User: demo (1 of 1, 0 complete) Password: Password138 (941 of 1000 complete)
2026-02-09 12:43:43 ACCOUNT CHECK: [smbnt] Host: 192.168.113.138 (1 of 1, 0 complete) User: demo (1 of 1, 0 complete) Password: Welcome138 (942 of 1000 complete)
2026-02-09 12:43:43 ACCOUNT CHECK: [smbnt] Host: 192.168.113.138 (1 of 1, 0 complete) User: demo (1 of 1, 0 complete) Password: Admin138 (943 of 1000 complete)
2026-02-09 12:43:43 ACCOUNT CHECK: [smbnt] Host: 192.168.113.138 (1 of 1, 0 complete) User: demo (1 of 1, 0 complete) Password: Password139 (944 of 1000 complete)
2026-02-09 12:43:43 ACCOUNT FOUND: [smbnt] Host: 192.168.113.138 User: demo Password: Password139 [SUCCESS (ADMIN$ - Access Denied)]
```

For logs on kali

➤ Install rsyslog

```
# apt install rsyslog -y
```

For log on Event log on windows:

```
Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4624} |
ForEach-Object {
    [PSCustomObject]@{
        TimeCreated = $_.TimeCreated
        Username    = $_.Properties[5].Value
        LogonType   = $_.Properties[8].Value
        SourceIP    = $_.Properties[18].Value
    }
} | Sort-Object TimeCreated -Descending
```

```

2/9/2026 9:46:09 AM DC$ 3 fe80::54fc:23d4:e007:75e9
2/9/2026 9:45:53 AM DC$ 3 ::1
2/9/2026 9:44:53 AM DC$ 3 ::1
2/9/2026 9:43:53 AM DC$ 3 ::1
2/9/2026 9:43:43 AM demo 3 192.168.113.136
2/9/2026 9:43:35 AM DC$ 3 fe80::54fc:23d4:e007:75e9
2/9/2026 9:42:53 AM DC$ 3 ::1
2/9/2026 9:41:53 AM DC$ 3 ::1
2/9/2026 9:41:24 AM demo 3 192.168.113.136
2/9/2026 9:41:20 AM DC$ 3 fe80::54fc:23d4:e007:75e9

```

AD Last logon details:

```

Untitled1.ps1* X  Untitled2.ps1*
1  cls
2  Get-ADUser -Filter * -Properties LastLogonDate |
3  Select Name, LastLogonDate | Sort LastLogonDate -Descending
4

```

Name	LastLogonDate
demo	2/9/2026 9:41:24 AM
jeetu	2/7/2026 9:38:12 AM
Administrator	2/7/2026 9:06:04 AM
Guest	
DefaultAccount	
krbtgt	

Task – Students to launch DNS based attacks and explore the results

Kali → Application → Ettercap

Config /etc/Ettercap/etter.conf file

Change uid and gid to "0"

```
[privs]
ec_uid = 0          # nobody is the default
ec_gid = 0          # nobody is the default
```

Uncomment redir lines

```
redir_command_on = "iptables -t nat -A PREROUTING -i %iface -p tcp -d %destination --dport %port -j REDIRECT --to-port %rport"
redir_command_off = "iptables -t nat -D PREROUTING -i %iface -p tcp -d %destination --dport %port -j REDIRECT --to-port %rport"

# pendant for IPv6 - Note that you need iptables v1.4.16 or newer to use IPv6 redirect
redir6_command_on = "ip6tables -t nat -A PREROUTING -i %iface -p tcp -d %destination --dport %port -j REDIRECT --to-port %rport"
redir6_command_off = "ip6tables -t nat -D PREROUTING -i %iface -p tcp -d %destination --dport %port -j REDIRECT --to-port %rport"
```

Save and exit

Now open /etc/Ettercap/etter.dns and add this to bottom

```
# vim:ts=8:noexpandtab
*.google.com      A      192.168.113.136
www.google.com    PTR     192.168.113.136
```

Install apache2

Go to /var/www/html and create index.html (and remove any other file)

```
<html>
<body>
<h1 align='center'>
You are hacked
</h1>
</body>
</html>
```

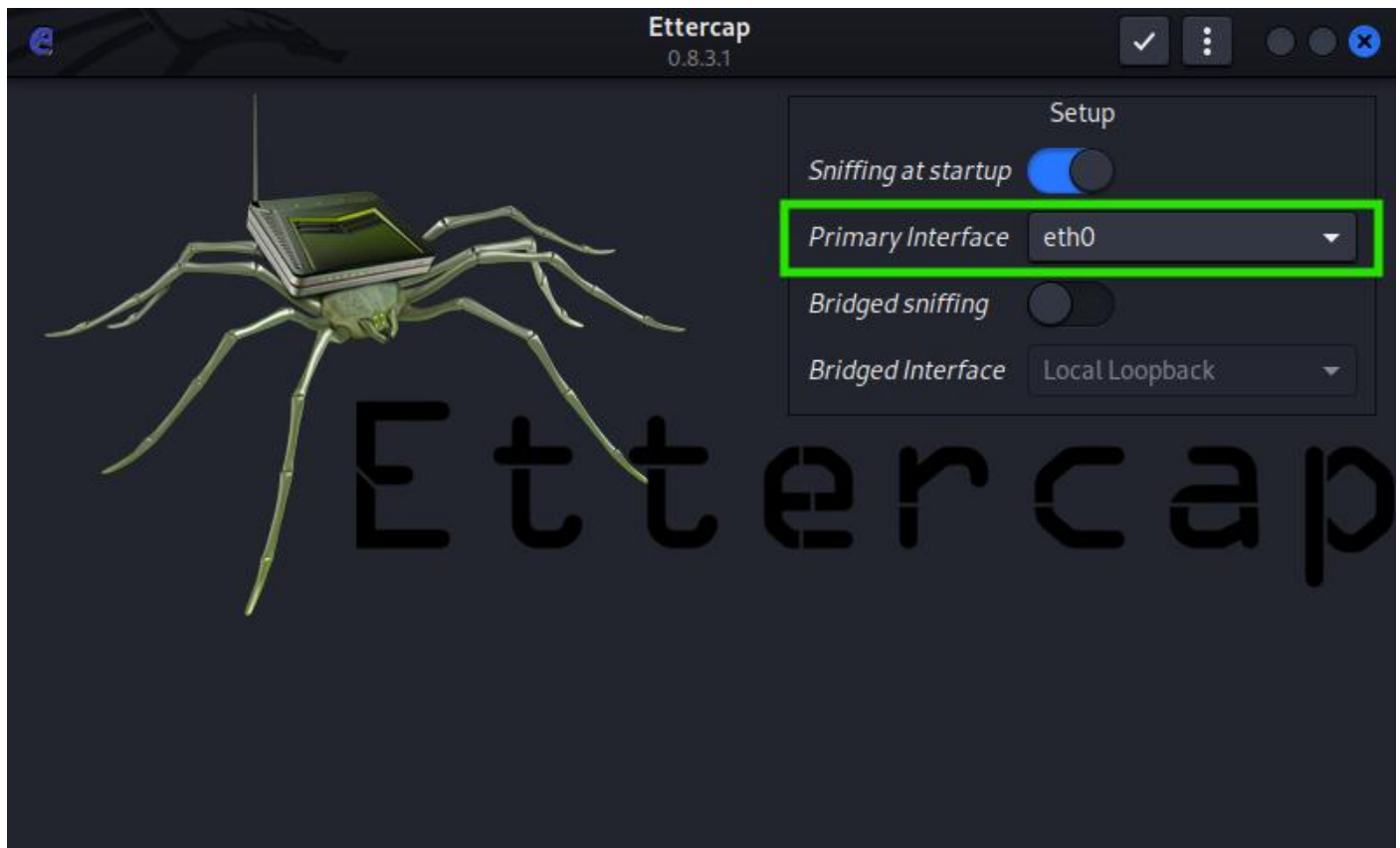
Start the ettercap graphical interface using below command

ettercap -G

Or search for Ettercap in the menu.

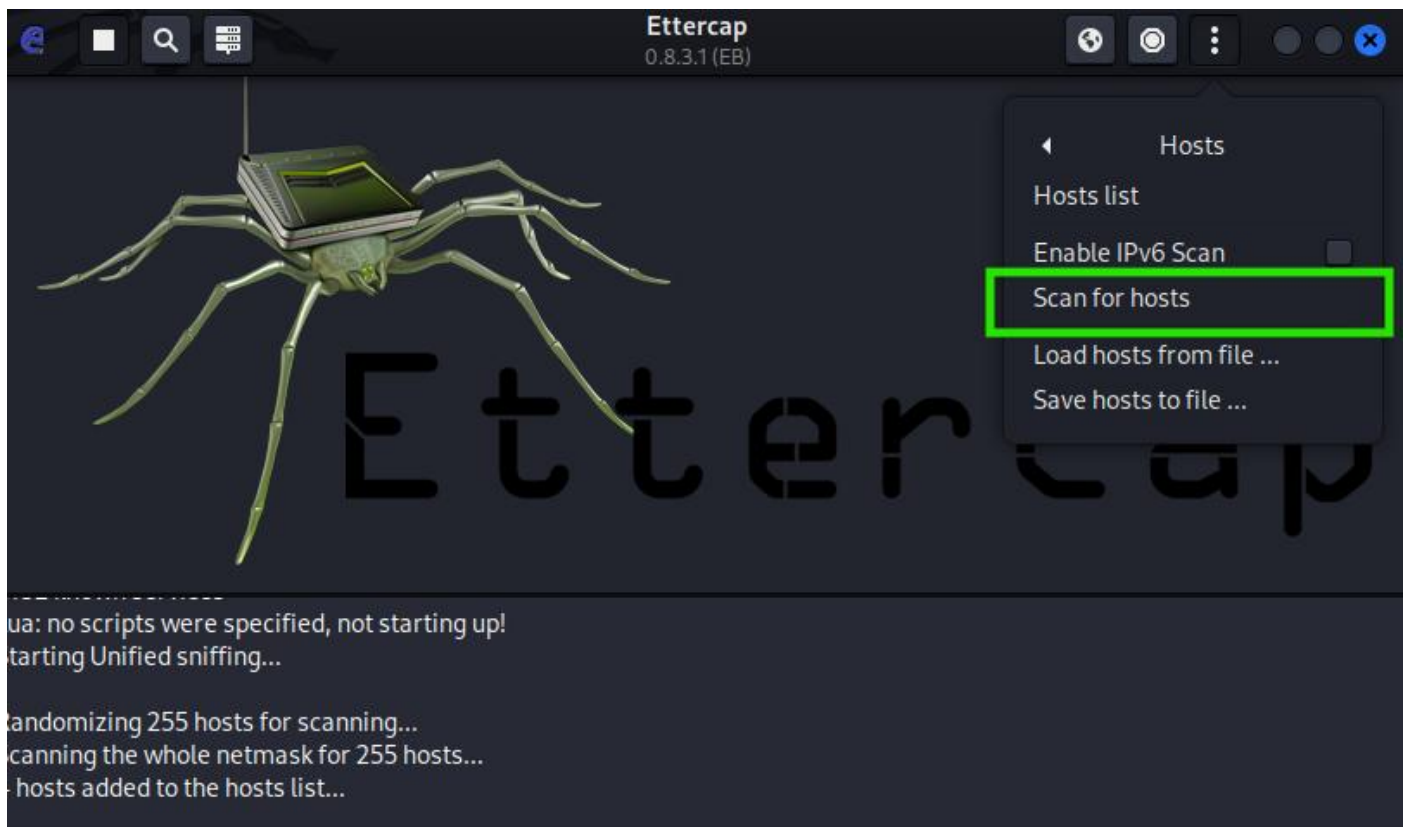
Select the Primary Interface (eth0, wlan0....)

Press enter or click to view image in full size



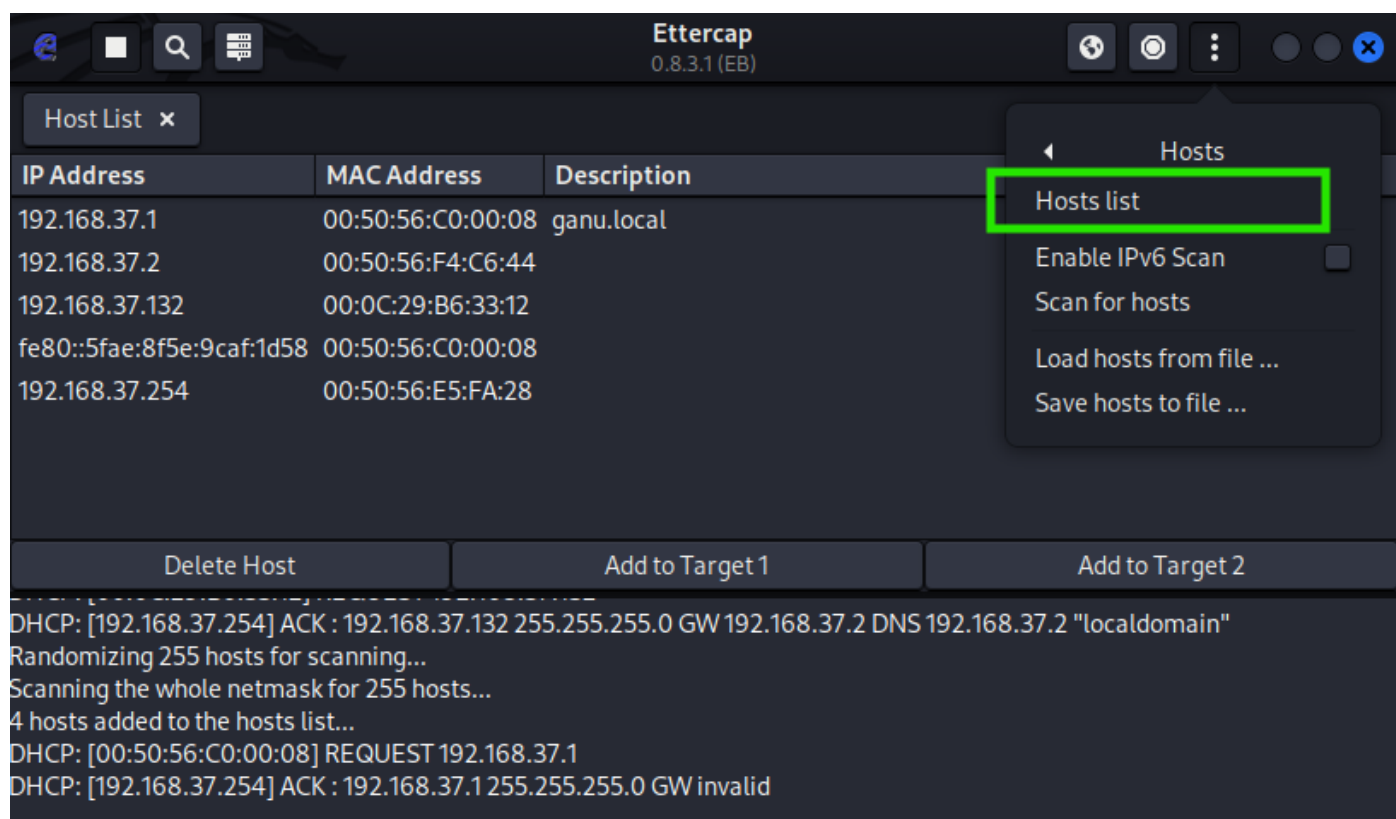
3. Scan the network click on the vertical three dots(.) then click on the **Hosts** after that click on **Scan for hosts**.

Press enter or click to view image in full size



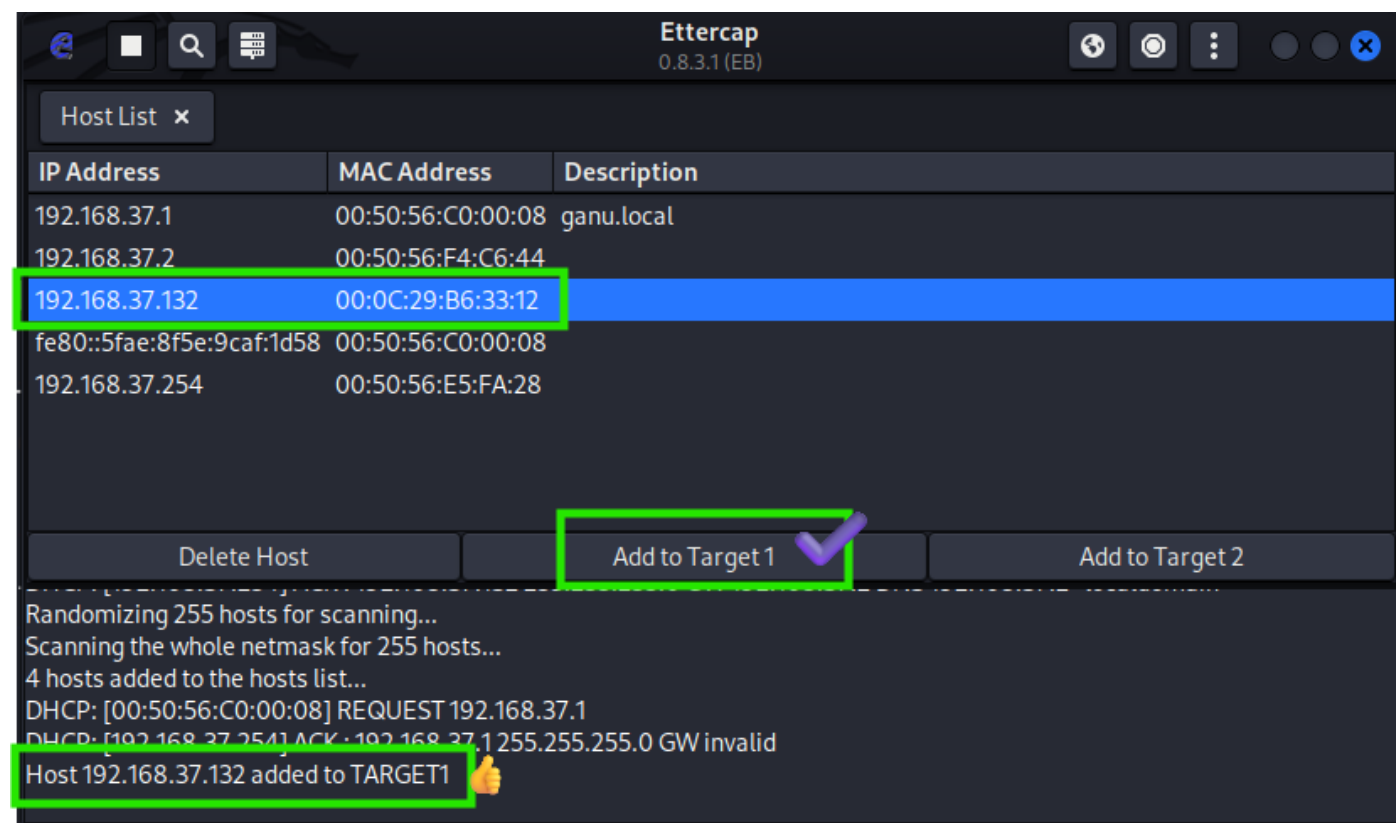
4. If scanning is completed then again click on the vertical three dots(.) then click on the **Hosts** after that click on **Hosts list**.

Press enter or click to view image in full size



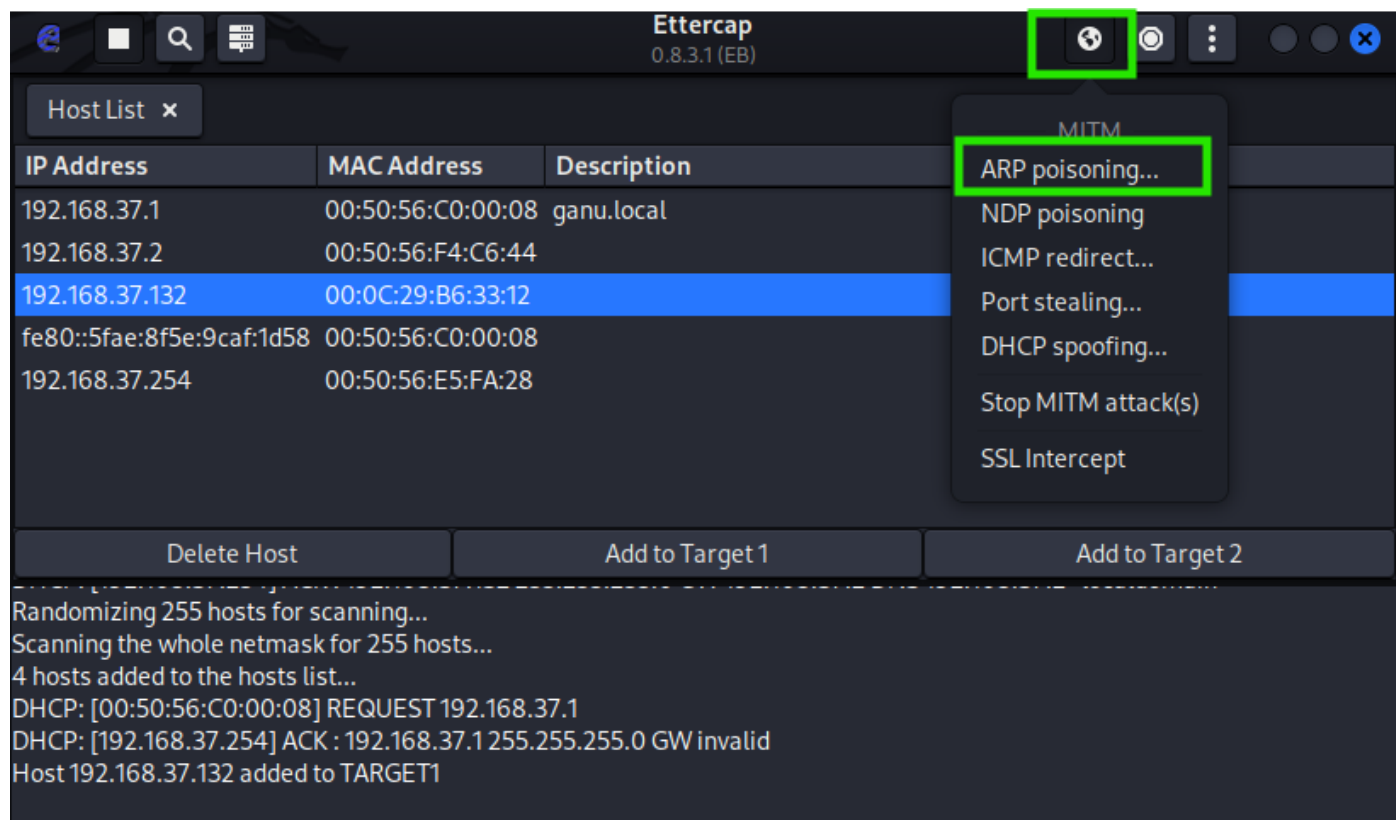
5. Click on the windows machine IP then click on the **Add Target1** see below your host is added.

Press enter or click to view image in full size



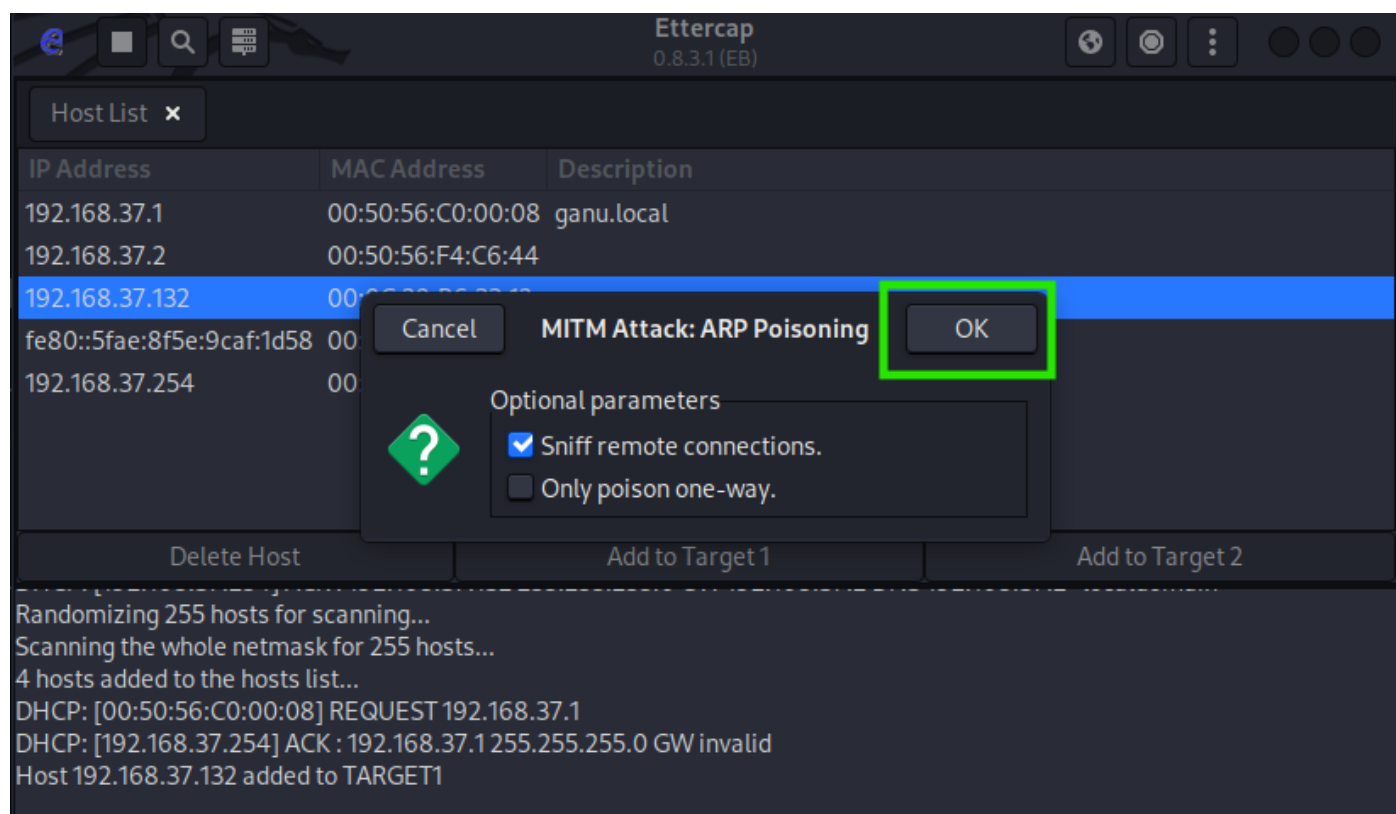
6. Select **Windows M/C IP** then click on **MITM menu >ARP Poisoning**.

Press enter or click to view image in full size



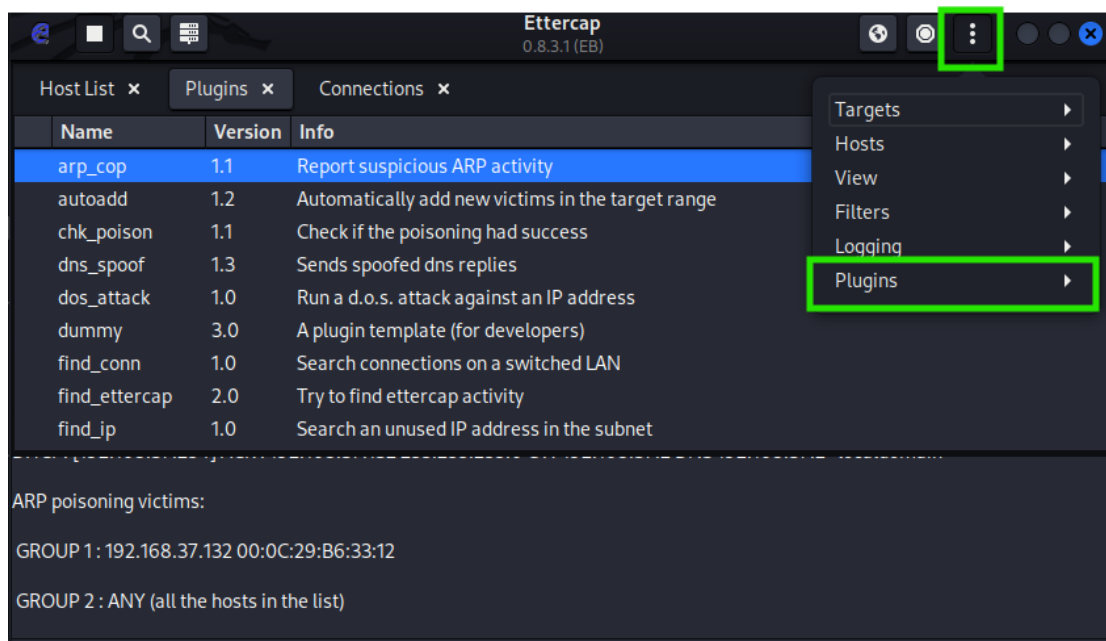
7. After that click on **OK** but **Sniff remote connection** must be enabled.

Press enter or click to view image in full size



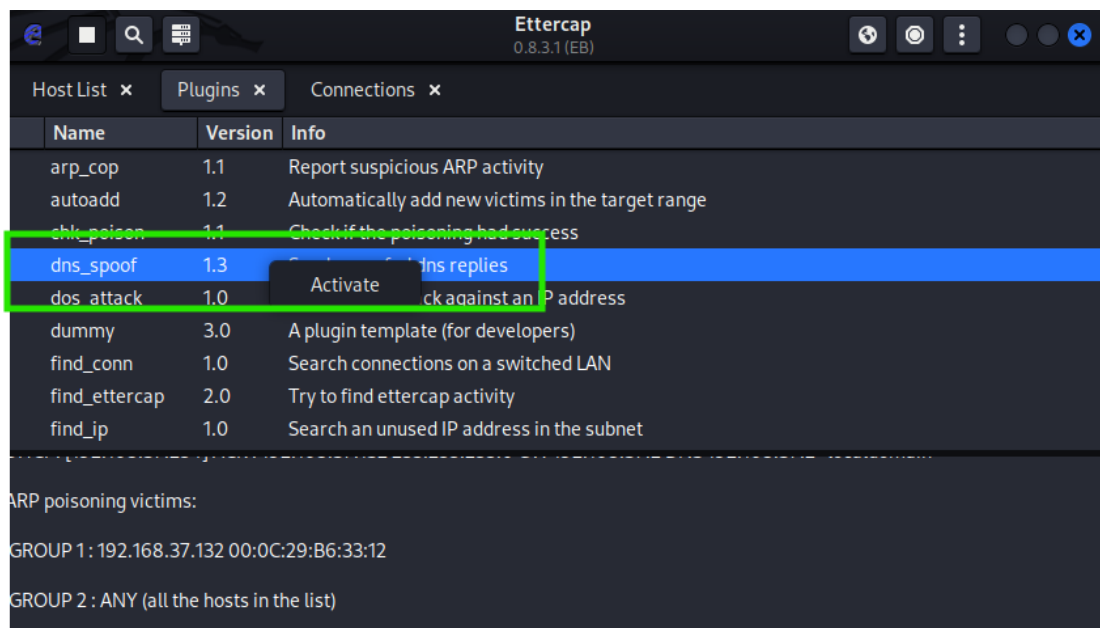
8. After ARP poisoning on victim the open the **plugins**

Press enter or click to view image in full size



9. After opening the plugin, click on **Manage Plugins**, then select **dns_spoof**. Right-click on **dns_spoof** and click **Activate**.

Press enter or click to view image in full size



Now go to windows and access google.com



You are hacked

URL:<https://medium.com/@ganeshshelke785/dns-poisoning-dns-spoofing-using-ettercap-and-setoolkit-35b2bfdbacd5>

Step 1 — Check the “From” vs “Return-Path”

Attackers spoof visible from address.

Red flags:

- From = x.com
- Return-path = y.com

Mismatch = spoofing attempt.

Step 2 — Analyze the “Received” Chain

Every mail server adds a Received entry.

Work bottom → top.

Validate:

- Source IP
- Sending mail server
- Geo location
- Known infrastructure

Red flags:

- First hop from residential IP
- Suspicious country
- Unknown relay server
- Broken or missing hops

Step 3 — SPF Authentication

SPF verifies sending server authorization.

Look for:

- spf=pass
- spf=fail
- spf=softfail
- spf=neutral

Red flags:

- SPF fail
- SPF softfail from external IP

Step 4 — DKIM Signature

Validates integrity and domain ownership.

Check:

- dkim=pass
- dkim=fail

Red flags:

- DKIM fail
- DKIM missing in corporate emails

Step 5 — DMARC Policy

Policy enforcement layer.

Look:

- dmarc=pass
- dmarc=fail

policy=reject/quarantine/none

Red flags:

- DMARC fail
- policy=none for large organizations

Step 6 — Message-ID Analysis

Legitimate Message-ID aligns with sender domain.

Example legit:

<12345@mail.microsoft.com>

Red flags:

- Random domain
- Public mail provider for corporate email
- Missing Message-ID

Step 7 — Reply-To Manipulation

Common phishing technique.

Red flags:

- From: finance@company.com
- Reply-To: gmail.com

Step 8 — X-Originating-IP

Identifies real sender.

Red flags:

- Residential ISP IP
- High-risk geo location
- TOR exit nodes

Step 9 — Attachment & MIME Clues

Check:

- Content-Type
- Encoded attachments
- Macro-enabled docs

Red flags:

- .html attachments
- .iso / .img
- password-protected zip

Step 1 — Prerequisites

1. Log in as Local Administrator.
2. Ensure latest Windows updates:
 - a. Settings → Update & Security → Windows Update → Check for updates
3. Download CIS Benchmark & GPO templates:
 - a. CIS Workbench → Windows 10 Benchmark
 - b. Microsoft Security Compliance Toolkit (for baseline GPOs)

Step 2 — Configure Account Policies

Open → Win+R → “secpol.msc”

Password Policy

Account Policies → Password Policy

Set:

Setting	Value
Enforce password history	24
Maximum password age	60 days
Minimum password age	1 day
Minimum password length	14
Complexity requirements	Enabled
Store passwords reversible encryption	Disabled

Account Lockout Policy:

Account Policies → Account Lockout Policy

Setting	Value
Account lockout threshold	5 attempts
Lockout duration	15 min
Reset counter after	15 min

Step 3 — Configure Local Policies

Open → secpol.msc → Local Policies

2.1 Audit Policy

Path → Local Policies → Audit Policy

Enable:

- Audit logon events — Success, Failure
- Audit account logon — Success, Failure
- Audit privilege use — Failure
- Audit policy change — Success
- Audit system events — Success, Failure
- Audit process tracking — Success
- Audit object access — Failure

2.2 User Rights Assignment

Local Policies → User Rights Assignment

Key CIS configurations:

Policy	Configuration
Access this computer from network	Administrators, Authenticated Users
Deny log on locally	Guests
Deny access to computer from network	Guests
Allow log on locally	Administrators, Users
Debug programs	Administrators only

2.3 Security Options

Path → Local Policies → Security Options

Important settings:

- Accounts: Administrator account status → Disabled
- Accounts: Guest account status → Disabled
- Interactive logon: Message text/title → Set warning banner
- Network security: LAN Manager authentication level → NTLMv2 only
- Microsoft network client: Digitally sign communications → Enabled
- SMB v1 → Disabled

Step 4 — Configure Windows Firewall

Open → wf.msc

Configure:

- Firewall state: ON (Domain, Private, Public)
- Inbound: Block by default
- Outbound: Allow (restrict as per org)

Enable logging → Monitoring → Firewall → Properties → Logging

Step 5 — Enable Advanced Audit Policy

Open → gpedit.msc

Enable:

- Logon/logoff
- Account logon
- Policy change
- Privilege use
- System
- Detailed tracking

Step 6 — Configure Microsoft Defender

Open → Windows Security

Antivirus

- Real-time protection → ON
- Cloud protection → ON
- Tamper protection → ON
- Automatic sample submission → ON

Attack Surface Reduction (PowerShell admin)

```
Set-MpPreference -AttackSurfaceReductionRules_Iids <rules> -AttackSurfaceReductionRules_Actions Enabled
```

Step 7 — Disable Unnecessary Services

Open → services.msc

Disable (if not needed):

- Remote Registry
- Xbox services
- Print Spooler (if no printer)
- Bluetooth services
- Fax

Step 8 — Secure RDP

If RDP enabled → System Properties → Remote

- Allow only Network Level Authentication
- Restrict to specific users
- Change default port (registry)
- Enable logging

Registry → HKLM\System\CurrentControlSet\Control\Terminal Server

Step 9 — Configure UAC

Control Panel → User Accounts → Change UAC settings

Set to → *Always notify*

Step 10 — Disable SMBv1

PowerShell (Admin) → `Disable-WindowsOptionalFeature -Online -FeatureName SMB1Protocol`

Step 11 — Configure Logging & Event Retention

Open → `eventvwr.msc`

For Security log:

- Max size: 1 GB+
- Retention: Do not overwrite

Step 12 — Configure BitLocker

Control Panel → BitLocker Drive Encryption

- Enable for OS drive
- TPM + PIN preferred

Step 13 — Browser Hardening

For Edge:

- SmartScreen → Enabled
- Block popups
- Disable password storage
- Disable extensions install for users

Step 14 — Apply CIS Registry Hardening

Use Microsoft baseline → `LGPO.exe /g <CIS-GPO-Folder>`

This applies:

- Credential Guard
- LSASS protection
- NTLM restrictions
- WinRM hardening

Step 15 — Validate Compliance

CIS CAT Tool

CIS Configuration Assessment Tool scans and reports compliance.

PowerShell verification

```
Get-MpComputerStatus  
auditpol /get /category:*  
net accounts
```

Step 16 — Continuous Monitoring (SOC-ready)

Monitor these event IDs:

Category	Event ID
Logon	4624, 4625
Privilege escalation	4672
New user created	4720
Group change	4732
Service installed	7045
PowerShell script	4104

Critical CIS Controls Not to Skip

These give maximum risk reduction:

- Disable SMBv1
- Enforce password policy
- Enable Defender + ASR
- Audit policy full logging
- Firewall strict inbound
- BitLocker
- LSASS protection
- UAC Always Notify

Task – Students will perform vulnerability scanning of a given vulnerable machine and export the results

Installing Nessus for networking monitoring:

Download link: <https://www.tenable.com/downloads/nessus?loginAttempted=true>

Selecting correct version:

1 Download and Install Nessus

Choose Download

Version

Nessus - 10.11.1



Platform

Linux - RHEL 8 - x86_64

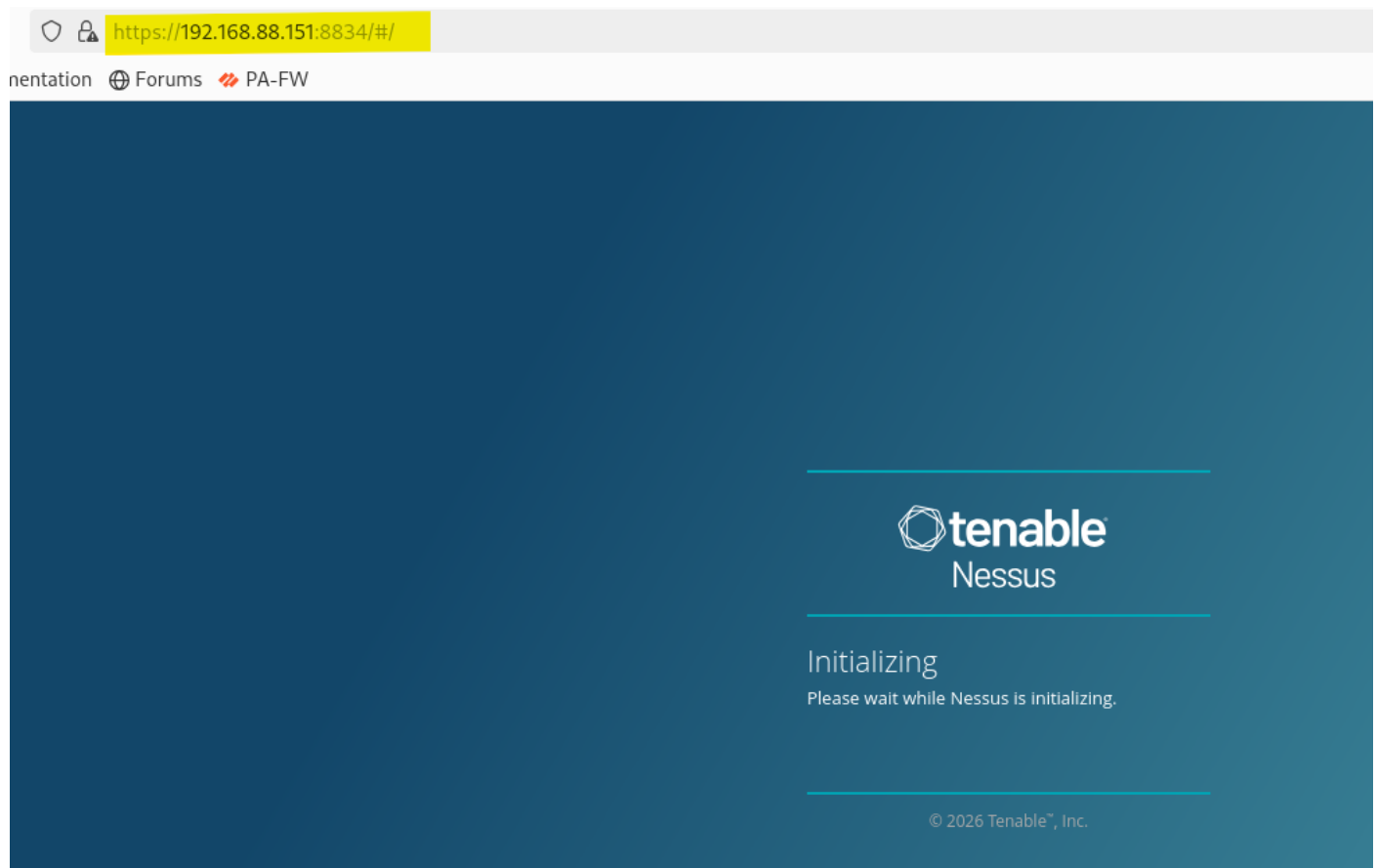


Download

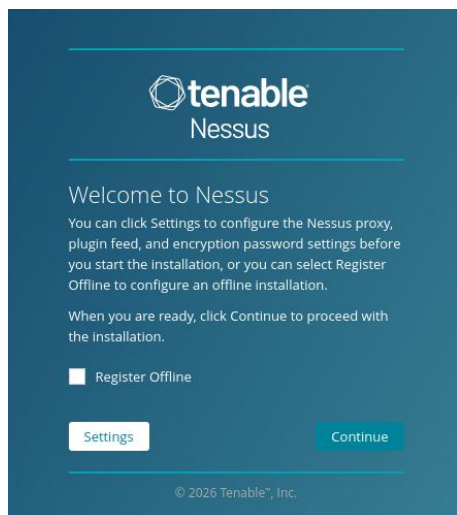
Checksum

- `sudo systemctl start nessusd`
- `sudo systemctl enable nessusd`

accessing Nessus on firefox <https://192.168.88.151:8834/>



Register Nessus Essential for small network



tenable
Nessus

Welcome to Nessus

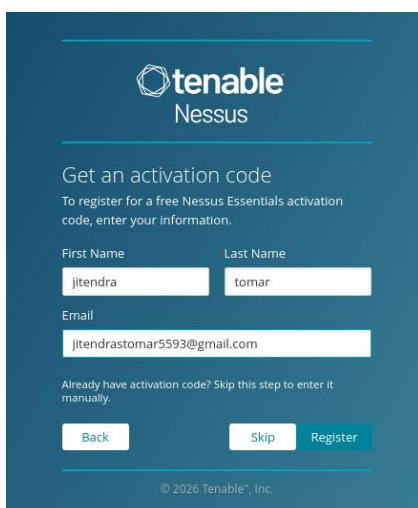
You can click Settings to configure the Nessus proxy, plugin feed, and encryption password settings before you start the installation, or you can select Register Offline to configure an offline installation.

When you are ready, click Continue to proceed with the installation.

☐ Register Offline

[Settings](#) [Continue](#)

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tenable
Nessus

Get an activation code

To register for a free Nessus Essentials activation code, enter your information.

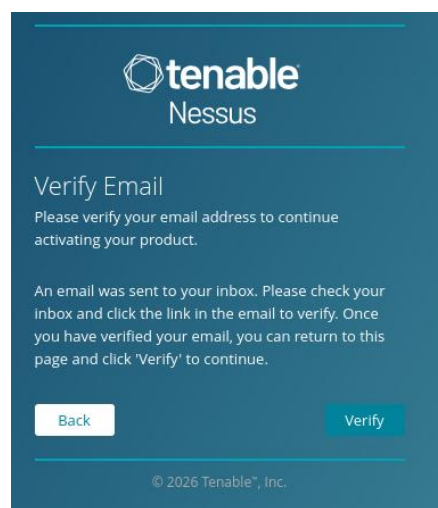
First Name Last Name

Email

Already have activation code? Skip this step to enter it manually.

[Back](#) [Skip](#) [Register](#)

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Nessus

Verify Email

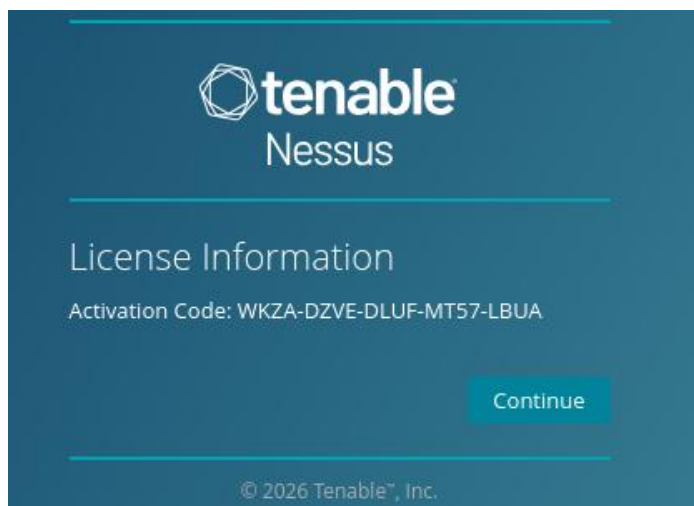
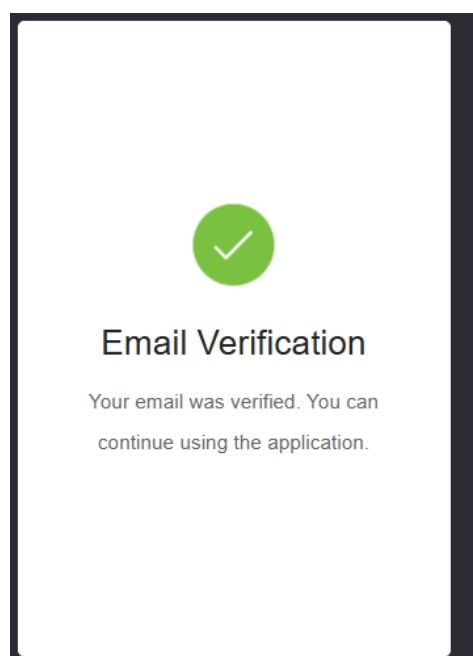
Please verify your email address to continue activating your product.

An email was sent to your inbox. Please check your inbox and click the link in the email to verify. Once you have verified your email, you can return to this page and click 'Verify' to continue.

[Back](#) [Verify](#)

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Verify over your email ID and click on verify button.



tenable
Nessus

License Information

Activation Code: WKZA-DZVE-DLUF-MT57-LBUA

[Continue](#)

© 2026 Tenable®, Inc.

Activation Code: WKZA-DZVE-DLUF-MT57-LBUA



Create a user account

Create a Nessus administrator user account. Use this username and password to log in to Nessus.

Username *

Password *

Back

Submit



Goto Scan → New Scan

CentOS Blog Documentation Forums PA-FW

tenable

Nessus Essentials

Scans

Settings

jeetu

FOLDERS

My Scans

All Scans

Trash

RESOURCES

Policies

Plugin Rules

My Scans

Import

New Folder

New Scan

This folder is empty. [Create a new scan.](#)

Choose Basic Network Scan

tenable

Nessus Essentials

Scans

Settings

FOLDERS

My Scans

All Scans

Trash

RESOURCES

Policies

Plugin Rules

Scan Templates

[Back to Scans](#)

Scanner

DISCOVERY



Host Discovery

A simple scan to discover live hosts and open ports.



Ping-Only Discovery

A simple scan to discover live hosts with minimal network traffic.

VULNERABILITIES



Basic Network Scan

A full system scan suitable for any host.



Credential Validation

Verify that host credential pairs for Windows & Unix successfully authenticate to scan targets



Advanced Scan

Configure a scan without using any recommendations.

JITENDRA SINGH TOMAR

69

After scanning,

tenable Nessus Essentials Scans Settings

demo_scan

Back to My Scans

Hosts 2 Vulnerabilities 90 History 1

Filter Search Hosts 2 Hosts

Host	Auth	Vulnerabilities	%
192.168.88.151	Pass	19 12 193	99%
192.168.88.155	Fail	43	100%

Scan Details

Policy: Basic Network Scan
Status: Running
Severity Base: CVSS v3.0
Scanner: Local Scanner
Start: Today at 3:19 PM

Vulnerabilities

Critical
High
Medium
Low
Info

Once you click the target IP address, it will list all the vulnerabilities:

tenable Nessus Essentials Scans Settings

demo_scan / 192.168.88.155

Back to Hosts

Vulnerabilities 22

Filter Search Vulnerabilities 22 Vulnerabilities

Sev	CVSS	VPR	EPSS	Name	Family	Count
MIXED	...	...	...	SMB (Multiple Issues)	Misc.	2
LOW	2.1 *	...	...	ICMP Timestamp Request Remote Date Disclosure	General	1
INFO	...	...	...	SMB (Multiple Issues)	Windows	6
INFO	...	...	...	HTTP (Multiple Issues)	Web Servers	5
INFO	...	...	...	DCE Services Enumeration	Windows	9
INFO	...	...	...	Nessus SYN scanner	Port scanners	5
INFO	...	...	...	Service Detection	Service detection	2
INFO	...	...	...	Common Platform Enumeration (CPE)	General	1
INFO	...	...	...	Device Type	General	1
INFO	...	...	...	Ethernet Card Manufacturer Detection	Misc.	1
INFO	...	...	...	Ethernet MAC Addresses	General	1

Host: 192.168.88.155

Host Details

IP: 192.168.88.155
MAC: 00:0C:29:49:A3:C0
OS: Microsoft Windows Server 2016 Datacenter Evaluation 14393
Start: Today at 3:19 PM
End: Today at 3:32 PM
Elapsed: 13 minutes
Auth: Fail

Vulnerabilities

Critical
High
Medium
Low
Info

Click on vulnerabilities

tenable Nessus Essentials Scans Settings

demo_scan

Back to My Scans

Hosts 2 Vulnerabilities 90 History 1

Filter Search Vulnerabilities 90 Vulnerabilities

Sev	CVSS	VPR	EPSS	Name	Family	Count
MIXED	...	...	...	CentOS (Multiple Issues)	CentOS Local Security Checks	22
MIXED	...	...	...	Nagios XI (Multiple Issues)	CGI abuses	4
HIGH	7.5	...	...	NFS Shares World Readable	RPC	1
MIXED	...	...	...	Apache Tomcat (Multiple Issues)	Web Servers	8
MIXED	...	...	...	SSL (Multiple Issues)	General	10
MIXED	...	...	...	SMB (Multiple Issues)	Misc.	2
LOW	2.1 *	...	...	ICMP Timestamp Request Remote Date Disclosure	General	1
INFO	...	...	...	HTTP (Multiple Issues)	Web Servers	16
INFO	...	...	...	SSH (Multiple Issues)	General	8
INFO	...	...	...	SMB (Multiple Issues)	Windows	6
INFO	...	...	...	Mysql (Multiple Issues)	Databases	4

Scan Details

Policy: Basic Network Scan
Status: Running
Severity Base: CVSS v3.0
Scanner: Local Scanner
Start: Today at 3:19 PM

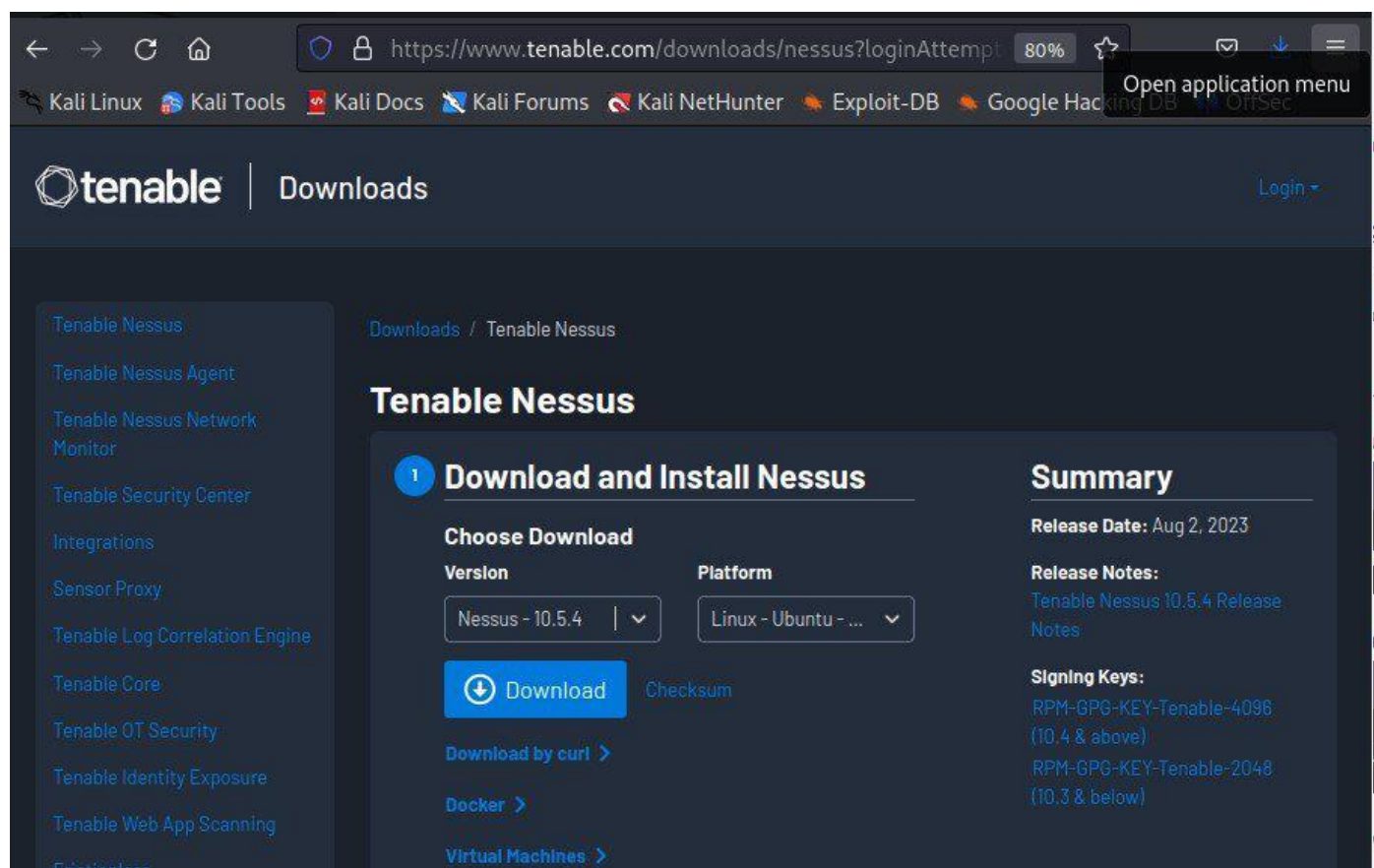
Vulnerabilities

Critical
High
Medium
Low
Info

Step 1. Download Nessus:

Visit the [Tenable Nessus website](https://www.tenable.com/downloads/nessus) and sign up for a Nessus Home edition account (if you don't have one):

After signing up, you'll receive a link to download the Nessus package and also receive an activation code. Download the appropriate package for your Linux distribution.



Download the appropriate package for your Linux distribution

Step 2. Install Nessus:

After downloading the Nessus package to your Downloads directory:

- **For Debian/Ubuntu:**

```
sudo dpkg -i /path/to/Nessus-<version>-<platform>.deb
```

```
(thinker@kali)-[~]
$ sudo dpkg -i /home/thinker/Downloads/Nessus-10.5.4-ubuntu1404_amd64.deb
[sudo] password for thinker:
Sorry, try again.
[sudo] password for thinker:
Selecting previously unselected package nessus.
(Reading database ... 397859 files and directories currently installed.)
Preparing to unpack .../Nessus-10.5.4-ubuntu1404_amd64.deb ...
Unpacking nessus (10.5.4) ...
Setting up nessus (10.5.4) ...
HMAC : (Module_Integrity) : Pass
SHA1 : (KAT_Digest) : Pass
SHA2 : (KAT_Digest) : Pass
SHA3 : (KAT_Digest) : Pass
TDES : (KAT_Cipher) : Pass
AES_GCM : (KAT_Cipher) : Pass
AES_ECB_Decrypt : (KAT_Cipher) : Pass
RSA : (KAT_Signature) : RNG : (Continuous_RNG_Test) : Pass
Pass
ECDSA : (PCT_Signature) : Pass
ECDSA : (PCT_Signature) : Pass
DSA : (PCT_Signature) : Pass
```

Debian Installation

- For CentOS/RedHat:

```
sudo rpm -ivh /path/to/Nessus-<version>-<platform>.rpm
```

Note: Replace '**<version>**' with the version number you downloaded, and '**<platform>**' with the appropriate platform identifier.

Step 3. Start the Nessus Service:

After installation, start the Nessus service:

```
sudo systemctl start nessusd
```

Verify that the service is running by running the following command.

```
sudo systemctl status nessusd
```

```
(thinker@kali)-[~]
$ sudo systemctl status nessusd
● nessusd.service - The Nessus Vulnerability Scanner
   Loaded: loaded (/lib/systemd/system/nessusd.service; disabled; preset: disabled)
   Active: active (running) since Sat 2023-08-12 01:23:38 MDT; 15min ago
     Main PID: 18954 (nessus-service)
        Tasks: 15 (limit: 2226)
      Memory: 113.8M
         CPU: 1min 25.655s
       CGroup: /system.slice/nessusd.service
              └─18954 /opt/nessus/sbin/nessus-service -q
                └─18956 nessusd -q
```

Running State-Disabled

If it is disabled then run the following command to enable it.

```
sudo systemctl enable nessusd
```



```
(thinker@kali)-[~]
$ sudo systemctl enable nessusd
Created symlink /etc/systemd/system/multi-user.target.wants/nessusd.service → /lib/systemd/system/nessusd.service.

(thinker@kali)-[~]
$
```

Now check whether it is enabled or not.

`sudo systemctl status nessusd`

```
(thinker@kali)-[~]
$ sudo systemctl status nessusd
● nessusd.service - The Nessus Vulnerability Scanner
   Loaded: loaded (/lib/systemd/system/nessusd.service; enabled; preset: disabled)
   Active: active (running) since Sat 2023-08-12 01:23:38 MDT; 24min ago
     Main PID: 18954 (nessus-service)
        Tasks: 15 (limit: 2226)
      Memory: 91.8M
         CPU: 1min 32.776s
    CGroup: /system.slice/nessusd.service
            └─18954 /opt/nessus/sbin/nessus-service -q
              └─18956 nessusd -q

Aug 12 01:23:38 kali systemd[1]: Started nessusd.service - The Nessus Vulnerability Scanner.
Aug 12 01:23:41 kali nessus-service[18956]: Cached 0 plugin libs in 0msec
Aug 12 01:23:41 kali nessus-service[18956]: Cached 0 plugin libs in 0msec

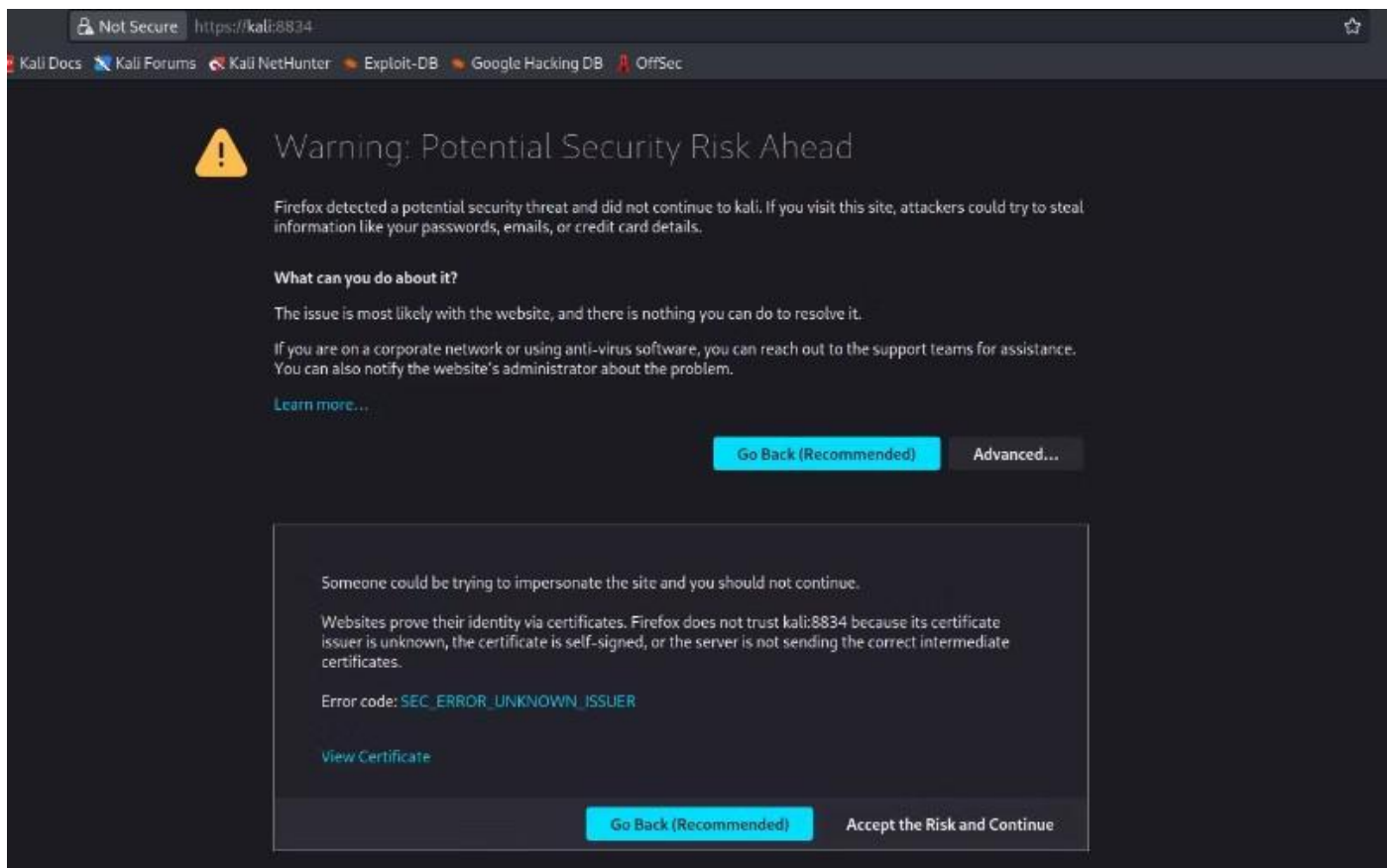
(thinker@kali)-[~]
$
```

Running State-Enabled

Step 4. Access the Nessus Web Interface:

Open a web browser and navigate to '<https://localhost:8834>'.

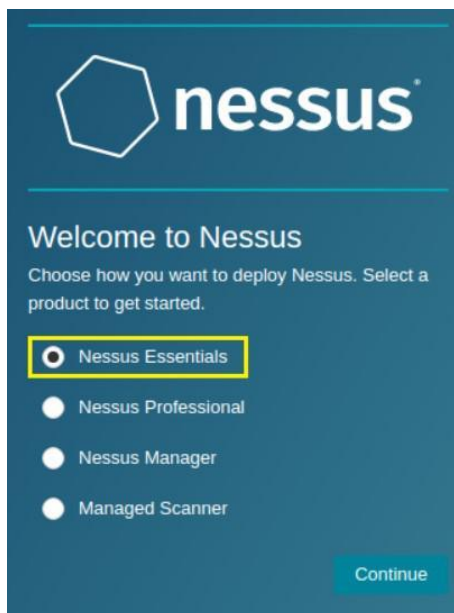
You might encounter a security warning due to the default self-signed SSL certificate used by Nessus. Accept the security risk and proceed.



Security Warning - accept the risk and proceed

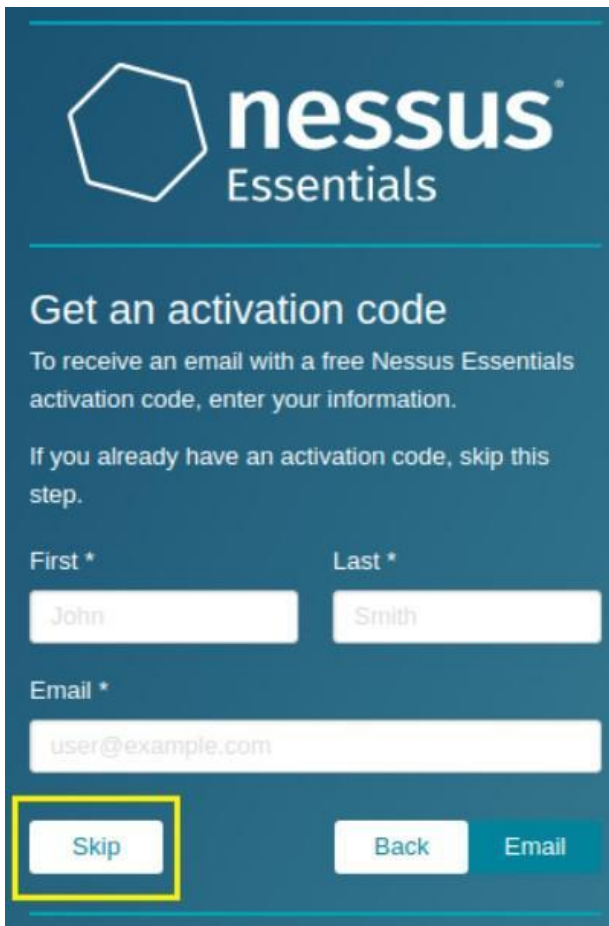
Step 5. Activate Nessus:

On the Welcome to Nessus screen, select the product version, **Nessus Essentials**.



Product Version

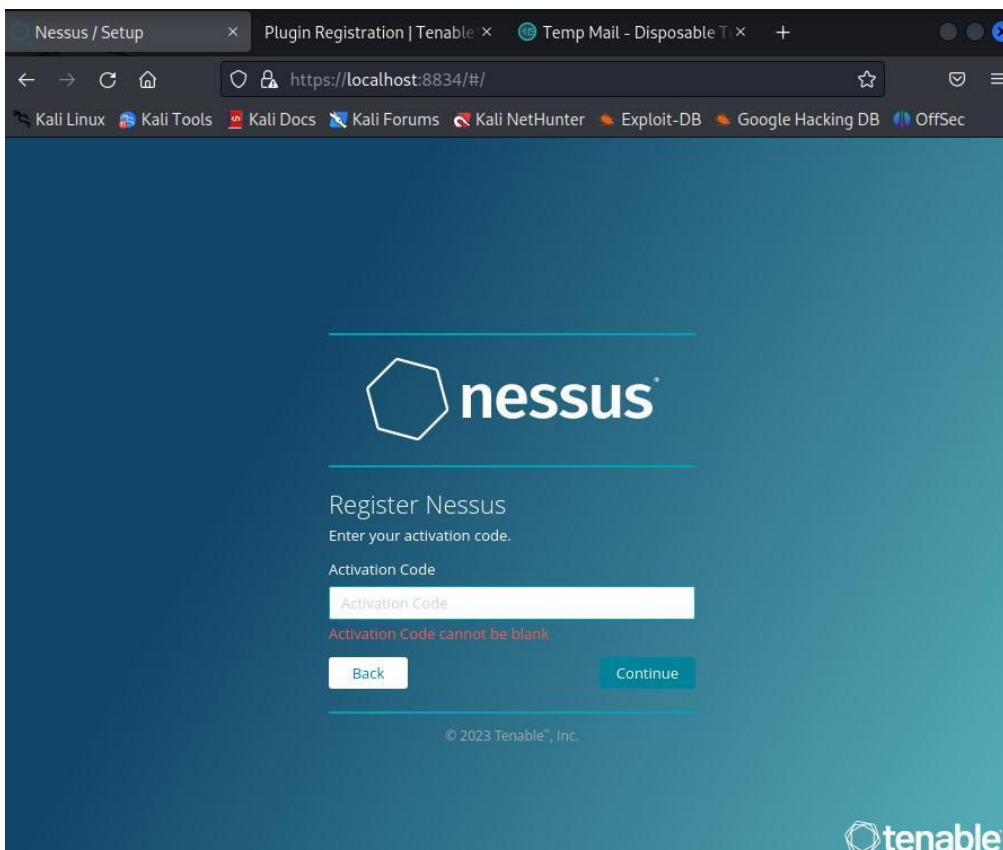
On the Get an activation code screen select skip, as you have previously obtained the activation code.



The image shows the Nessus Essentials registration page. At the top is the Nessus logo and the text "nessus Essentials". Below this is the heading "Get an activation code" followed by the instruction "To receive an email with a free Nessus Essentials activation code, enter your information." and a note "If you already have an activation code, skip this step." The form contains three input fields: "First *" with the value "John", "Last *" with the value "Smith", and "Email *" with the value "user@example.com". At the bottom are three buttons: "Skip" (highlighted with a yellow box), "Back", and "Email".

activation code

Enter the code to activate Nessus.

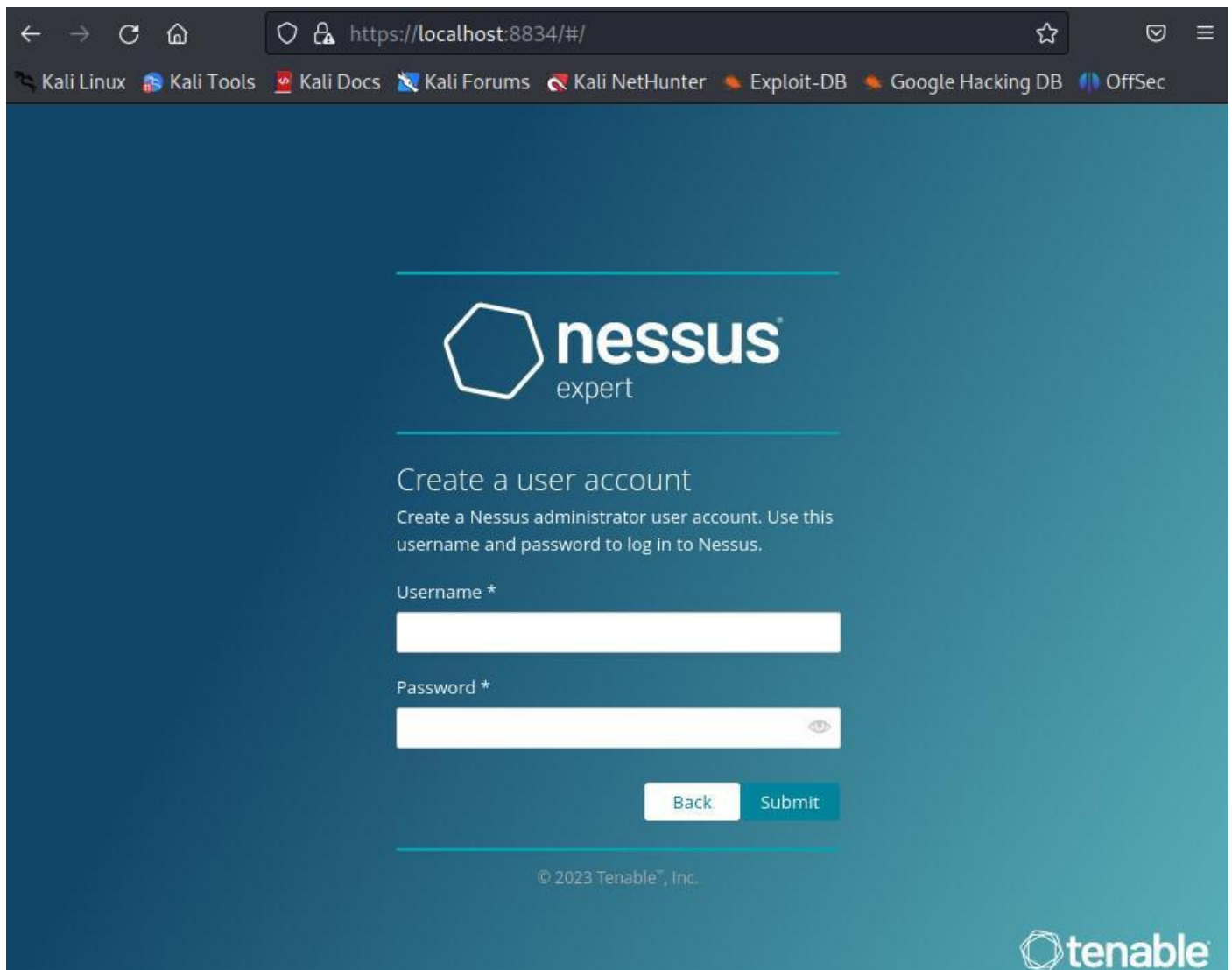


The image shows a web browser window displaying the Nessus registration page. The browser's address bar shows "https://localhost:8834/#/". The page has the Nessus logo and the heading "Register Nessus" with the instruction "Enter your activation code." Below this is an "Activation Code" input field. A red error message "Activation Code cannot be blank" is displayed below the input field. At the bottom are "Back" and "Continue" buttons. The footer includes the copyright "© 2023 Tenable, Inc." and the Tenable logo.

Enter the activation code

Step 6. Set Up a Username and Password:

After activation, Create a Nessus user account (**username and password**). This account will be used to access the Nessus web interface.



The screenshot shows a web browser window with the URL `https://localhost:8834/#/`. The browser's address bar and tabs are visible at the top. The main content area has a dark blue background with the Nessus Expert logo. Below the logo, the heading 'Create a user account' is displayed, followed by the instruction: 'Create a Nessus administrator user account. Use this username and password to log in to Nessus.' There are two input fields: 'Username *' and 'Password *'. The 'Password *' field includes a toggle icon for password visibility. At the bottom of the form are 'Back' and 'Submit' buttons. The footer contains the copyright notice '© 2023 Tenable™, Inc.' and the Tenable logo.

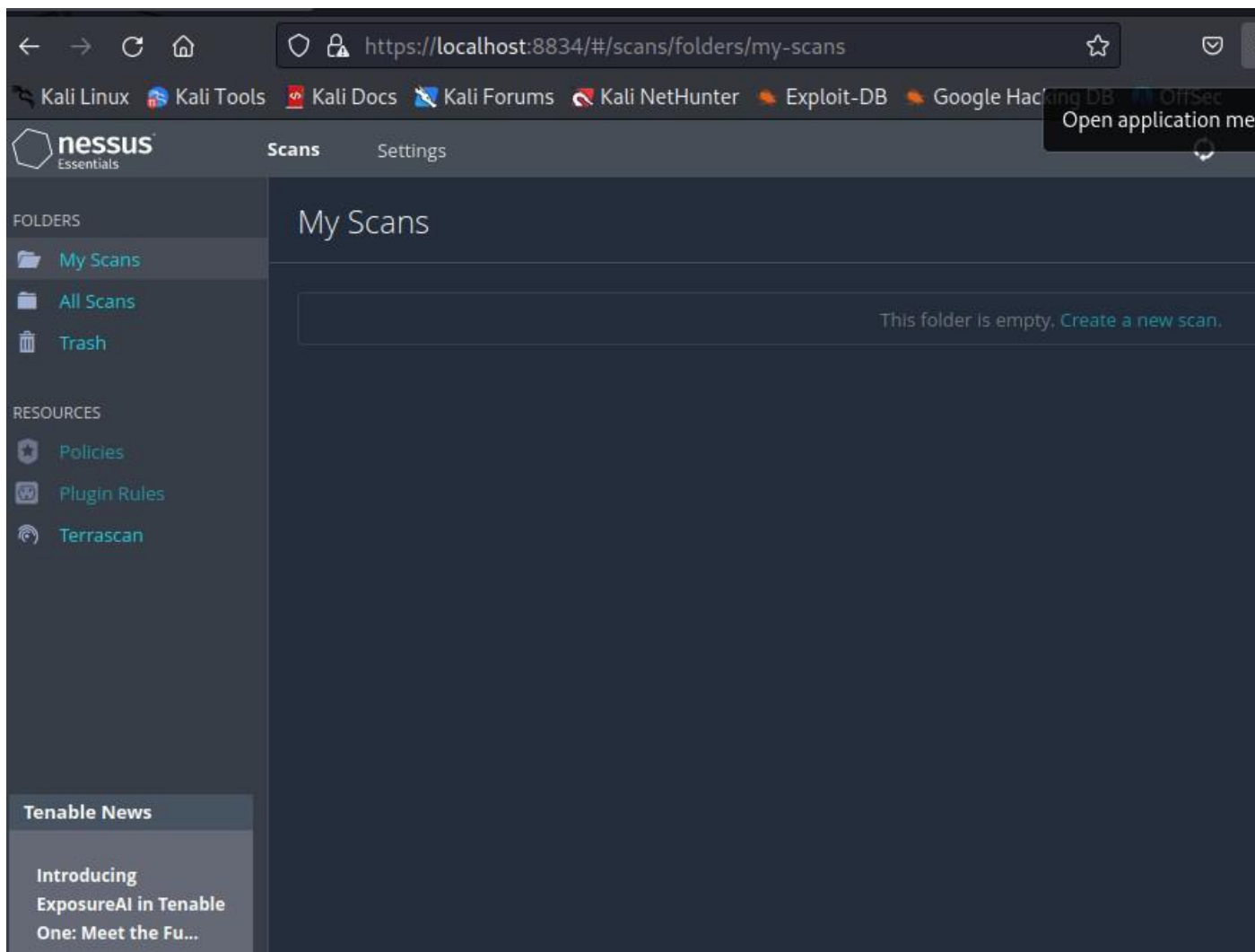
Setup Nessus User Account

Step 7. Log in to Nessus:

Log in to the Nessus web interface using the account you just created.

Step 8. Finish the Setup:

That's it! You now have Nessus installed and running on your Linux system. You can use the Nessus web interface to start scanning your network for vulnerabilities.



Nessus Web Interface

Let's go over two practical examples of using Nessus for vulnerability scanning.

How to Use Nessus on Linux?

Example 1: Basic scan

In this example, we will perform an initial vulnerability scan on the target system using Nessus.

Step 1: Start Nessus

Verify that you have turned on the Nessus service with the following command:

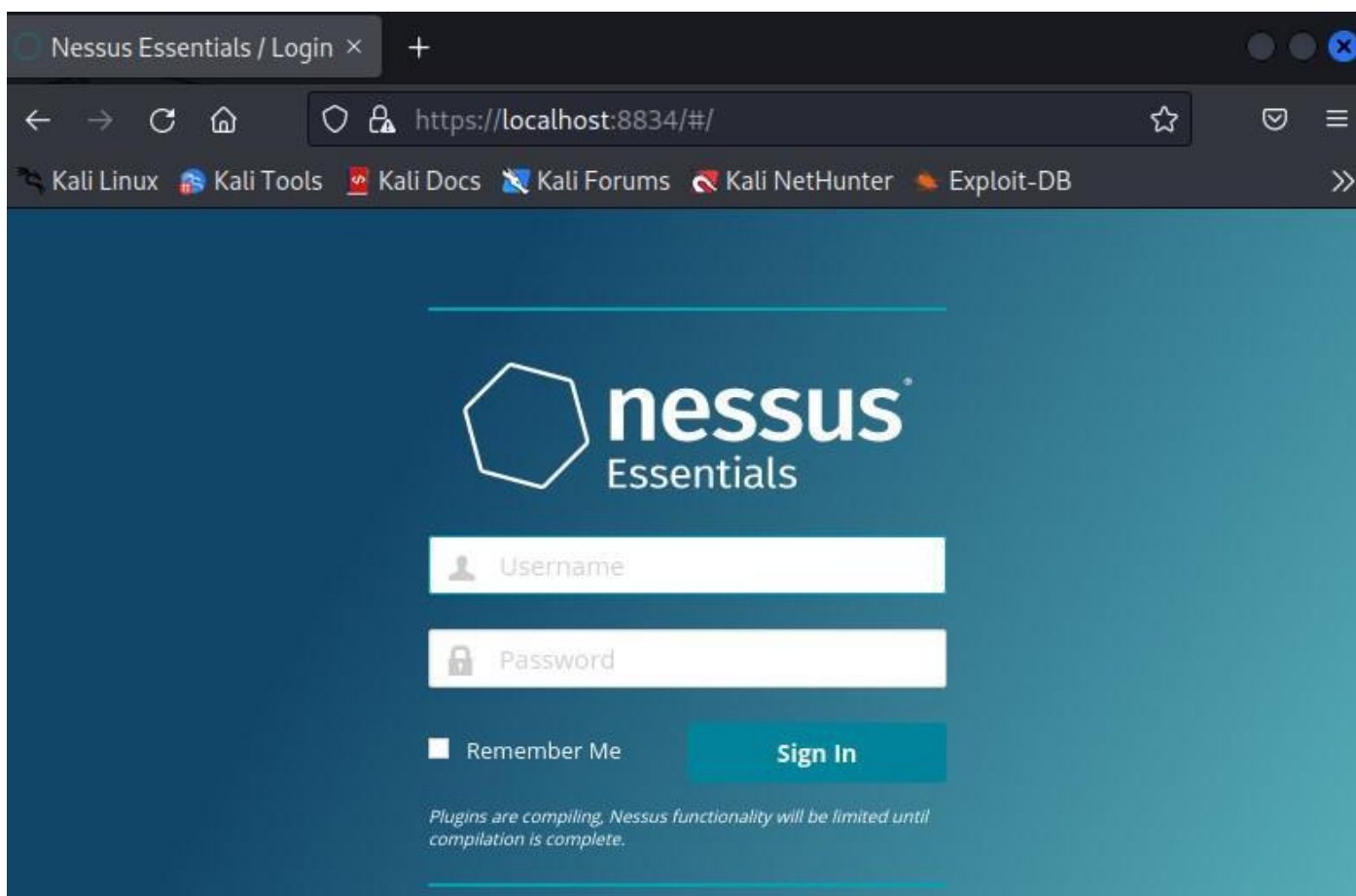
```
sudo systemctl start nessusd
```

```
thinker@kali: ~  
File Actions Edit View Help  
—(thinker@kali)-[~]  
$ ls  
Desktop Documents Downloads Music Pictures Public Templates Videos  
—(thinker@kali)-[~]  
$ sudo systemctl start nessusd  
[sudo] password for thinker:  
—(thinker@kali)-[~]  
$
```

Start Nessus

Step 2: Go to the Nessus Web Connection

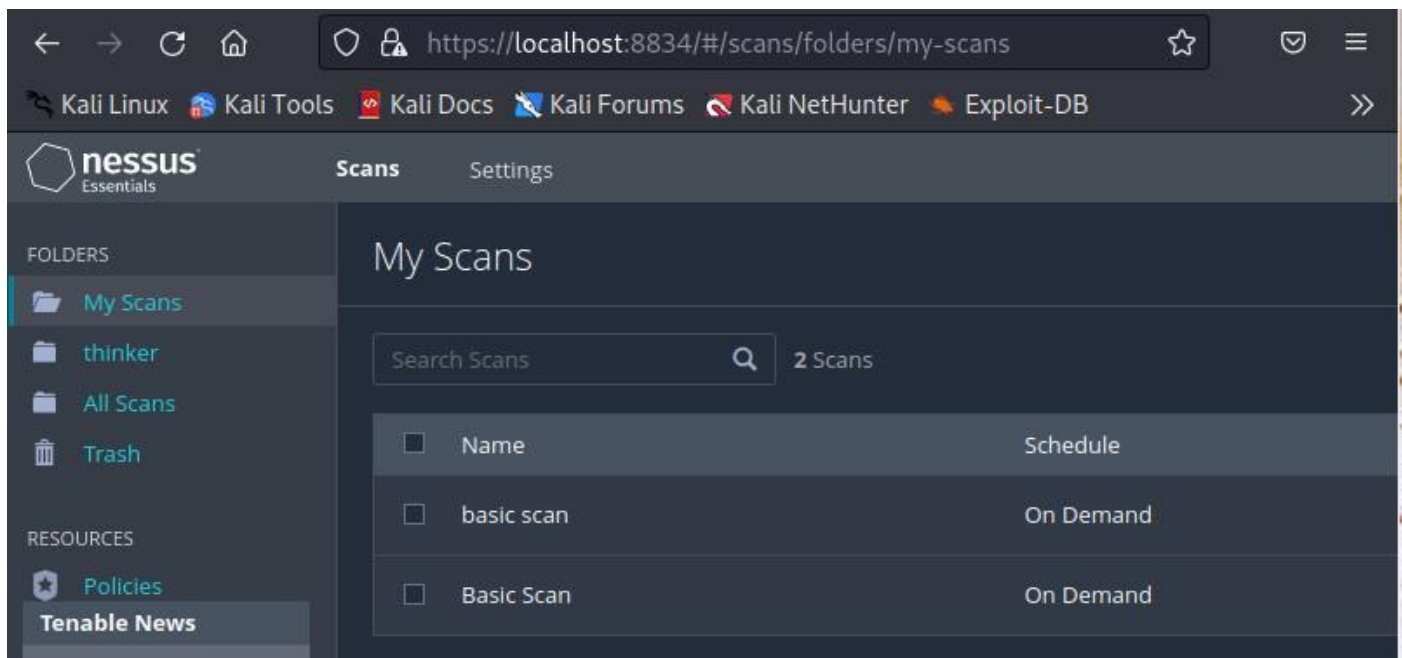
Open a web browser and navigate to **https://localhost:8834**. Log in with your Nessus username and password.



Log In Page

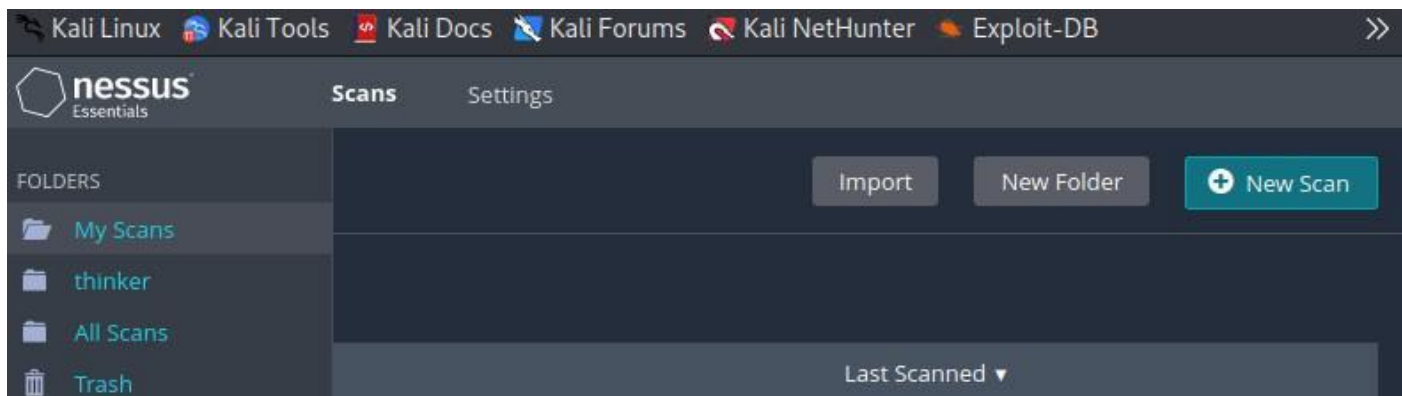
Step 3: Create a New Scan

1. Click on the "Scans" tab in the top navigation bar.



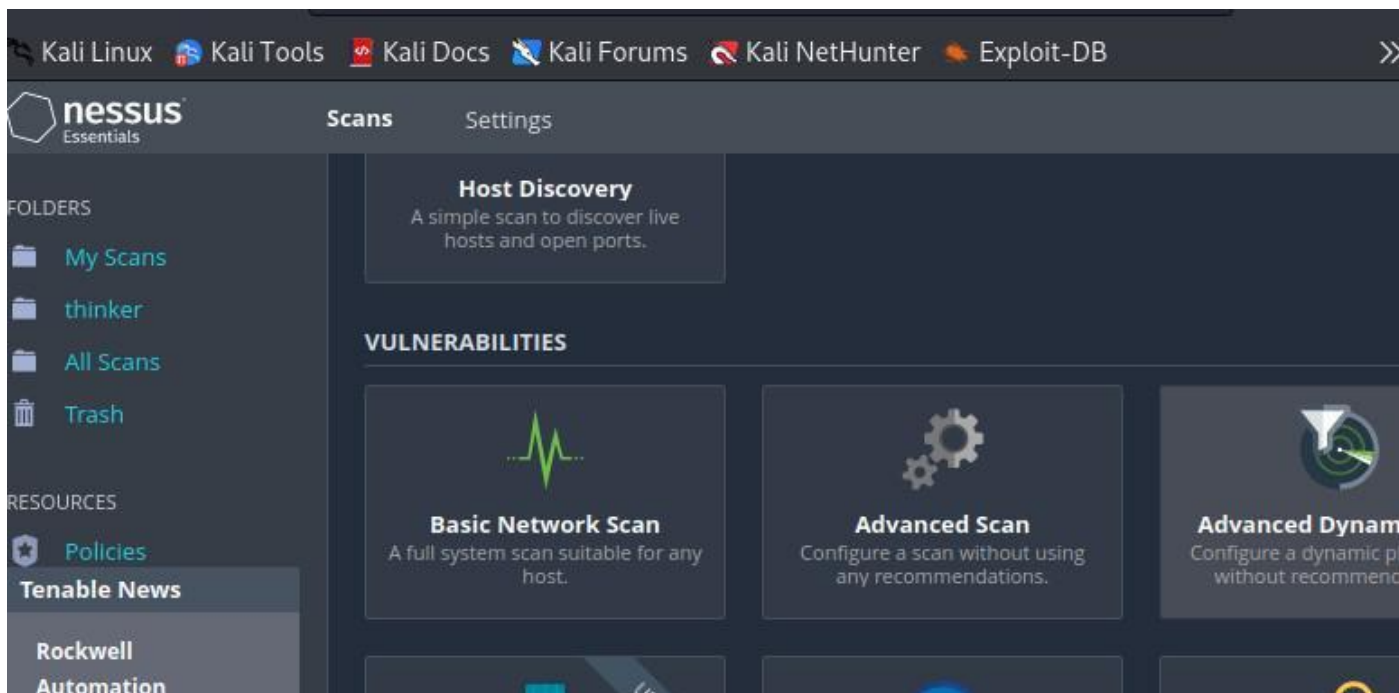
Scan Tab

2. Click the "New Scan" button.



Click on New Scan

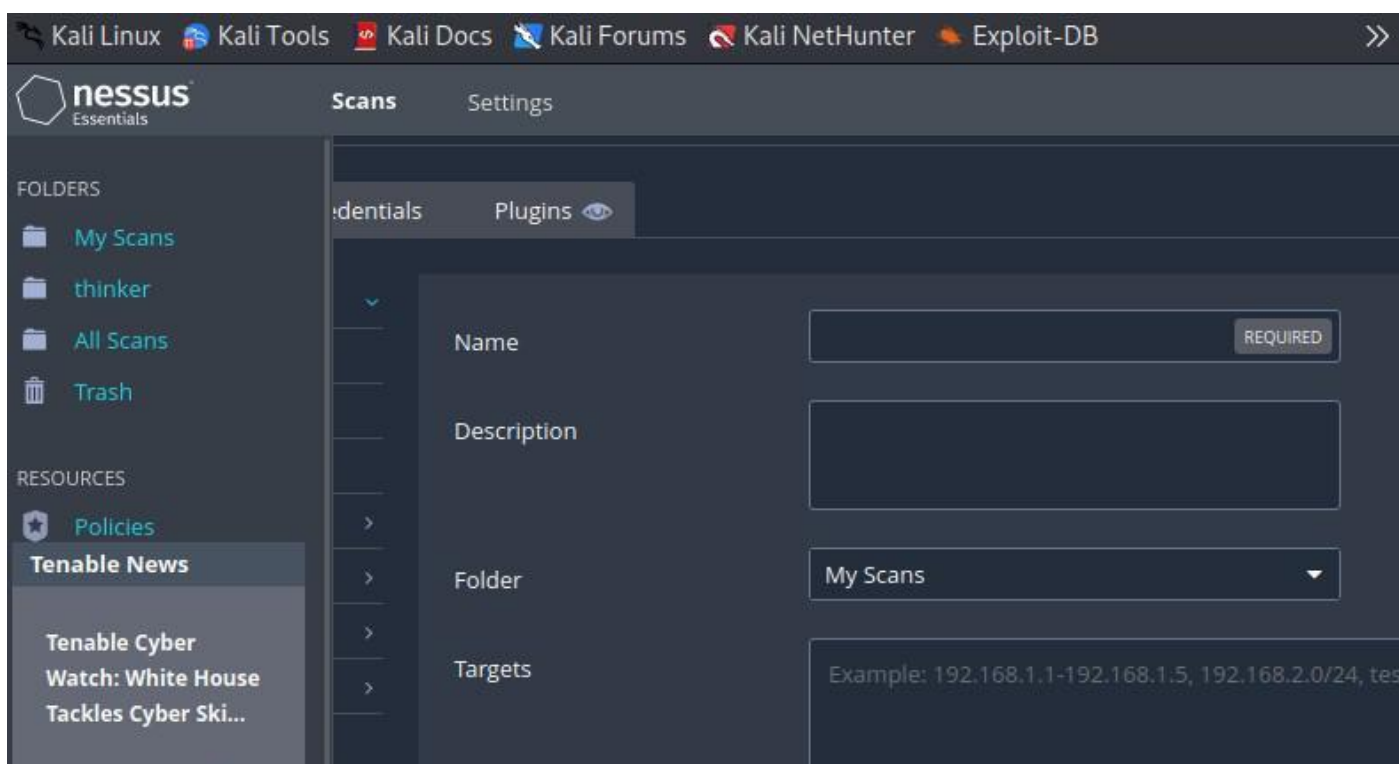
3. Select the "Basic Network Scan" template.



Templates

Step 4: Configure the Scan

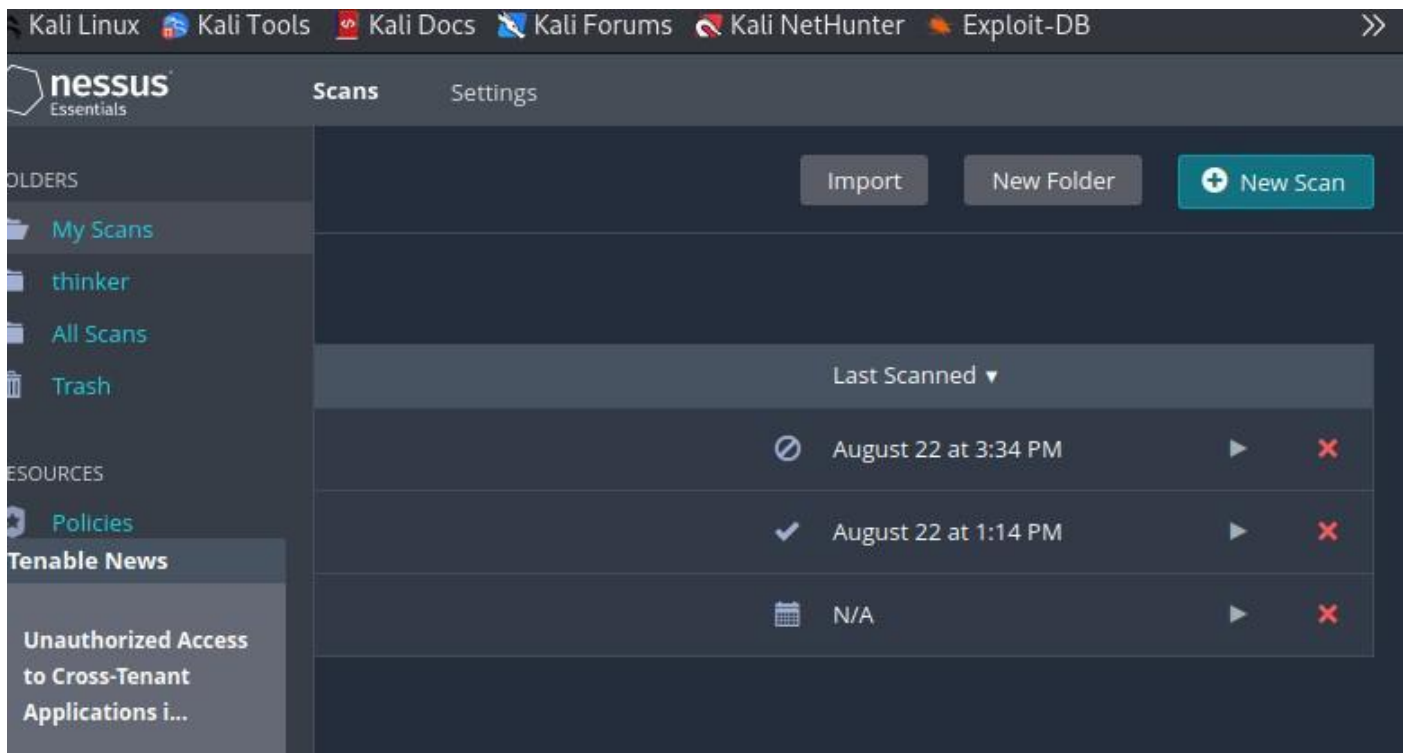
In the "Targets" section, enter the [IP address](#) or hostname of the target system you want to scan. You can leave other settings at their default values for this basic example.



Enter the scan name and target IP address

Step 5: Start the Scan

Click the "Save" button to save the scan configuration, then click the "Launch" button to start the scan.



Click on launch button

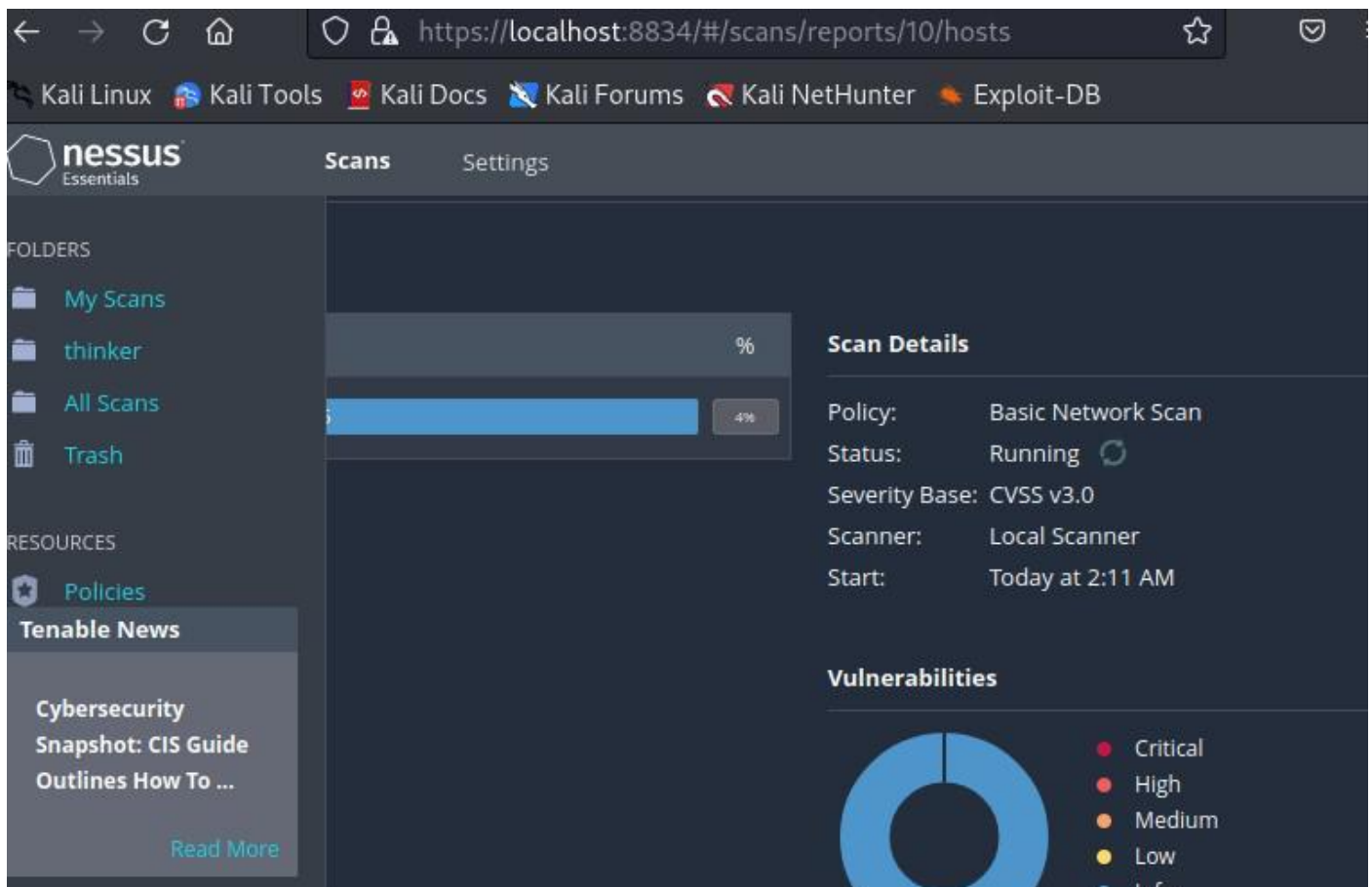
In this example, we started a basic network scan using Nessus. We provided the target's IP address or hostname and used the "**Basic Network Scan**" template. This scan looks for common vulnerabilities in the target system.

Example 2: Create a Report

In this example, we will create a report from the results of a previous scan.

Step 1: Wait for the scan to complete

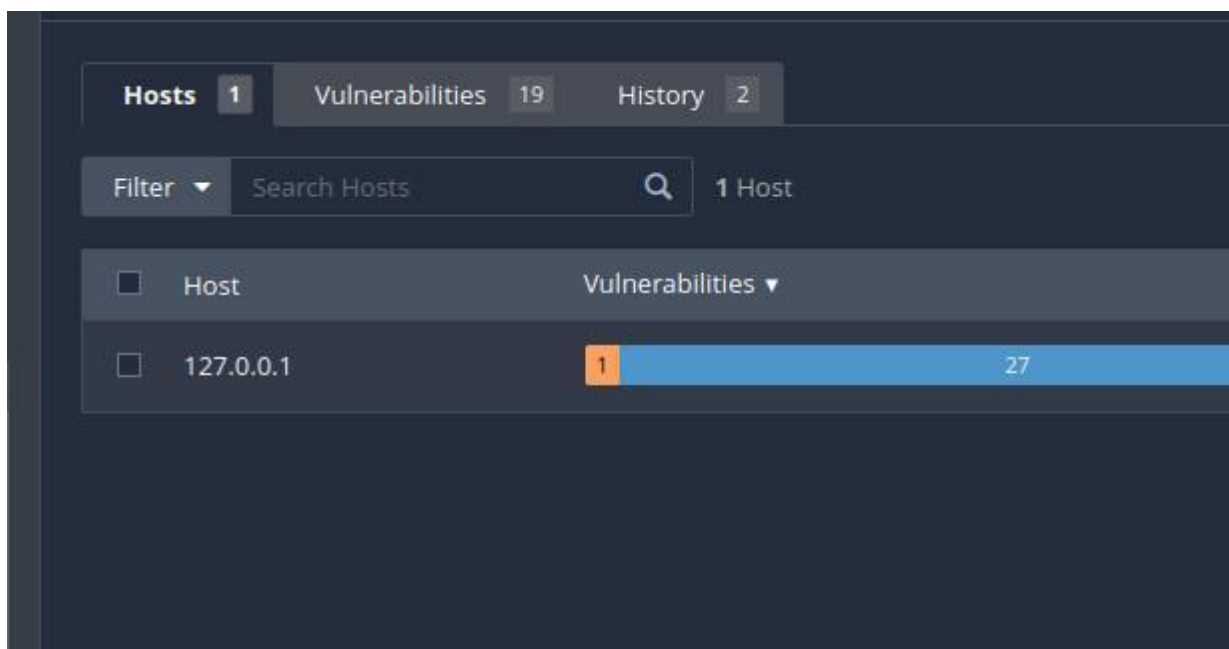
Wait for the scan to complete. Depending on the size of the target network, this can take some time.



Running Scan

Step 2: Access Scan Results

Once the scan is complete, go to the "Scans" tab. Click on the scan you performed earlier ("Basic Scan").



Scan tab

Step 3: Generate a Report

1. In the scan details page, click the "Export" button.
2. Select the format for the report (e.g., PDF, HTML).

3. Choose the report template (e.g., "Executive Summary").
4. Click the "Generate" button.

Generate Report

Report Format: ☐ HTML ☒ PDF ☐ CSV

Select a Report Template:

SYSTEM

Complete List of Vulnerabilities by Host

Detailed Vulnerabilities By Host

Detailed Vulnerabilities By Plugin

Vulnerability Operations

Template Description:

This report provides a summary list of vulnerabilities detected in the scan.

Filters Applied:

None

Formatting Options:

☒ Include page breaks between vulnerability results

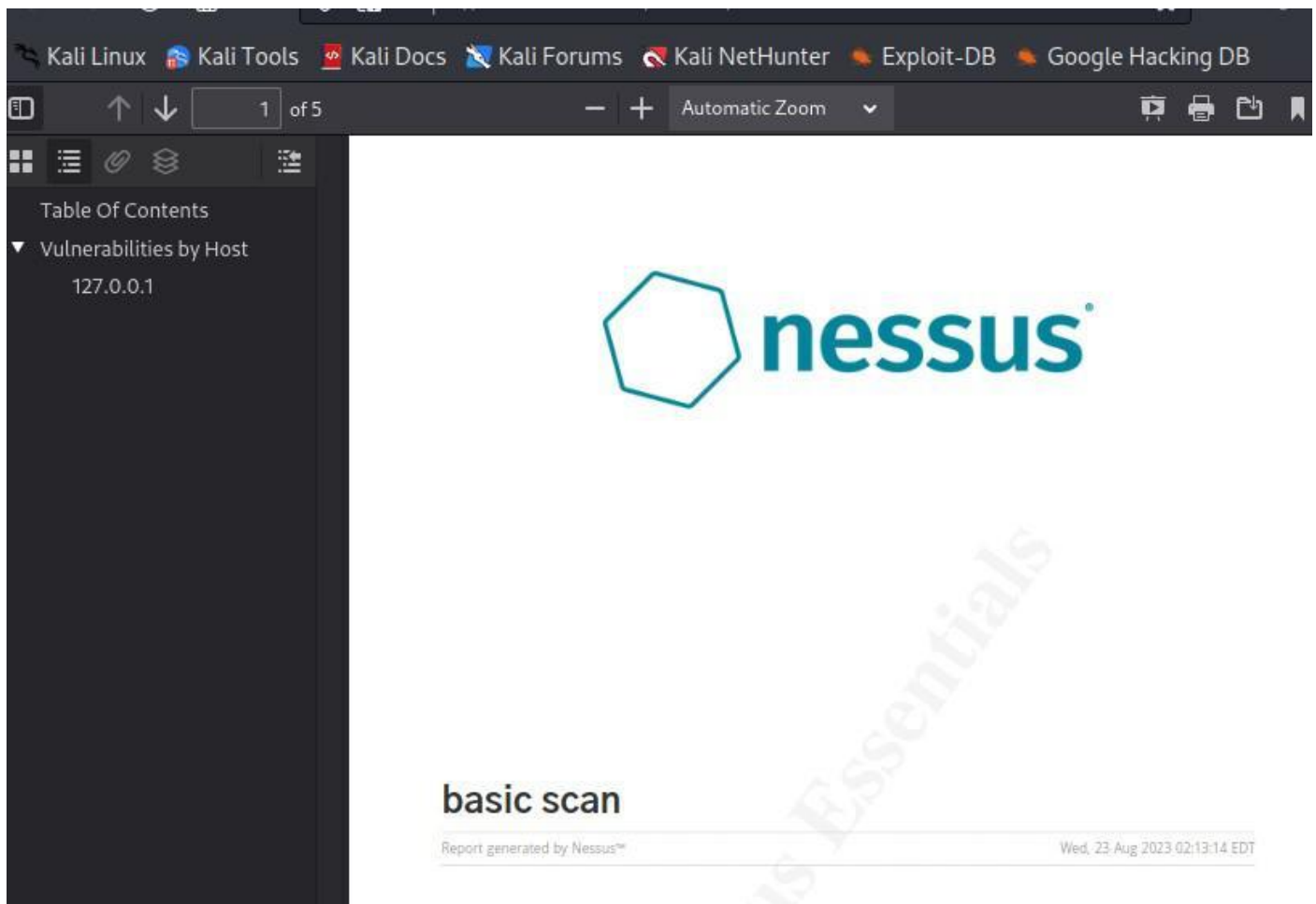
Generate Report

Cancel

Generate a Report

Step 4: Download the Report

Once the report is generated, you'll see a link to download it. Click the link to download the report in the selected format.



In this example, we get the scan results and generate a report. Nessus offers a variety of report templates, such as "Executive Summary" and "Full Audit".

Task – Students build a playbook using Flexible IR / Osquery

Download link: <https://osquery.io/downloads/official/5.21.0>

Commands:

- `curl -L https://pkg.osquery.io/rpm/GPG | sudo tee /etc/pki/rpm-gpg/RPM-GPG-KEY-osquery`
- `yum install yum-utils -y`
- `sudo yum-config-manager --add-repo https://pkg.osquery.io/rpm/osquery-s3-rpm.repo`
- `sudo yum-config-manager --enable osquery-s3-rpm-repo`
- `sudo yum install osquery`

Verify osquery is working

```
# osqueryi
```

You should enter the interactive shell:

```
Using a virtual database. Need help, type '.help'
```

```
osquery>
```

Test a simple query:

```
# select * from processes limit 5;
```

```
If data returns → installation is good.
```

2) Start with core investigation queries

System inventory

```
# select * from system_info;
```

Logged-in users

```
# select * from logged_in_users;
```

Running processes

```
# select pid, name, path from processes;
```

Installed software

```
# select * from programs;
```

Open network connections

```
# select pid, address, port from listening_ports;
```

Autoruns / persistence

```
# select * from startup_items;
```

3) Threat hunting use cases

Suspicious processes

```
# select name, cmdline from processes
where name like '%powershell%';
```

External connections

select * from process_open_sockets where remote_address not like '192.168%';

Category	Command	Purpose	Example
Service management	systemctl status osqueryd	Check osquery daemon status	Verify if service is running
	systemctl start osqueryd	Start osquery daemon	Start background monitoring
	systemctl stop osqueryd	Stop daemon	Stop before config change
	systemctl restart osqueryd	Restart service	Apply new config
	systemctl enable osqueryd	Enable at boot	Persistence after reboot
Interactive shell	osqueryi	Launch interactive console	Run live queries
	.tables	List all available tables	Discover data sources
	.schema <table>	View schema of table	.schema processes
	.help	Show help menu	CLI assistance
System information	SELECT * FROM system_info;	Host OS & hardware info	IR baseline
	SELECT * FROM uptime;	System uptime	Detect reboots
	SELECT hostname, cpu_brand FROM system_info;	Host identity	Asset validation
User investigation	SELECT * FROM users;	Local users list	Account audit
	SELECT * FROM logged_in_users;	Active sessions	Live access detection
	SELECT * FROM last;	Login history	Persistence check
Process investigation	SELECT * FROM processes;	Running processes	Malware hunt
	SELECT name,pid,path FROM processes;	Key process details	Suspicious binary check
	SELECT * FROM process_open_files WHERE pid=<PID>;	Files used by process	Data exfil analysis
File integrity / artifacts	SELECT * FROM file WHERE path='/etc/passwd';	Check file metadata	Integrity verification
	SELECT * FROM hash WHERE path='/bin/bash';	File hash	IOC comparison
	SELECT * FROM yara WHERE path='/home';	Scan with YARA	Threat detection
Network investigation	SELECT * FROM listening_ports;	Open ports	Backdoor detection
	SELECT * FROM process_open_sockets;	Process-network mapping	C2 analysis
	SELECT * FROM interface_addresses;	IP addresses	Network inventory
	SELECT * FROM arp_cache;	ARP entries	Lateral movement
Persistence checks	SELECT * FROM crontab;	Cron jobs	Scheduled malware
	SELECT * FROM startup_items;	Startup programs	Persistence vectors
	SELECT * FROM services;	System services	Rogue services
Log & security	SELECT * FROM sudoers;	Sudo access	Privilege audit
	SELECT * FROM groups;	Group memberships	Admin enumeration
Package auditing	SELECT * FROM rpm_packages;	Installed packages	Unauthorized software
	SELECT name,version FROM rpm_packages;	Package versions	Vulnerability mapping
	SELECT * FROM kernel_modules;	Loaded modules	Rootkit detection

<i>Kernel & hardware</i>	SELECT * FROM block_devices;	Disk devices	Rogue storage
<i>Configuration paths</i>	/etc/osquery/osquery.conf	Main config file	Query packs & logging
	/var/log/osquery/	Logs directory	Incident evidence

High-value IR query examples:

Scenario	Query
Suspicious processes	SELECT pid,name,path,cmdline FROM processes WHERE name LIKE '%ssh%';
Unauthorized users	SELECT * FROM users WHERE uid >= 1000;
External connections	SELECT * FROM process_open_sockets WHERE remote_address NOT LIKE '127.%';
Persistence hunt	SELECT * FROM crontab WHERE command NOT LIKE '%system%';
Recently modified binaries	SELECT * FROM file WHERE directory='/usr/bin' ORDER BY mtime DESC LIMIT 20;

Osquery IR Cheat Sheet (Top Queries Used by Analysts)

1) Host triage

Objective	Query
Identify host details	SELECT hostname, os_version, cpu_brand FROM system_info;
System uptime	SELECT * FROM uptime;
Kernel version	SELECT version FROM kernel_info;

2) User & access investigation

Objective	Query
Local users	SELECT * FROM users;
Active sessions	SELECT * FROM logged_in_users;
Login history	SELECT * FROM last;
Sudo privileges	SELECT * FROM sudoers;
Admin groups	SELECT * FROM groups WHERE groupname='wheel';

3) Process & malware hunting

Objective	Query
Running processes	SELECT pid,name,path,cmdline FROM processes;
Suspicious binaries	SELECT * FROM processes WHERE path NOT LIKE '/usr/%';

Objective	Query
Parent-child chain	SELECT pid,parent,name,path FROM processes;
Open files by process	SELECT * FROM process_open_files;

4) Persistence detection

Objective	Query
Cron jobs	SELECT * FROM crontab;
Services	SELECT name,status,path FROM services;
Startup programs	SELECT * FROM startup_items;

5) Network triage

Objective	Query
Listening ports	SELECT * FROM listening_ports;
Active sockets	SELECT * FROM process_open_sockets;
External connections	SELECT pid,remote_address,remote_port FROM process_open_sockets WHERE remote_address NOT LIKE '127.%';
IP addresses	SELECT * FROM interface_addresses;

6) File integrity & artifacts

Objective	Query
File hash	SELECT * FROM hash WHERE path='/bin/bash';
Modified binaries	SELECT path,mtime FROM file WHERE directory='/usr/bin' ORDER BY mtime DESC LIMIT 20;
YARA scan	SELECT * FROM yara WHERE path='/home';

7) Package & kernel audit

Objective	Query
Installed packages	SELECT * FROM rpm_packages;
Kernel modules	SELECT * FROM kernel_modules;
Block devices	SELECT * FROM block_devices;

Osquery IR Playbook mapped to ATT&CK-style phases

Initial access validation

Goal	Query
------	-------

Check new users `SELECT * FROM users WHERE uid >= 1000;`

Suspicious logins `SELECT * FROM last WHERE host NOT LIKE '192.%';`

Execution detection

Goal	Query
------	-------

Suspicious commands `SELECT pid,cmdline FROM processes WHERE cmdline LIKE '%wget%';`

Binary from temp dirs `SELECT * FROM processes WHERE path LIKE '/tmp/%';`

Persistence discovery

Goal	Query
------	-------

Cron persistence `SELECT * FROM crontab WHERE command NOT LIKE '%system%';`

Rogue service `SELECT * FROM services WHERE status!='running';`

Privilege escalation

Goal	Query
------	-------

Sudo abuse `SELECT * FROM sudoers;`

Wheel group users `SELECT * FROM user_groups WHERE groupname='wheel';`

Defense evasion

Goal	Query
------	-------

Disabled security tools `SELECT * FROM processes WHERE name LIKE '%audit%';`

Hidden files `SELECT * FROM file WHERE filename LIKE '.*%';`

Credential access

Goal	Query
------	-------

SSH keys `SELECT * FROM file WHERE path LIKE '/home/%/.ssh/%';`

Shadow access `SELECT * FROM process_open_files WHERE path='/etc/shadow';`

Discovery

Goal	Query
------	-------

Network interfaces `SELECT * FROM interface_details;`

ARP enumeration `SELECT * FROM arp_cache;`

Lateral movement

Goal	Query
------	-------

Remote connections	SELECT * FROM process_open_sockets WHERE remote_address!='127.0.0.1';
--------------------	---

SSH processes	SELECT * FROM processes WHERE name='ssh';
---------------	---

Command & control

Goal	Query
------	-------

External IP traffic	SELECT * FROM process_open_sockets WHERE remote_port NOT IN (80,443);
---------------------	---

Exfiltration indicators

Goal	Query
------	-------

Large file access	SELECT * FROM process_open_files WHERE size > 100000000;
-------------------	--

Network investigation

- Using Wireshark
 - `tcpdump -i eth0 -w capture.pcap`
- Use NMAP
 - Identify SYN scans in wireshark

Host investigation

On windows, install Sysmon

- Download URL: [Sysmon - Sysinternals | Microsoft Learn](#)
- Commands: [Sysmon: How to install, upgrade, and uninstall](#)

Windows host investigation

- To check Sysmon logs,
 - Event viewer → Application and service logs → Microsoft → Windows → Sysmon

Linux host investigation

- `last`
- `who`
- `ps -ef`
- `netstat -tulpn`
 - investigate unauthorized login.

Application investigation – collecting logs

Windows logs (Windows 10)

A. Export Windows Event Logs manually

Use Event Viewer:

1. Open **Event Viewer**
2. Go to:
 - Windows Logs → Security
 - Windows Logs → System
 - Windows Logs → Application
3. Right-click → **Save All Events As**
4. Save as .evtx or .csv

B. Export via PowerShell (recommended)

For automation and periodic collection:

Export Security log

```
wevtutil epl Security C:\logs\Security.evtx
```

Export System log

```
wevtutil epl System C:\logs\System.evtx
```

Export Application log

```
wevtutil epl Application C:\logs\Application.evtx
```

Windows Firewall logs

Enable firewall logging first

1. Windows Defender Firewall → Advanced Settings
2. Properties → Logging tab
3. Enable:
 - Log dropped packets
 - Log successful connections
4. Default log location:
C:\Windows\System32\LogFiles\Firewall\pfirewall.log

Just copy:

- copy C:\Windows\System32\LogFiles\Firewall\pfirewall.log C:\logs\

CentOS logs

System logs:

- /var/log/messages
- /var/log/secure
- /var/log/cron
- /var/log/audit/audit.log

Export:

- cp /var/log/messages /tmp/logs/
- cp /var/log/secure /tmp/logs/

Firewall logs on CentOS:

- /var/log/messages
- journalctl -u firewalld > /tmp/firewalld.log

Apache logs (CentOS):

- /var/log/httpd/access_log
- /var/log/httpd/error_log

Export:

- cp /var/log/httpd/access_log /tmp/logs/
- cp /var/log/httpd/error_log /tmp/logs/